S130 Design Rule Manual

Spec No. 002-22690

September 29, 2021
DRAFT—Approved for Customer Distribution

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Revision History

Release Date	Page	Description
September 29, 2021	–	Rev. **: Initial release (draft).

1 Introduction

This document describes circuit design guidelines, including physical design rules, for the SkyWater S130 process technology. These guidelines are based on the manufacturing capability and operation principles of S130 integrated circuits. The products manufactured in this technology are designed in the 130 nm space, where the drawn design is the same size as the final product, with no optical shrink occurring in the mask shop.

SkyWater process and design engineers are responsible for defining and revising the document content. They must review and approve all content changes proposed by other parties. Contact SkyWater Technology to propose changes.

Note: This document is a DRAFT, and all content is subject to change at any time without notice. Existing content might contain technical errors and omissions and should not be relied upon for accuracy and completeness.

1.1 Technology Features

Although SkyWater developed the S130 technology primarily for programmable system-on-a-chip (PSoC) products, the following features can be adapted to support many mixed-signal applications:

- Power supply voltage (V_{DD}): 1.8 V, 3.3 V, or 5.0 V
- Cross-well (n+ in a p-well to p+ in an n-well) spacing: 0.47 µm core, 0.52 µm periphery
- Gate material: n-doped polysilicon
- Minimum physical gate length: 0.15 µm (1.8 V) or 0.50 µm (5.0 V)
- Gate-oxide thickness: 32 Å for 1.8 V field-effect transistors (FETs), 110 Å for 5.0 V and higher FETs
- I_{DSAT} ($V_{GS} = V_{DS}$ = 1.2 V): 500 µA/µm (NMOS), 190 µA/µm (PMOS)
- Triple well on a p-type substrate for device optimization and noise immunity
- Multiple threshold voltage (V_t) options for 1.8 V devices
- 12 V and 20 V devices for high-voltage applications
- Optional metal-insulator-metal (MIM) capacitors: 2.0 fF/µm^2
- Inverter gate delay (fanout = 1): 32 ps
- Single polysilicon layer with nonsalicided source and drain
- Titanium nitride (TiN) local interconnect
- Aluminum-based, five-metal-layer (5LM) stack

1.2 Supported Devices

Table 1-1 on page 11 describes the devices currently supported in the S130 technology.

Table 1-1. Supported Devices (Page 1 of 3)

Cell Name	Device Description
NMOS Devices	
nmos	1.8 V NMOS device
nmos_lvt	Low-V_t 1.8 V NMOS device
nmos_v5	5 V NMOS device
nmos_nat_v3	Native 3.3 V NMOS device
nmos_nat_v5	Native 5 V NMOS device
nmos_de_v12	5–12 V drain-extended NMOS device
nmos_de_v20	5–20 V drain-extended NMOS device
nmos_de_iso_v20	Isolated 5–20 V drain-extended NMOS device
nmos_nat_v20	Native 5–20 V drain-extended NMOS device
nmos_zvt_v20	Zero-V_t 5–20 V drain-extended NMOS device
PMOS Devices	
pmos	1.8 V PMOS device
pmos_lvt	Low-V_t 1.8 V PMOS device
pmos_hvt	High-V_t 1.8 V PMOS device
pmos_v5	5 V PMOS device
pmos_de_v12	5–12 V drain-extended PMOS device
pmos_de_v20	5–20 V drain-extended PMOS device
Resistors	
rpoly	Poly resistor
rpoly_hp	P+ poly resistor
rpoly_hp2k	P- poly resistor
rmet	li resistor
	met1 resistor
	met2 resistor
	met3 resistor
	met4 resistor
	met5 resistor
rndiff	N-type diffusion resistor
rndiff_v5	5 V n-type diffusion resistor
rpdiff	P-type diffusion resistor
rpdiff_v5	5 V p-type diffusion resistor
rpwell	Isolated p-well resistor

Table 1-1. Supported Devices (Page 2 of 3)

Cell Name	Device Description
Capacitors	
cm3m4	MIM capacitor over met3
cm4m5	MIM capacitor over met4
Diodes	
dnsd_pw	N+ diffusion / p-well diode
dnsd_pw_v5	5 V n+ diffusion / p-well diode
dnsd_pw_lvt	Low-V_t n+ diffusion / p-well diode
dnsd_pw_nat	Native n+ diffusion / p-well diode
dpsd_nw	P+ diffusion / n-well diode
dpsd_nw_v5	5 V p+ diffusion / n-well diode
dpsd_nw_lvt	Low-V_t p+ diffusion / n-well diode
dpsd_nw_hvt	High-V_t p+ diffusion / n-well diode
dipw_dnw	Isolated p-well / deep n-well diode
dipw_dnw_v5	5 V isolated p-well / deep n-well diode
dipw_dnw_v12	12 V isolated p-well / deep n-well diode
dipw_dnw_v20	20 V isolated p-well / deep n-well diode
ddnw_sub	Deep n-well / p- substrate diode
ddnw_sub_v5	5 V deep n-well / p- substrate diode
ddnw_sub_v12	12 V deep n-well / p- substrate diode
ddnw_sub_v20	20 V deep n-well / p- substrate diode
Bipolar Junction Transistors	
pnp	Parasitic pnp bipolar junction transistor
pnp_5x	Large parasitic pnp bipolar junction transistor
npn_1x1_v5	5 V gated parasitic npn bipolar junction transistor
npn_1x1	1 µm × 1 µm emitter parasitic npn bipolar junction transistor
npn_1x2	1 µm × 2 µm emitter parasitic npn bipolar junction transistor
Electrostatic Discharge (ESD) Protection Devices	
nmos_esd	NMOS device for ESD protection
nmos_esd_v5	5 V NMOS device for ESD protection
nmos_esd_nat_v5	Native 5 V NMOS device for ESD protection
pmos_esd_v5	5 V PMOS device for ESD protection
dnsd_pw_esd	N+ p-well diode for ESD protection
dnsd_pw_esd_v5	5 V n+ p-well diode for ESD protection

Table 1-1. Supported Devices (Page 3 of 3)

Cell Name	Device Description
Electrostatic Discharge (ESD) Protection Devices (continued)	
dpsd_nw_esd	P+ n-well diode for ESD protection
dpsd_nw_esd_v5	5 V p+ n-well diode for ESD protection
Input/Output (I/O) Pads	
pad_bond	I/O bond pad
pad_probe	I/O probe pad
pad_microprobe	I/O microprobe pad
Seal-Ring Structure	
sealring	Hermetic seal ring

1.3 Related Documents

This document is intended to be used in conjunction with the SkyWater engineering specs listed in *Table 1-2*. Listed documents excluded from the S130 process design kit (PDK) are available from SkyWater Technology by request unless otherwise noted.

Table 1-2. Related Documents

Spec No.	Notes	Document Title
00-00064		*Record Retention Policy and Procedures*
01-70004		*S8 and C9 Technology Family Stress Relief, Bond Pad, E-test Pad, Scribeline Rules*
001-01905		*S8 Technology Family Topographical Design Rules*
001-02735		*S8 Platform Model Requirements and Gap Analysis*
001-04067	1	*S8 Technology Family Objective Specification*
001-04469		*S8 Miscellaneous Topological Design Rules (MTDR)*
002-21908		*S8 Electrical Design Rules (EDR)*
002-22691		*S130 PDK User Guide*

1. Access to this document is restricted to a limited number of SkyWater employees only.

2 Layout Information

2.1 Drawing and Mask Layers

Table 2-1 describes the drawing and mask layer sequence for SkyWater S130 circuit designs. Layers listed in the first column are drawn layers unless otherwise noted. *Figure 2-1* on page 16 shows a cross section of the metal layer sequence in the aluminum-based 5LM stack.

Table 2-1. Drawing and Mask Layer Sequence (Page 1 of 3)

Drawing Layer Name	Description	Mask Acronym	GDSII		Mask Layer	
			Layer No.	Data Type	Name	Tone
diff	Field-oxide region for active devices, regardless of device type or well type; electric charge type is opposite that of the well in which it's placed.	FOM	65	20	cfom	Clear
dnwell	Deep n-well region that reduces substrate noise coupling in mixed-signal designs.	DNM	64	18	cdnm	Opaque
pwbm	Region to be blocked from a p-well implant, such as a drain-extended MOS device region.	PWBM	19	44	cpwbm	Clear
pwde	P-well drain-extended implant region.	PWDEM	124	20	cpwdem	Opaque
nwell	N-type well diffused on a p-type substrate active area; may be placed inside a deep n-well.	NWM	64	20	cnwm	Clear
hvtp	High-threshold voltage (V_t) 1.8 V PMOS device implant region.	HVTPM	78	44	chvtpm	Clear
lvtn	Low-V_t NMOS device implant region.	LVTNM	125	44	clvtnm	Opaque
hvtr	High-V_t RF PMOS transistor implant region.	HVTRM	18	20	chvtrm	Clear
ncm	N-core device implant region for nonvolatile static random-access memory (NVSRAM); not currently supported in this technology.	NCM	92	44	cncm	Clear
tunm	Tunnel oxide implant region, such as a silicon-oxide-nitride-oxide-silicon (SONOS) device region.	TUNM	80	20	ctunm	Opaque
thkox	High-voltage (5 V or higher), thick gate oxide region or device.	LVOM	75	21	clvom	Clear
rpm	Region that protects p-type poly resistors from n-type poly implants; drawn over p+ poly and created over p-poly.	RPM	86	20	crpm	Clear
rrpm	Created layer used to implant 300 Ω/sq. p+ poly resistors.	RRPM	102	20	crrpm	Opaque
urpm	Drawn layer used to implant 2000 Ω/sq. p- poly resistors.	URPM	79	20	curpm	Opaque

1. The RRAM cell is not yet qualified; it will be available in a future process design kit (PDK) release.

Table 2-1. Drawing and Mask Layer Sequence (Page 2 of 3)

Drawing Layer Name	Description	Mask Acronym	GDSII		Mask Layer	
			Layer No.	Data Type	Name	Tone
poly	Polysilicon region used to define all field-effect transistor (FET) gates, poly resistors, gate oxide capacitors, and poly-bound emitters of npn bipolar junction transistors.	P1M	66	20	cp1m	Clear
ntm	Created layer used to define n-tip or n-type lightly doped drain (LDD) implant regions; blocks all non-implanted regions, including PMOS and high-voltage NMOS device and SONOS memory regions.	NTM	26	20	cntm	Clear
hvntm	Created layer used to define high-voltage n-tip or n-type LDD implant regions; enables LDD implants into 5.0 V NMOS devices. A drawn layer is also provided for eliminating design rule checking (DRC) gaps and slivers.	HVNTM	125	20	chvntm	Opaque
ldntm	LDD implant region inside SONOS memory arrays only.	LDNTM	11	44	cldntm	Opaque
npc	Region that enables selective nitride removal in preparation for local interconnection contact (licon) placement on poly.	NPCM	95	20	cnpc	Opaque
psdm	P+ implant region.	PSDM	94	20	cpsdm	Clear
nsdm	N+ implant region.	NSDM	93	44	cnsdm	Clear
licon	Local interconnect contact to diffusion and poly regions; uses a conventional tungsten plug to the silicon.	LICM1	66	44	clicm1	Opaque
li	Local interconnect layer between licon and mcon; typically used for short runs.	LI1M	67	20	cli1m	Clear
mcon	Metal contact between li and met1.	CTM1	67	44	cctm1	Opaque
met1	First metal layer between mcon and via1.	MM1	68	20	cmm1	Clear
via1	First via contact between met1 and met2.	VIM	68	44	cviam	Opaque
r1c	A resistive random-access memory (RRAM) cell between met1 and met2.[1]	R1C	201	20	r1c	Clear
r1v	Top of contact to RRAM cell (when via is enclosed by r1c) or top of connection between met1 and met2 (when via is not enclosed by r1c).[1]	R1V	202	20	r1v	Opaque
met2	Second metal layer between via1 and via2.	MM2	69	20	cmm2	Clear
via2	Second via contact between met2 and met3.	VIM2	69	44	cviam2	Opaque
capm	Metal-insulator-metal (MIM) capacitor between met3 and met4; capacitance set by dielectric between capm and met3.	CAPM	89	44	ccapm	Clear
met3	Third metal layer between via2 and via3.	MM3	70	20	cmm3	Clear
via3	Third via contact between met3 and met4.	VIM3	70	44	cviam3	Opaque
cap2m	MIM capacitor between met4 and met5; capacitance set by dielectric between cap2m and met4.	CAP2M	97	44	ccap2m	Clear

1. The RRAM cell is not yet qualified; it will be available in a future process design kit (PDK) release.

Table 2-1. Drawing and Mask Layer Sequence (Page 3 of 3)

Drawing Layer Name	Description	Mask Acronym	GDSII		Mask Layer	
			Layer No.	Data Type	Name	Tone
met4	Fourth metal layer between via3 and via4.	MM4	71	20	cmm4	Clear
via4	Fourth via contact between met4 and met5.	VIM4	71	44	cviam4	Opaque
met5	Fifth (top) metal layer above via4.	MM5	72	20	cmm5	Clear
nsm	Nitride seal mask used as hermetic seal ring to prevent ionic contamination and crack propagation.	NSM	61	20	cnsm	Opaque
pad	Pad connection region.	PDM	76	20	cpdm	Opaque
target	Laser fuse region; not currently supported in this technology.	–	76	44	–	–
pmm	Polyimide region.	PMM	85	44	cpmm	Opaque
rdl	Redistribution layer that connects a customer's top metal to bump terminals.	CU1M	74	20	ccu1m	Opaque
pmm2	Alternate polyimide region.	PMM2	77	20	cpmm2	Opaque
ubm	Under solder bump metal layer.	UBM	127	21	cubm	Opaque
bump	Solder bump region.	BUMP	127	22	cbump	Opaque

1. The RRAM cell is not yet qualified; it will be available in a future process design kit (PDK) release.

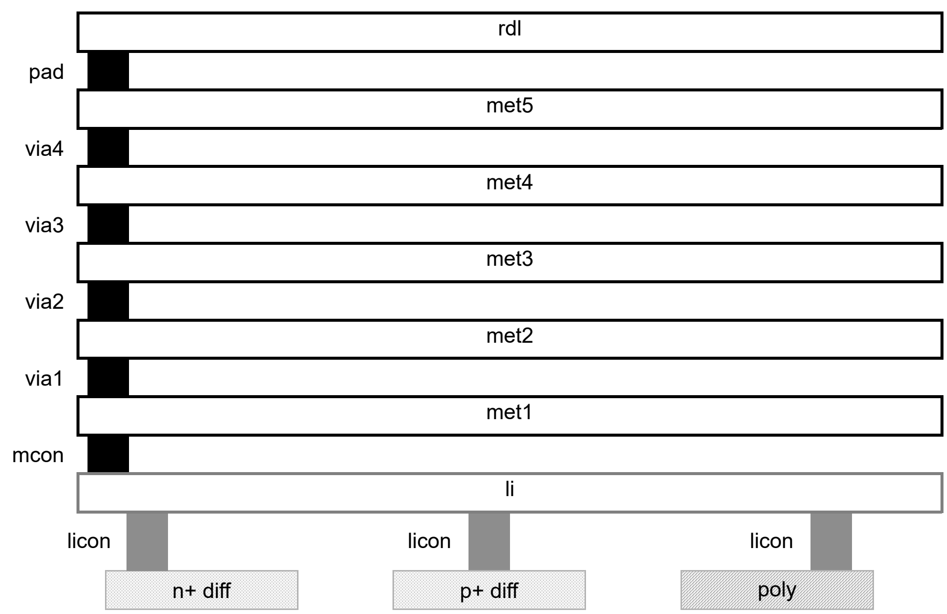


Figure 2-1. 5LM Stack Cross Section

2.2 Marker Layers

Table 2-2 lists the marker layers manually placed during layout to identify specific devices or voltage regions for passing design rule checking (DRC), created layer DRC (CLDRC), or layout versus schematic (LVS) checking. Marked areas are checked against specific design rules.

Table 2-2. Marker Layers (Page 1 of 2)

Marker Layer Name	Purpose	Description	GDSII	
			Layer No.	Data Type
areaid	analog	Identifies an analog circuit area that restricts diffusion geometries to rectangular shapes and prohibits other shapes such as T- or L-shaped geometries.	81	79
	core	Identifies the memory core for DRC; permits tighter design rules for exempt cells. Replaces “coreid.”	81	2
	critCorner	Identifies a critical corner in the sealring pcell for metal stress relief DRC (see *Figure 2-2* on page 19).	81	51
	critSid	Identifies a critical side in the sealring pcell for metal stress relief DRC (see *Figure 2-2).*	81	52
	deadZon	Identifies a dead zone, an area with no active circuitry, in the sealring pcell for metal stress relief DRC (see *Figure 2-2)*.	81	50
	dieCut	Identifies the location of a die within the frame.	81	11
	diode	Identifies a diode for LVS checking.	81	23
	esd	Identifies an electrostatic discharge (ESD) protection device for DRC.	81	19
	etest	Identifies an area used in an e-test module for DRC.	81	101
	extendedDrain	Identifies an extended-drain device for LVS checking.	81	57
	fabBlock	Identifies Nikon marks, optical proximity correction (OPC) calibration structures, revision letters, critical dimension (CD) bars, and so on; block size is defined by the areaid:moduleCut boundary layer.	81	82
	frame	Identifies a pad in the frame for DRC.	81	3
	frameRect	Identifies the frame boundary for DRC and CLDRC; also used as a fill delimiter.	81	12
	guard_boundary	Identifies a guard ring for DRC; not currently used.	81	73
	injection	Identifies a circuit susceptible to injection; marked area must be more than 100 µm from a signal pad diffusion for DRC.	81	17
	low_vt	Identifies a 20 V diode (dnwdiodehv_psub); this marker layer is not currently used.	81	108
	lowTapDensity	Identifies an area with a low density of substrate taps AND a low risk of latchup; marked area must be more than 50 µm from a signal pad diffusion for DRC.	81	14
	lvNative	Identifies a 3.3 V native NMOS device for LVS checking.	81	60

Table 2-2. Marker Layers (Page 2 of 2)

Marker Layer Name	Purpose	Description	GDSII	
			Layer No.	Data Type
areaid	moduleCut	For SkyWater use only. Identifies the location of an e-test module within the frame.	81	10
	notCritSide	Identifies an area where critical side stress rules are not checked in DRC; this marker layer is not supported.	81	15
	opcDrop	Identifies an area where OPC is not automatic within fab blocks and lithography calibration structures.	81	54
	pad_gnd	Identifies ground pad type for latchup checking.	81	72
	pad_io	Identifies signal pad type for latchup checking.	81	70
	pad_pwr	Identifies power pad type for latchup checking.	81	71
	padLength	Identifies pad orientation and length for DRC.	81	67
	photo	Identifies a deep n-well photodiode for DRC.	81	81
	rdlprobepad	Identifies a probe pad on the optional rdl drawing layer.	81	27
	rfdiode	Identifies an RF diode that requires series resistance extraction for LVS checking.	81	125
	seal	Identifies the seal ring; also used as a fill delimiter.	81	1
	standardc	Identifies a cell in the standard cell library for DRC.	81	4
	substrateCut	Identifies an isolated substrate region when two different resistively connected substrate regions are spaced more than 100 µm apart, for latchup, LVS, and soft connection checking.	81	53
diff	res	Identifies a diffusion resistor for DRC and LVS checking.	65	13
li	res	Identifies a local interconnect resistor for DRC and LVS checking.	67	13
LVS_exclude	–	Identifies an area excluded from LVS checking.	84	44
met1	res	Identifies a met1 resistor for DRC and LVS checking.	68	13
met2	res	Identifies a met2 resistor for DRC and LVS checking.	69	13
met3	res	Identifies a met3 resistor for DRC and LVS checking.	70	13
met4	res	Identifies a met4 resistor for DRC and LVS checking.	71	13
met5	res	Identifies a met5 resistor for DRC and LVS checking.	72	13
npn	–	Identifies an npn bipolar junction transistor for DRC and LVS checking.	82	20
pnp	–	Identifies a pnp bipolar junction transistor for DRC and LVS checking.	82	44
poly	res	Identifies a poly resistor for DRC and LVS checking.	66	13
pwell	res	Identifies a pwell resistor for DRC and LVS checking.	64	13
v5	–	Identifies a 5 V diffusion or well for DRC and LVS checking.	75	20
v12	–	Identifies a 12 V diffusion or well for DRC and LVS checking.	74	21
v20	–	Identifies a 20 V diffusion or well for DRC and LVS checking.	74	22

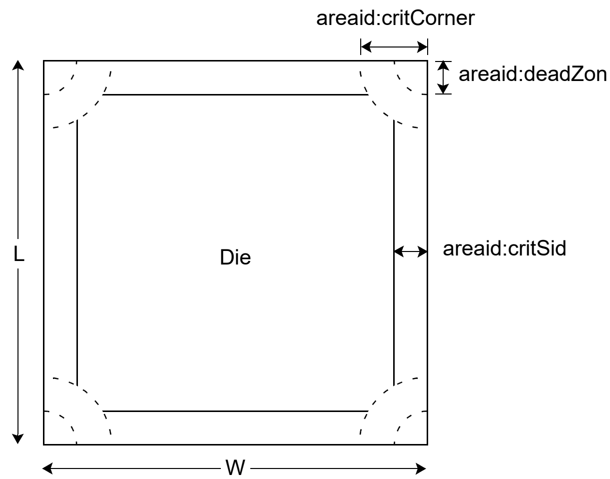


Figure 2-2. High-Stress Regions in “sealring” Pcell

2.3 Device Tables

The devices listed in *Table 2-3 Device by Drawing Layer* and *Table 2-4 Device by Marker Layer,* beginning on page 20, are described in *Table 1-1 Supported Devices* on page 11.

Table 2-3. Device by Drawing Layer (Page 1 of 3)

Cell Name	diff	dnwell	pwbm	pwde	nwell	hvtp	lvtn	tunm	thkox	rpm	rrpm	urpm	poly	ldntm	npc	nsdm	psdm	licon	li	capm	cap2m	met1	met2	met3	met4	met5	nsm	pad	rdl
nmos	**D**		No	No	No	No	No	No	No	No	No	No	**D**	No	No	**D**	No										No		
nmos_lvt	**D**		No	No	No	No	**D**	No	No	No	No	No	**D**	No	No	**D**	No										No		
nmos_v5	**D**		No	No	No	No	No	No	**D**	No	No	No	**D**	No	No	**D**	No										No		
nmos_nat_v3	**D**		No	No	No	No	**D**	No	**D**	No	No	No	**D**	No	No	**D**	No										No		
nmos_nat_v5	**D**		No	No	No	No	**D**	No	**D**	No	No	No	**D**	No	No	**D**	No										No		
nmos_de_v12	**D**	No	No	No	**D**	No	No	No	**D**	No	No	No	**D**	No	No	**D**	No										No		
nmos_de_v20	**D**	**D**	**D**	No	**D**	No	**D**	No	**D**	No	No	No	**D**	No	No	**D**	No										No		
nmos_de_iso_v20	**D**	**D**	**D**	No	**D**	No	**D**	No	**D**	No	No	No	**D**	No	No	**D**	No										No		
nmos_nat_v20	**D**	**D**	**D**	No	**D**	No	**D**	No	**D**	No	No	No	**D**	No	No	**D**	No										No		
nmos_zvt_v20	**D**	**D**	**D**	No	**D**	No	**D**	No	**D**	No	No	No	**D**	No	No	**D**	No										No		
pmos	**D**		No	No	**D**	**D**	No	No	No	No	No	No	**D**	No	No	No	**D**										No		
pmos_lvt	**D**		No	No	**D**	No	**D**	No	No	No	No	No	**D**	No	No	No	**D**										No		
pmos_hvt	**D**		No	No	**D**	**D**	**C**	No	No	No	No	No	**D**	No	No	No	**D**										No		
pmos_v5	**D**		No	No	**D**	No	No	No	**D**	No	No	No	**D**	No	No	No	**D**										No		
pmos_de_v12	**D**	**D**	No	No	**D**	No	No	No	**D**	No	No	No	**D**	No	No	No	**D**										No		
pmos_de_v20	**D**	**D**	**D**	**D**	**D**	No	**D**	No	**D**	No	No	No	**D**	No	No	No	**D**										No		
rpoly	No							No		No	No	No	**D**	No			No										No		
rpoly_hp	No							No		**D**	**C**	No	**D**	No	**D**	No	**D**										No		
rpoly_hp2k	No							No		**D**	No	**D**	**D**	No	**D**	No	**D**										No		
rmet (on li)																		No	**D**								No		
rmet (on met1)																						**D**					No		
rmet (on met2)																							**D**				No		
rmet (on met3)																								**D**			No		
rmet (on met4)																									**D**		No		
rmet (on met5)																										**D**	No		
rndiff	**D**		No	No	No	No	No	No	No	No	No	No	No	No	No	**D**	No										No		
rndiff_v5	**D**		No	No	No	No	No	No	**D**	No	No	No	No	No	No	**D**	No										No		
rpdiff	**D**		No	No	**D**	No	No	No	No	No	No	No	No	No	No	No	**D**										No		
rpdiff_v5	**D**		No	No	**D**	No	No	No	**D**	No	No	No	No	No	No	No	**D**										No		
rpwell		**D**	No	No	No	No	No	No	No	No	No	No	No	No	No												No		

Key:
C = Layer must be created for the device
D = Layer must be drawn for the device
No = Layer must not overlap the device
Blank = Layer may overlap the device

Table 2-3. Device by Drawing Layer (Page 2 of 3)

Cell Name	diff	dnwell	pwbm	pwde	nwell	hvtp	lvtn	tunm	thkox	rpm	rrpm	urpm	poly	ldntm	npc	nsdm	psdm	licon	li	capm	cap2m	met1	met2	met3	met4	met5	nsm	pad	rdl
cm3m4																				**D**				**D**			No		
cm4m5																					**D**				**D**		No		
dnsd_pw	**D**		No	No	No	No	No	No	No	No	No	No	No	No	No	**D**	No										No		
dnsd_pw_v5	**D**		No	No	No	No	No	No	**D**	No	No	No	No	No	No	**D**	No										No		
dnsd_pw_lvt	**D**		No	No			**D**	No		No	No	No	No	No	No	**D**	No										No		
dnsd_pw_nat	**D**		No	No			**D**	No	**D**	No	No	No	No	No	No		No										No		
dpsd_nw	**D**		No	No	**D**	No	No	No	No	No	No	No	No	No	No	No	**D**										No		
dpsd_nw_v5	**D**		No	No	**D**	No	No	No	**D**	No	No	No	No	No	No	No	**D**										No		
dpsd_nw_lvt	**D**		No	No	**D**	No	**D**	No	**D**	No	No	No	No	No	No	No	**D**										No		
dpsd_nw_hvt	**D**		No	No	**D**	**D**	**C**	No	No	No	No	No	No	No	No	No	**D**										No		
dipw_dnw		**D**	No	No	No			No		No	No	No															No		
dipw_dnw_v5		**D**	No	No	No			No		No	No	No															No		
dipw_dnw_v12		**D**	No	No	No			No		No	No	No															No		
dipw_dnw_v20		**D**	No	No	No			No		No	No	No															No		
ddnw_sub		**D**	No	No				No		No	No	No															No		
ddnw_sub_v5		**D**	No	No				No		No	No	No															No		
ddnw_sub_v12		**D**	No	**D**				No		No	No	No															No		
ddnw_sub_v20		**D**	**D**					No		No	No	No															No		
pnp	**D**	No	No	No	**D**		No	No	No	No	No	No	No	No		**D**	**D**										No		
pnp_5x	**D**	No	No	No	**D**		No	No	No	No	No	No	No	No		**D**	**D**										No		
npn_1x1_v5	**D**	**D**	No	No	**D**		No	No	**D**	No	No	No	**D**			**D**	**D**										No		
npn_1x1	**D**	**D**	No	No	**D**		No	No	No	No	No	No	No			**D**	**D**										No		
npn_1x2	**D**	**D**	No	No	**D**		No	No	No	No	No	No	No			**D**	**D**										No		
nmos_esd	**D**		No	No	No	No	No	No	No	No	No	No	**D**	No	No	**D**	No										No		
nmos_esd_v5	**D**		No	No	No	No	No	No	**D**	No	No	No	**D**	No	No	**D**	No										No		
nmos_esd_nat_v5	**D**		No	No	No	No	**D**	No	**D**	No	No	No	**D**	No	No	**D**	No										No		
pmos_esd_v5	**D**		No	No	**D**	No	No	No	**D**	No	No	No	**D**	No	No	No	**D**										No		
dnsd_pw_esd	**D**		No		No	No	No	No	No	No	No	No	No			**D**	No										No		
dpsd_nw_esd	**D**		No		**D**	No	No	No	No	No	No	No	No			No	**D**										No		

Key:

C = Layer must be created for the device
D = Layer must be drawn for the device
No = Layer must not overlap the device
Blank = Layer may overlap the device

Table 2-3. Device by Drawing Layer (Page 3 of 3)

Cell Name	diff	dnwell	pwbm	pwde	nwell	hvtp	lvtn	tunm	thkox	rpm	rrpm	urpm	poly	ldntm	npc	nsdm	psdm	licon	li	capm	cap2m	met1	met2	met3	met4	met5	nsm	pad	rdl
dnsd_pw_esd_v5	**D**		No		No	No	No	No	**D**	No	No	No	No			**D**	No										No		
dpsd_nw_esd_v5	**D**		No		**D**	No	No	No	**D**	No	No	No	No			No	**D**										No		
pad_bond																									No	**D**	No	**D**	
pad_probe																									No	**D**	No	**D**	
pad_microprobe																									No	**D**	No	**D**	
sealring	**D**	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	**D**	No	No

Key:
C = Layer must be created for the device
D = Layer must be drawn for the device
No = Layer must not overlap the device
Blank = Layer may overlap the device

Table 2-4. Device by Marker Layer (Page 1 of 3)

Cell Name	areaid:core	areaid:diode	areaid:esd	areaid:extendedDrain	areaid:lvNative	areaid:photo	areaid:seal	areaid:substrateCut	diff:res	li:res	met1:res	met2:res	met3:res	met4:fuse	met4:res	met5:res	npn	pnp	poly:model	poly:res	pwell:res	v5	v12	v20
nmos	No	No		No	No	No	No		No	No				No			No	No	No	No	No	No	No	No
nmos_lvt	No	No	No	No	No	No	No		No	No				No			No	No	No	No	No	No	No	No
nmos_v5	No	No		No	No	No	No		No	No				No			No	No	No	No	No	**D**	No	No
nmos_nat_v3	No	No	No	No	**D**	No	No		No	No				No			No	No	No	No	No	**D**	No	No
nmos_nat_v5	No	No	No	No	No	No	No		No	No				No			No	No	No	No	No	**D**	No	No
nmos_de_v12	No	No	No	**D**	No	No	No		No	No				No			No	No	No	No	No	No	**D**	No
nmos_de_v20	No	No	No	**D**	No	No	No		No	No				No			No	No	No	No	No	No	No	**D**
nmos_de_iso_v20	No	No	No	**D**	No	No	No		No	No				No			No	No	No	No	No	No	No	**D**
nmos_nat_v20	No	No	No	**D**	No	No	No		No	No				No			No	No	No	No	No	No	No	**D**
nmos_zvt_v20	No	No	No	**D**	No	No	No		No	No				No			No	No	No	No	No	No	No	**D**
pmos	No	No		No	No	No	No		No	No				No			No	No	No	No	No	No	No	No
pmos_lvt	No	No	No	No	No	No	No		No	No				No			No	No	No	No	No	No	No	No
pmos_hvt	No	No	No	No	No	No	No		No	No				No			No	No	No	No	No	No	No	No
pmos_v5	No	No		No	No	No	No		No	No				No			No	No	No	No	No	**D**	No	No

Key:
D = Layer must be drawn for the device
No = Layer must not overlap the device
Blank = Layer may overlap the device

Table 2-4. Device by Marker Layer (Page 2 of 3)

Cell Name	areaid:core	areaid:diode	areaid:esd	areaid:extendedDrain	areaid:lvNative	areaid:photo	areaid:seal	areaid:substrateCut	diff:res	li:res	met1:res	met2:res	met3:res	met4:fuse	met4:res	met5:res	npn	pnp	poly:model	poly:res	pwell:res	v5	v12	v20
pmos_de_v12	No	No	No	**D**	No	No	No		No	No				No			No	No	No	No	No	No	**D**	No
pmos_de_v20	No	No	No	**D**	No	No	No		No	No				No			No	No	No	No	No	No	No	**D**
rpoly	No	No		No	No	No	No		No	No				No			No	No	No	**D**	No			
rpoly_hp	No	No		No	No	No	No		No	No				No			No	No	No	**D**	No			
rpoly_hp2k	No	No		No	No	No	No		No	No				No			No	No	No	**D**	No			
rmet (on li)	No			No		No	No			**D**				No					No	No				
rmet (on met1)				No			No				**D**			No					No					
rmet (on met2)				No			No					**D**		No					No					
rmet (on met3)				No			No						**D**	No					No					
rmet (on met4)				No			No							No	**D**				No					
rmet (on met5)				No			No							No		**D**			No					
rndiff	No	No		No	No	No	No		**D**	No				No			No	No	No	No		No	No	No
rndiff_v5	No	No		No	No	No	No		**D**	No				No			No	No	No	No		**D**	No	No
rpdiff	No	No		No	No	No	No		**D**	No				No			No	No	No	No		No	No	No
rpdiff_v5	No	No		No	No	No	No		**D**	No				No			No	No	No	No		**D**	No	No
rpwell	No			No	No	No	No		No	No				No			No	No	No	No	**D**	No	No	No
cm3m4	No			No	No	No	No							No					No			No	No	No
cm4m5	No			No	No	No	No							No					No			No	No	No
dnsd_pw		**D**		No	No	No	No		No	No				No			No	No	No	No	No	No	No	No
dnsd_pw_v5	No	**D**		No	No	No	No		No	No				No			No	No	No	No	No	**D**	No	No
dnsd_pw_lvt		**D**		No		No	No							No					No	No	No	No	No	No
dnsd_pw_nat		**D**		No		No	No							No					No	No	No	No	No	No
dpsd_nw		**D**		No	No	No	No		No	No				No			No	No	No	No	No	No	No	No
dpsd_nw_v5	No	**D**		No	No	No	No		No	No				No			No	No	No	No	No	**D**	No	No
dpsd_nw_lvt	No	**D**		No	No	No	No		No	No				No			No	No	No	No	No	No	No	No
dpsd_nw_hvt	No			No	No	No	No		No					No			No	No	No	No	No	No	No	No
dipw_dnw	No			No	No	No	No		No					No			No	No	No	No	No	No	No	No

Key:
D = Layer must be drawn for the device   No = Layer must not overlap the device   Blank = Layer may overlap the device

Table 2-4. Device by Marker Layer (Page 3 of 3)

Cell Name	areaid:core	areaid:diode	areaid:esd	areaid:extendedDrain	areaid:lvNative	areaid:photo	areaid:seal	areaid:substrateCut	diff:res	li:res	met1:res	met2:res	met3:res	met4:fuse	met4:res	met5:res	npn	pnp	poly:model	poly:res	pwell:res	v5	v12	v20
dipw_dnw_v5	No			No	No	No	No		No					No			No	No	No	No	No	**D**	No	No
dipw_dnw_v12	No			No	No	No	No		No					No			No	No	No	No	No	No	**D**	No
dipw_dnw_v20	No			No	No	No	No		No					No			No	No	No	No	No	No	No	**D**
ddnw_sub	No			No	No	No	No		No					No			No	No	No	No	No	No	No	No
ddnw_sub_v5	No			No	No	No	No		No					No			No	No	No	No	No	**D**	No	No
ddnw_sub_v12	No			No	No	No	No		No					No			No	No	No	No	No	No	**D**	No
ddnw_sub_v20	No			No	No	No	No		No					No			No	No	No	No	No	No	No	**D**
pnp	No	No	No	No	No	No	No		No	No				No			No	**D**	No	No	No	No	No	No
pnp_5x	No	No	No	No	No	No	No		No	No				No			No	**D**	No	No	No	No	No	No
npn_1x1_v5	No	No	No	No	No	No	No		No	No				No			**D**	No	No	No	No	**D**	No	No
npn_1x1	No	No	No	No	No	No	No		No	No				No			**D**	No	No	No	No	No	No	No
npn_1x2	No	No	No	No	No	No	No		No	No				No			**D**	No	No	No	No	No	No	No
nmos_esd	No	No	**D**	No	No	No	No										No	No				No	No	No
nmos_esd_v5					No		No		No	No				No			No	No	No	No	No	**D**	No	No
nmos_esd_nat_v5	No	No	**D**	No	No	No	No		No	No				No			No	No	No	No	No	**D**	No	No
pmos_esd_v5	No	No	**D**	No	No	No	No		No	No				No			No	No	No	No	No	**D**	No	No
dnsd_pw_esd	No	**D**	**D**	No	No	No	No		No	No				No			No	No	No	No	No	No	No	No
dpsd_nw_esd	No	**D**	**D**	No	No	No	No		No	No				No			No	No	No	No	No	No	No	No
dnsd_pw_esd_v5	No	**D**	**D**	No	No	No	No		No	No				No			No	No	No	No	No	**D**	No	No
dpsd_nw_esd_v5	No	**D**	**D**	No	No	No	No		No	No				No			No	No	No	No	No	**D**	No	No
pad_bond							No							No										
pad_probe							No							No										
pad_microprobe							No							No										
sealring	No	No	No	No	No	No		No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No

Key:
D = Layer must be drawn for the device   No = Layer must not overlap the device   Blank = Layer may overlap the device

3 Physical Design Rules

The physical design rules described in this section supersede all S130 (formerly S8) design rules described elsewhere, including those described in the specs listed in *Section 1.3 Related Documents* on page 13.

3.1 General Guidelines

The default layout grid is 0.005 µm. To minimize design rule checking (DRC) errors, use a layout grid with 0.005 µm vertices unless otherwise directed. For more information regarding the layout grid and comprehensive descriptions of additional design requirements and recommended guidelines, refer to the “Setting Up the Layout Environment” subsection in the “S130 Basic PDK Information” section in the *S130 Process Design Kit User Guide,* Spec No. 002-22691.

3.2 Design Rule Terminology

Table 3-1 defines the DRC terms used in the design rule descriptions that follow.

Table 3-1. DRC Rule Term Definitions (Page 1 of 4)

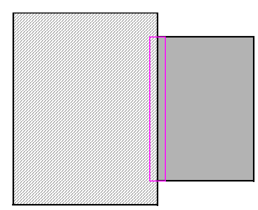
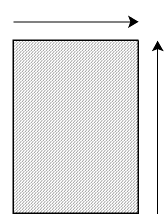
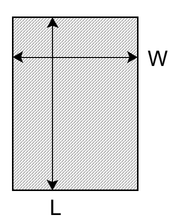
Rule Term and Definition	Illustration
Abut Outside edge of a shape touches the outside edge of another shape.	
Adjacent sides Two sides of a shape that share a common corner.	
Area Total sum area of a shape, for example, W × L for a rectangle.	

Table 3-1. DRC Rule Term Definitions (Page 2 of 4)

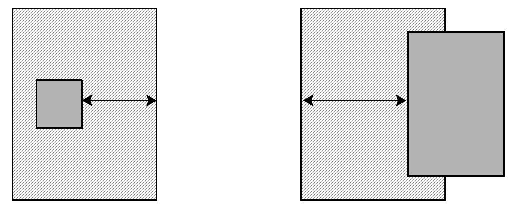
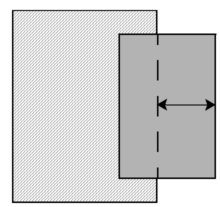
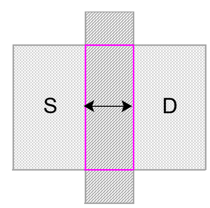
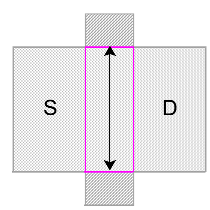
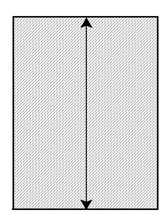
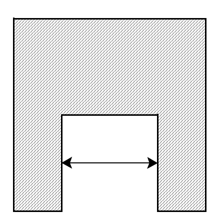
Rule Term and Definition	Illustration
Enclosure Distance from the inside edge of a shape to the outside edge of an interior shape.	
Extension Distance past the inside edge of a shape that straddles a shape to the outside edge of the shape it straddles.	
Gate length Distance between the inside edges of poly-silicon in the same direction as the source and drain.	
Gate width Distance between the inside edges of diffusion in the opposite direction of the source and drain.	
Length (L) Longest distance from the inside edge of a shape to the inside edge of the same shape; opposite of width.	
Notch Distance from the outside edge of a shape to the outside edge of the same shape.	

Table 3-1. DRC Rule Term Definitions (Page 3 of 4)

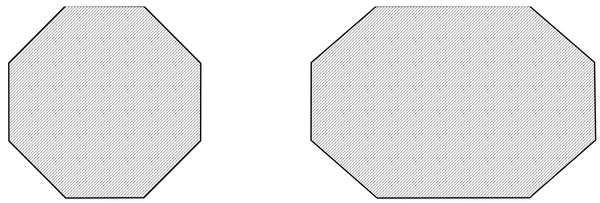
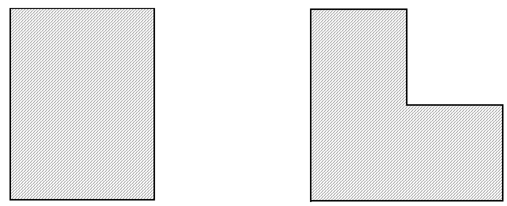
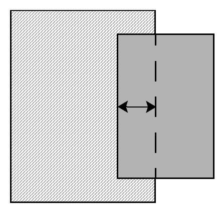
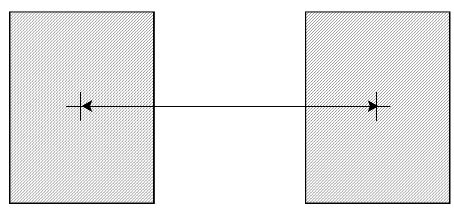
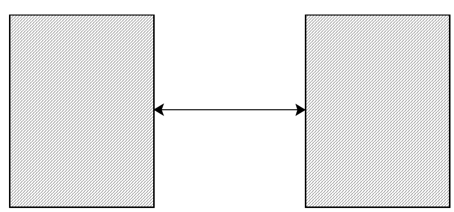
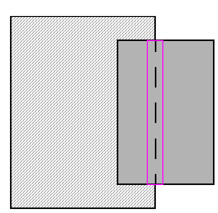
Rule Term and Definition	Illustration
Octagonal Shape with 45°angles or integer multiples of 45° angles.	
Orthogonal Shape with 90°angles or integer multiples of 90° angles.	
Overlap Distance from the inside edge of a shape to the inside edge of another shape.	
Pitch Distance from the center of a shape to the center of another shape.	
Space Distance from the outside edge of a shape to the outside edge of another shape.	
Straddle Segment shared by two overlapping shapes, or the presence of one shape overlapping another.	

Table 3-1. DRC Rule Term Definitions (Page 4 of 4)

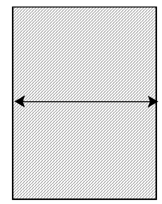
Rule Term and Definition	Illustration
Width (W) Shortest distance from the inside edge of a shape to the inside edge of the same shape; opposite of length.	

Table 3-2 defines the computer-aided design (CAD) terms used in the design rule descriptions that follow.

Table 3-2. CAD Terminology (Page 1 of 2)

Term	Definition
at_risk_non_Vcc_nwell	nwell containing pdiff connected to a 3.3 V minimum external power supply, where nwell is not connected to the same power supply
bondpad	pad outside areaid:frameRect
bus_stack	Stack with width ≥ 25 µm
chip_extent	Holes inside areaid:seal OR areaid:seal
core_poly_gap	Space between two exactly opposite poly line ends, where the width of both poly lines is ≤ 0.15 µm, inside areaid:core
de_fet_gate	poly AND areaid:extendedDrain
de_nfet_drain	(isolated_tap AND areaid:extendedDrain) overlapping nwell
de_nfet_gate	de_fet_gate overlapping (diff NOT dnwell)
de_nfet_source	(diff AND areaid:extendedDrain) overlapping de_nfet_gate
de_pfet_drain	(isolated_tap AND areaid:extendedDrain) not overlapping nwell
de_pfet_gate	de_fet_gate overlapping (diff AND dnwell)
de_pfet_source	(diff AND areaid:extendedDrain) overlapping de_pfet_gate
diff_fill	Shapes added to chip for chemical-mechanical planarization (CMP) on the diff:fill layer
drain	(diff NOT poly) abutting (diff AND poly)
drain_diff	diff not abutting poly in 20 V drain-extended devices
esd_nwell_tap	ntap coincident with nwell such that ntap and nwell are surrounded by and abutting ndiff along all edges, within areaid:esd
fuse	met4:fuse drawing layer used to identify fuses
fuse_shield	2.4 µm × 0.5 µm met4:drawiing rectangle between fuse and periphery, not overlapping contacts or vias
gate	poly AND diff
gate_poly	poly overlapping gate
hvGate	gate abutting hv_source/drain

Table 3-2. CAD Terminology (Page 2 of 2)

Term	Definition
hvPoly	poly that is electrically connected to hv_source/drain or to another high-voltage feature (like another hvPoly) through an interconnect
hv_source/drain	(diff NOT poly) overlapping (v12 OR v20)
isolated_pwell	(NOT nwell) AND dnwell, or pwell inside areaid:substrateCut
isolated_tap	tap that does not abut diff
li_fill	Shapes added to chip for CMP on the li:fill layer
metX_fill	Shapes added to chip for CMP on the metX:fill layer
ndiff	(nsdm AND diff) NOT nwell
nfet	gate NOT nwell
non-isolated_fuse_edge	Long edge of fuse spaced < 4 µm from met4
non_Vcc_nwell	nwell or dnwell connected to (a) a power supply pad through a resistance > 10 Ω, (b) V_{SS}, or (c) a signal pad
ntap	(nsdm AND diff) AND nwell
pdiff	(psdm AND diff) AND nwell
pfet	gate AND nwell
poly_fill	Shapes added to chip for CMP on the poly:fill layer
poly_licon	licon that overlaps poly
prec_resistor	poly AND (poly:res AND [rpm OR urpm] AND psdm)
ptap	(psdm AND diff) NOT nwell
ptub	Isolated p+ substrate formed inside dnwell inside an nwell ring
pwell	Data extent NOT (nwell OR ptub OR areaid:substrateCut)
scribe_line	areaid:frameRect NOT areaid:dieCut (outer edge of areaid:seal)
shv	diff or poly inside v12 or v20, or connected signal or power pad
shv_net	A voltage net connected to 12 V or 20 V power supply or signal pad
slotted_licon	0.19 µm × 2.0 µm local interconnect contact
source	(diff NOT poly) abutting (diff AND poly)
source_diff	(diff NOT poly) abutting poly in 20 V drain-extended devices
stack	metX connected by a via to met(X + 1)
tap	ntap OR ptap
tap_licon	tap AND licon
var_channel	poly AND diff AND (nwell NOT v5) NOT areaid:core
wide_metX	metX dimensions wider and longer than 3.0 µm
wide_pad	Pad with width > 100 µm

3.3 Core Design Rules

Core design rules are used specifically for SkyWater-provided memory cell designs. They are identified and checked using the areaid:core marker layer. Core rule values are typically tighter than corresponding periphery rule values. For memory cell designs, follow the periphery design rules unless a corresponding core design rule exists.

Note: Use of core design rules inside any device or memory cell not provided by SkyWater must be pre-approved by SkyWater Technology.

3.3.1 N-Core Implant Design Rules

These design rules are used to define the threshold voltage adjustment implant region for a 1.8 V NMOS device in a nonvolatile static random-access memory (NVSRAM) core.

Note: The S130 process design kit (PDK) does not currently support NVSRAM. Contact SkyWater Technology for more information.

Table 3-3. N-Core Implant Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
ncm.ANG.3		ncm edge must be octagonal.		–
ncm.AR.1		ncm minimum area (μm^2).	≥	0.265
ncm.AR.2		ncm hole minimum area (μm^2).	≥	0.265
ncm.ENC.1		ncm inside areaid:core (not exempt) minimum enclosure of pdiff.	≥	0.235
ncm.OVL.1	1	ncm inside areaid:core (not exempt) must not overlap ndiff inside periphery.		–
ncm.OVL.2		ncm inside areaid:core (not exempt) must not overlap ndiff.		–
ncm.SP.1	**2**	ncm minimum space and notch.	≥	0.38
ncm.SP.2		ncm inside areaid:core (not exempt) minimum space to ndiff.	≥	0.235
ncm.SP.3		ncm inside areaid:core (not exempt) minimum space to nwell outside areaid:core.	≥	0.38
ncm.WID.1	**2**	ncm inside areaid:core minimum width.	≥	0.38

1. This rule is not checked within "*_tech_CD_top*" cells.
2. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.3.2 Lightly Doped N-Tip Implant Design Rules

These design rules are used to define a lightly doped n-tip implant region inside the memory core, exclusively within a silicon-oxide-nitride-oxide-silicon (SONOS) memory array.

Table 3-4. Lightly Doped N-Tip Implant Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
ldntm.ANG.3		ldntm edge must be octagonal.		–
ldntm.ENC.1		ldntm minimum enclosure of ndiff.	≥	0.18
ldntm.ENC.2		ldntm inside areaid:core minimum enclosure of nfet.	≥	0.125
ldntm.OVL.1	1	ldntm outside areaid:core is prohibited.		–
ldntm.SP.1	**2**	ldntm inside areaid:core minimum space and notch.	≥	0.70
ldntm.SP.2	1	ldntm inside areaid:core minimum space to pdiff.	≥	0.18
ldntm.WID.1	**2**	ldntm inside areaid:core minimum width.	≥	0.70
NC	3	ldntm straddling ndiff is permitted.		–

1. This rule is not checked within "*_tech_CD_top*" cells.
2. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
3. This rule is not checked during DRC; use it as a guideline.

3.3.3 Core High-V_t PMOS Implant Design Rule

This design rule is used to define the threshold voltage adjustment implant region for a high-V_t 1.8 V PMOS device inside areaid:core.

Table 3-5. Core High-V_t PMOS Implant Design Rule

Rule No.	Description	Operator	Value (µm)
hvtp.ENC.2	hvtp inside areaid:core minimum/maximum enclosure of nwell.	==	0

3.3.4 Core Low-V_t NMOS/PMOS Implant Blocking Design Rules

This design rule is used to define the region that blocks threshold voltage adjustment implants for low-V_t 1.8 V NMOS and PMOS devices inside areaid:core.

Table 3-6. Core Low-V_t NMOS/PMOS Implant Blocking Design Rule

Rule No.	Description	Operator	Value (µm)
lvtn.SP.5	lvtn minimum space to nwell inside areaid:core.	≥	0.38

3.3.5 Core Thick Gate Oxide Design Rule

This design rule is used to define thick gate oxide inside areaid:core.

Table 3-7. Core Thick Gate Oxide Design Rule

Rule No.	Description	Operator	Value (μm)
thkox.WID.2	diff inside thkox and areaid:core minimum width.	≥	0.15

3.3.6 Core Diffusion Design Rules

These design rules are used to define an active region or a contact to the substrate inside areaid:core.

Table 3-8. Core Diffusion Design Rules

Rule No.	Notes	Description	Operator	Value (μm)
diff.ENC.3		nwell minimum enclosure of pdiff inside areaid:core.	≥	0.15
diff.ENC.4		nwell minimum enclosure of ntap inside areaid:core.	≥	0.15
diff.ENC.5		nwell minimum enclosure of pdiff adjacent sides inside areaid:core.	≥	0.18
diff.ENC.6		pwell minimum enclosure of ndiff adjacent sides inside areaid:core.	≥	0.34
diff.SP.5		ndiff inside areaid:core minimum space to nwell.	≥	0.32
diff.WID.3	**1**	diff inside areaid:core minimum width.	≥	0.14
diff.WID.6		tap inside areaid:core minimum width.	≥	0.14
diff.WID.11		gate inside areaid:core minimum width.	≥	0.14
diff.WID.12		tap abutting and between diff inside areaid:core minimum width.	≥	0.38

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.3.7 Core Poly Design Rules

These design rules are used to define field-effect transistor (FET) gates, interconnects, resistors, capacitor plates, and poly-bound emitters in npn bipolar junction transistors inside areaid:core.

Table 3-9. Core Poly Design Rules

Rule No.	Notes	Description	Operator	Value (μm)
poly.SP.3		poly not for core_poly_gap minimum spacing.	≥	0.175
poly.SP.4	**1**	poly for core_poly_gap minimum spacing.	≥	0.16
poly.SP.9		poly inside areaid:core minimum space to diff.	≥	0.03
poly.SP.10		poly inside areaid:core minimum space to tap.	≥	0.03

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.3.8 Tunnel Implant Mask Design Rules

These design rules are used to define the tunnel implant mask region for SONOS FETs.

Table 3-10. Tunnel Implant Mask Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
tunm.ANG.3		tunm edge must be octagonal.		–
tunm.AR.1		tunm minimum area (µm^2).	≥	0.672
tunm.CON.1		gate must not straddle tunm.		–
tunm.CON.2	1	tunm must be inside dnwell.		–
tunm.CON.3		tunm must be inside areaid:core.		–
tunm.ENC.1		tunm minimum extension beyond gate.	≥	0.095
tunm.SP.1	**2**	tunm minimum space and notch.	≥	0.50
tunm.SP.2		tunm minimum space to gate outside tunm.	≥	0.095
tunm.WID.1	**2**	tunm minimum width.	≥	0.41

1. This rule is not checked within "*_tech_CD_top*" cells.
2. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.3.9 Core N+ Source/Drain Implant Design Rules

These design rules are used to define the opening for an n+ source or drain implant inside areaid:core.

Table 3-11. Core N+ Source/Drain Implant Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
nsdm.ENC.3		nsdm minimum enclosure of ntap inside areaid:core.	≥	0.13
nsdm.SP.4	**1**	nsdm inside areaid:core (opposite parallel edges only) minimum space and notch.	≥	0.38
nsdm.SP.5		nsdm inside areaid:core minimum space and notch.	≥	0.29
nsdm.WID.3	**1**	nsdm inside areaid:core minimum width.	≥	0.29
nsdm.WID.4		nsdm (opposite parallel edges only) minimum width.	≥	0.38

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.3.10 Core P+ Source/Drain Implant Design Rules

These design rules are used to define the opening for a p+ source or drain implant inside areaid:core.

Table 3-12. Core P+ Source/Drain Implant Design Rules

Rule No.		Description	Operator	Value (µm)
psdm.ENC.3		psdm minimum enclosure of ptap inside areaid:core.	≥	0.12
psdm.SP.4	**1**	psdm inside areaid:core (opposite parallel edges only) minimum space and notch.	≥	0.38
psdm.SP.5		psdm inside areaid:core minimum space and notch.	≥	0.29
psdm.WID.3	**1**	psdm inside areaid:core minimum width.	≥	0.29
psdm.WID.4		psdm (opposite parallel edges only) minimum width.	≥	0.38

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.3.11 Core Local Interconnect Contact Design Rules

These design rules are used to define the contact between poly or diff and the local interconnect (li) layer inside areaid:core.

Table 3-13. Core Local Interconnect Contact Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
licon.CON.12		psdm overlapping poly_licon is prohibited.		–
licon.ENC.8	**1**	npc minimum enclosure of poly_licon inside areaid:core.	≥	0.045
licon.SP.13		poly_licon minimum space to diff.	≥	0.130
licon.SP.14		licon (non-bar) minimum spacing.	≥	0.165
NC	2	licon straddling diff is permitted.		–

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
2. This rule is not checked during DRC; use it as a guideline.

3.3.12 Core Local Interconnect Design Rules

These design rules are used to define the local interconnect (li) layer to diff and poly inside areaid:core.

Table 3-14. Core Local Interconnect Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
li_core.ANG.3		li edge inside areaid:core must be octagonal.		–
li.SP.3	**1**	li inside areaid:core minimum space and notch.	≥	0.14
li.WID.3	**1**	li inside areaid:core minimum width.	≥	0.14

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.3.13 Core Second Metal Layer Design Rule

These design rules are used to define the second layer of metal interconnects, buses, and so on inside areaid:core.

Table 3-15. Core Second Metal Layer Design Rule

Rule No.	Notes	Description	Operator	Value (µm)
met2.ENC.3	**1**, 2	met2 minimum enclosure of via1 inside areaid:core.	≥	0.04

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
2. This rule applies to an aluminum (Al) physical design flow only.

3.3.14 OPC Mask Add and Drop Design Rules

These design rules are used to define optical proximity correction (OPC) mask additions and subtractions inside the memory core.

Table 3-16. OPC Mask Add and Drop Design Rules

Rule No.	Description	Operator	Value (µm)
FOMadd.CON.3	areaid:core must enclose FOMadd.		–
FOMdrop_noSeal.CON.3	areaid:core must enclose FOMdrop_noSeal.		–
HVTPMadd.CON.3	areaid:core must enclose HVTPMadd.		–
HVTPMdrop.CON.3	areaid:core must enclose HVTPMdrop.		–
LI1Madd.CON.3	areaid:core must enclose LI1Madd.		–
LI1Mdrop.CON.3	areaid:core must enclose LI1Mdrop.		–
LICM1add.CON.3	areaid:core must enclose LICM1add.		–
LICM1drop.CON.3	areaid:core must enclose LICM1drop.		–
LVTNMadd.CON.3	areaid:core must enclose LVTNMadd.		–
LVTNMdrop.CON.3	areaid:core must enclose LVTNMdrop.		–
NSDMadd.CON.3	areaid:core must enclose NSDMadd.		–
NSDMdrop.CON.3	areaid:core must enclose NSDMdrop.		–
NTMadd.CON.3	areaid:core must enclose NTMadd.		–
NTMdrop.CON.3	areaid:core must enclose NTMdrop.		–
P1Madd.CON.3	areaid:core must enclose P1Madd.		–
P1Mdrop.CON.3	areaid:core must enclose P1Mdrop.		–
PSDMadd.CON.3	areaid:core must enclose PSDMadd.		–
PSDMdrop.CON.3	areaid:core must enclose PSDMdrop.		–

3.4 Periphery Design Rules

Periphery design rules are used in the “periphery,” that is, everywhere outside a memory cell core. These design rules might be superseded by specialized design rules, for example, electrostatic discharge (ESD) protection, latchup, or assembly design rules, when rule conflicts arise.

3.4.1 Deep N-Well Design Rules

These design rules are used to define a deep n-well (DNW) for isolating a p-well and improving noise immunity.

Table 3-17. Deep N-Well Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
dnwell.ANG.3		dnwell edge must be octagonal.		–
dnwell.CON.1		dnwell must not overlap the pnp marker layer.		–
dnwell.CON.2		pdiff must not straddle dnwell.		–
dnwell.CON.3		dnwell must not straddle the areaid:substrateCut layer.		–
dnwell.CON.4		dnwell must interact with nwell.		–
dnwell.SP.1	**1**	dnwell minimum space and notch (exempt inside v20).	≥	6.3
dnwell.SP.2		dnwell inside v20 on the same net minimum spacing.	≥	2.5
dnwell.SP.3		dnwell inside v20 on different nets minimum spacing.	≥	12.0
dnwell.SP.4		dnwell inside v20 minimum space to dnwell outside v20.	≥	12.0
dnwell.SP.5		dnwell inside v20 minimum space to nwell outside v20.	≥	9.5
dnwell.WID.1	**1**	dnwell minimum width.	≥	3.0

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

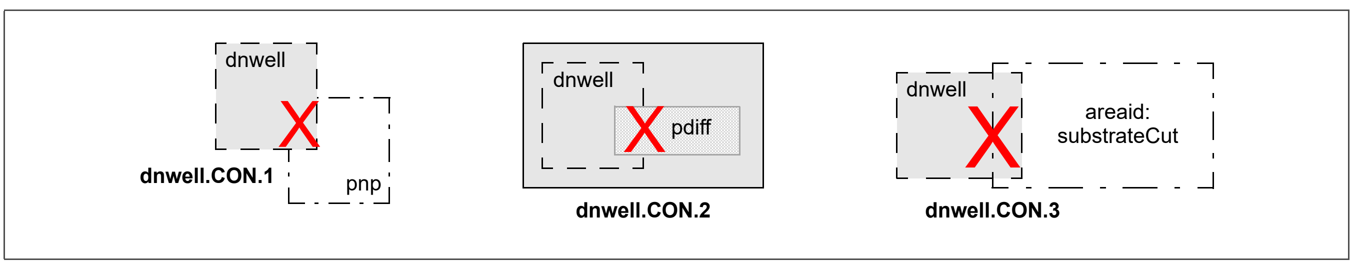


Figure 3-1. Deep N-Well Layout (Page 1 of 2)

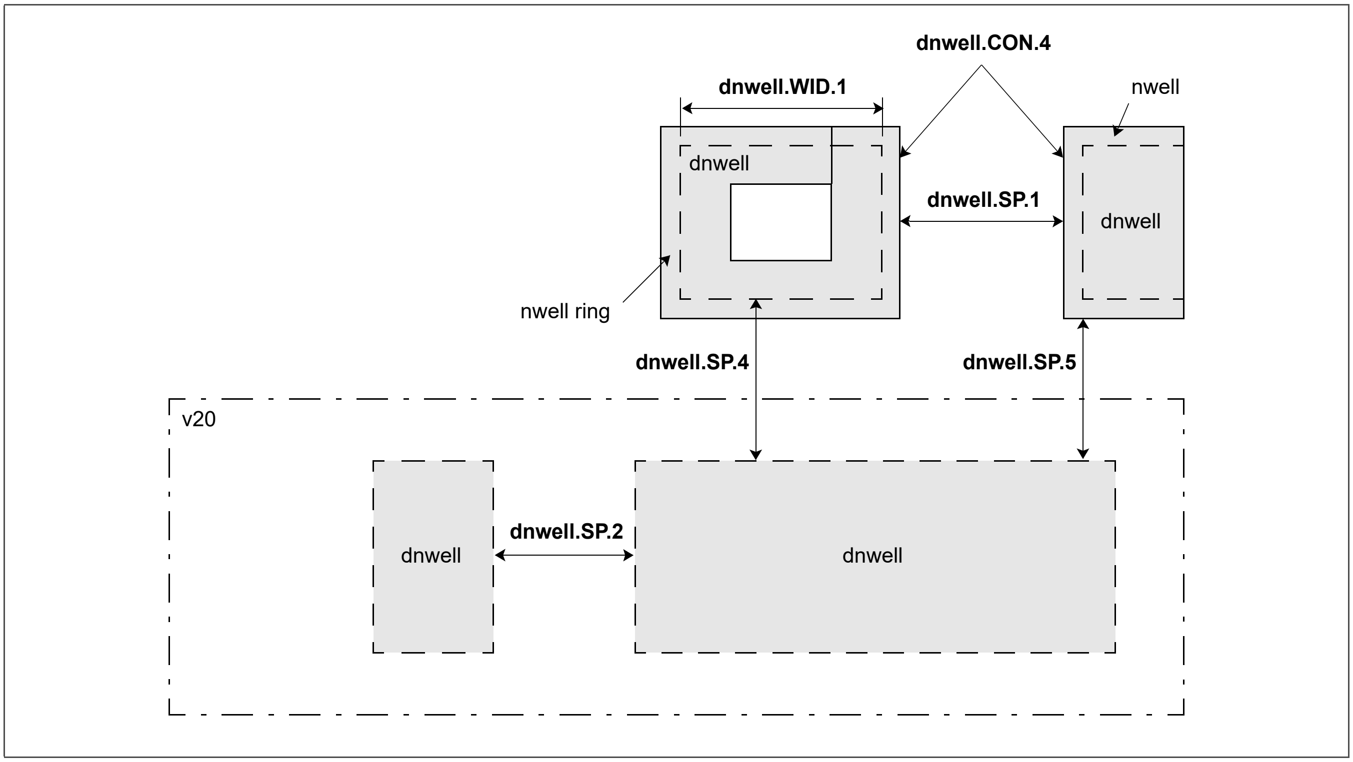


Figure 3-1. Deep N-Well Layout (Page 2 of 2)

3.4.2 N-Well Design Rules

These design rules are used to define an n-well implant region.

Note: nwell voltage is derived from the highest overlap of ([v5, v12, or v20] over nwell).

Table 3-18. N-Well Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
nwell.ANG.3		nwell edge must be octagonal.		–
nwell.CON.1	1	nwell of a different voltage must be on a different net outside “*_tech_CD_top*” cells.		–
nwell.CON.4		nwell connected to 12 V source or drain must be inside v12.		–
nwell.CON.5		nwell connected to 20 V source or drain must be inside v20.		–
nwell.ENC.1	1	nwell minimum enclosure of dnwell (exempt inside v20 and/or “*_tech_CD_top*” cells).	≥	0.4
nwell.ENC.2	1	dnwell minimum enclosure of nwell hole outside v20 and/or “*_tech_CD_top*” cells.	≥	1.03
nwell.ENC.3		dnwell must enclose nwell inside v20.		–
nwell.OVL.1		nwell must be connected to at least one ntap.		–
nwell.SP.1	**2**	1.8 V nwell minimum space and notch.	≥	1.27
nwell.SP.2		1.8 V nwell minimum space to 5 V nwell.	≥	1.27
nwell.SP.3		1.8 V nwell minimum space to 12 V nwell.	≥	2.0
nwell.SP.4		1.8 V nwell minimum space to 20 V nwell.	≥	2.5
nwell.SP.5		5 V nwell minimum space and notch.	≥	1.27
nwell.SP.6		5 V nwell minimum space to 12 V nwell.	≥	2.0
nwell.SP.7		5 V nwell minimum space to 20 V nwell.	≥	2.5
nwell.SP.8		12 V nwell minimum space and notch.	≥	2.0
nwell.SP.9		12 V nwell minimum space to 20 V nwell.	≥	2.5
nwell.SP.10		20 V nwell minimum space and notch.	≥	2.5
nwell.SP.11	1, 3	nwell minimum space to dnwell on different nets outside “*_tech_CD_top*” cells.	≥	4.5
nwell.WID.1	**2**	nwell minimum width.	≥	0.84

1. This rule is not checked within “*_tech_CD_top*” cells.
2. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
3. During DRC, nwell and dnwell spacing is checked on separate nets. To prevent violations when the net names are to be joined at an upper hierarchy level, select “Connect all nets by name” or specify “Connect nets named” in the Siemens Calibre Setup > DRC Options menu. For Cadence PVS > DRC Options, select “Connect all by name” or specify “Connect only names.”

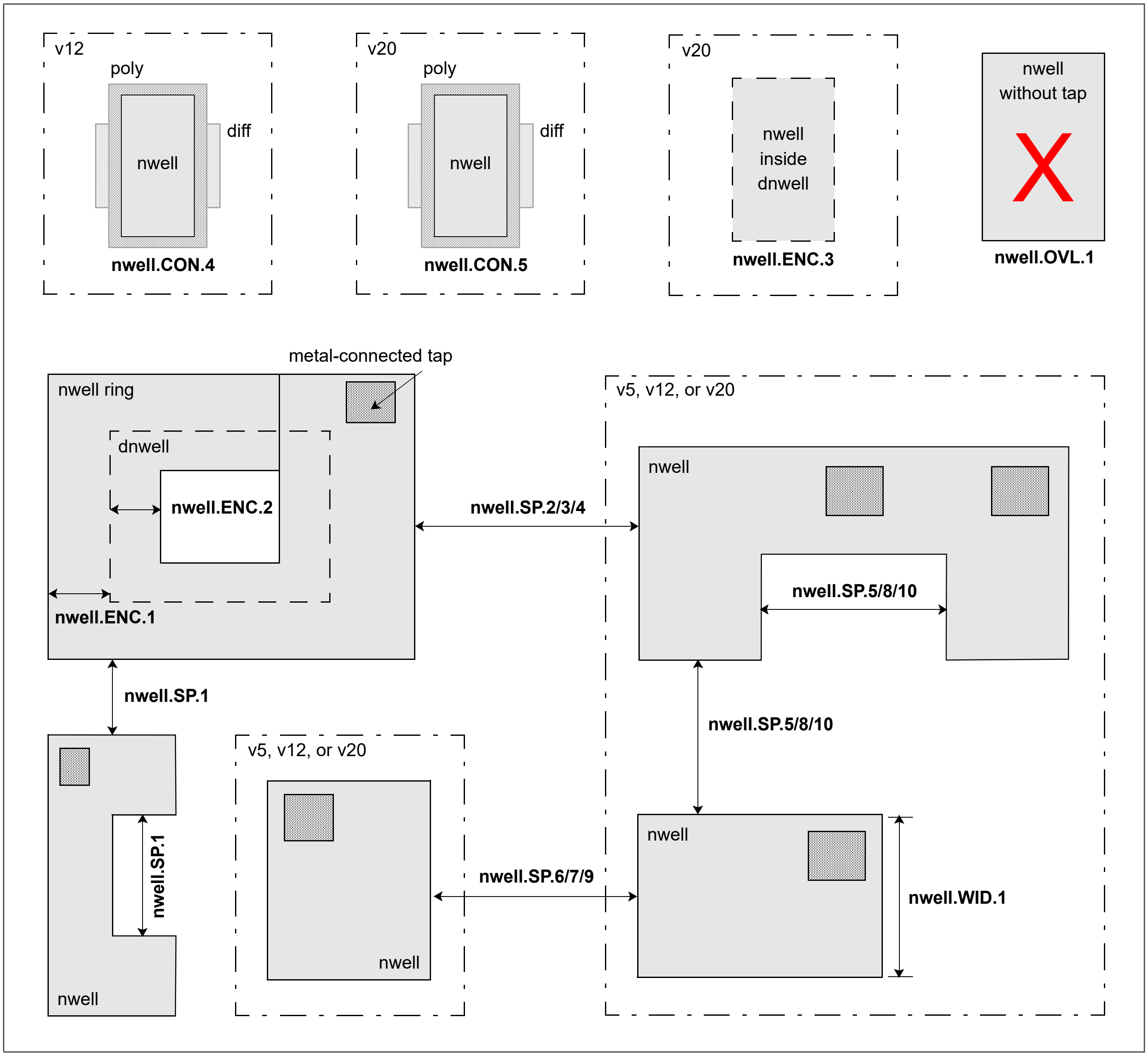


Figure 3-2. N-Well Layout

3.4.3 P-Well Drain-Extended Implant Design Rules

These design rules are used to define a p-well drain-extended (pwde) implant region.

Note: These design rules are currently used exclusively for 20 V devices.

Table 3-19. P-Well Drain-Extended Implant Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
pwde.ENC.1		v20 must enclose pwde.		–
pwde.ENC.2		pwbm minimum enclosure of pwde.	≥	0
pwde.ENC.3		dnwell must enclose pwde.		–
pwde.ENC.4		dnwell inside v20 minimum enclosure of pwde.	≥	1.00
pwde.SP.1	**1**	pwde inside v20 minimum spacing.	≥	1.27
pwde.WID.1	**1**	pwde minimum width.	≥	0.84

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

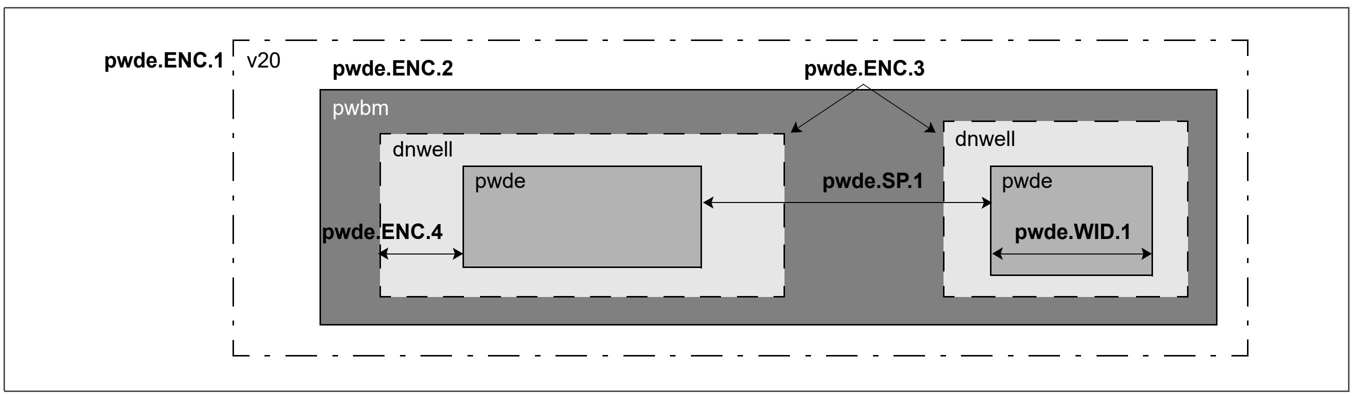


Figure 3-3. P-Well Drain-Extended Implant Layout

3.4.4 P-Well Block Mask Design Rules

These design rules are used to define a region to be blocked from a p-well implant (pwbm).

Note: These design rules are currently used exclusively for 20 V devices.

Table 3-20. P-Well Block Mask Design Rules

Rule No.	Description	Operator	Value (µm)
pwbm.ENC.1	pwbm must enclose dnwell inside v20 (exempt for pwbm hole inside dnwell).		–
pwbm.ENC.2	pwbm minimum enclosure of dnwell inside v20 (exempt for pwbm hole inside dnwell).	≥	0
pwbm.SP.1	pwbm inside v20 minimum space and notch.	≥	1.27
pwbm.SP.2	pwbm hole inside v20 minimum spacing.	≥	0.84
pwbm.WID.1	pwbm minimum width.	≥	0.84

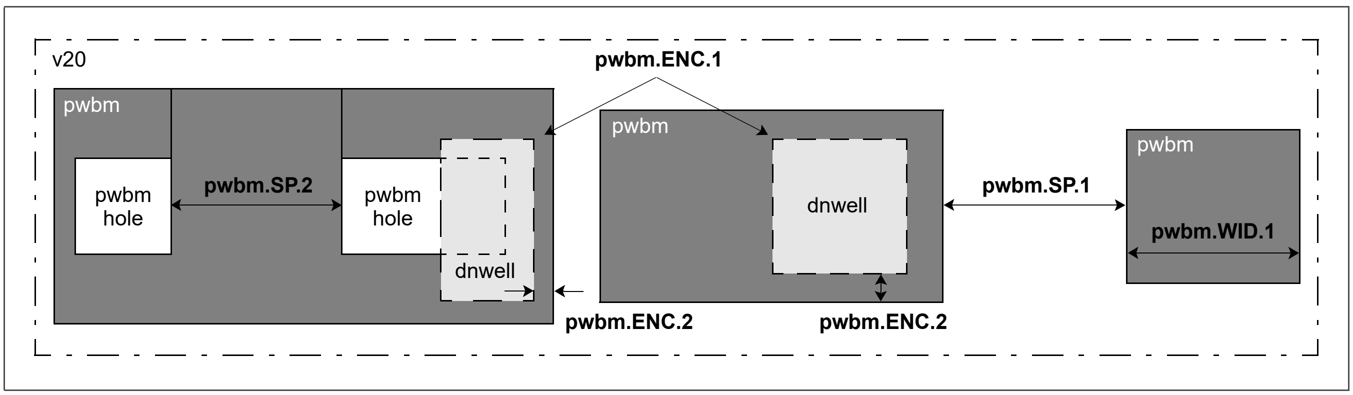


Figure 3-4. P-Well Block Mask Layout

3.4.5 Thick Gate Oxide Design Rules

These design rules are used to define thick gate oxide for 5 V, 12 V, or 20 V devices.

Table 3-21. Thick Gate Oxide Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
thkox.ANG.3		thkox edge must be octagonal.		–
thkox.OVL.1		thkox must not overlap tunm.		–
thkox.SP.1	**1**, 2	thkox inside periphery minimum space and notch.	≥	0.7
thkox.SP.2		Non-abutting thkox minimum space to nwell.	≥	0.7
thkox.WARN.1		thkox must interact with v5, v12, or v20.		–
thkox.WID.1	**1**, 2	thkox inside periphery minimum width.	≥	0.6

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
2. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.

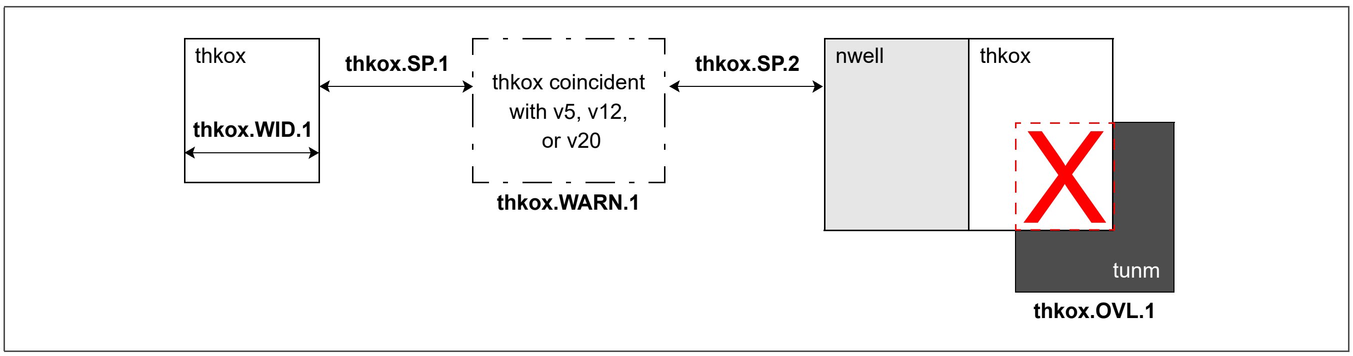


Figure 3-5. Thick Gate Oxide Layout

3.4.6 v5 Design Rules

These design rules are used to define the v5 marker layer.

Table 3-22. v5 Design Rules

Rule No.	Description	Operator	Value (µm)
v5.CON.9	v5 over diff must overlap thkox.		–
v5.OVL.1	v5 must not overlap areaid:core.		–
v5.OVL.2	gate inside v5 must overlap thkox.		–
v5.OVL.4	v5 must not straddle source or drain inside v5.		–
v5.OVL.5	v5 must not straddle gate inside v5.		–
v5.OVL.6	v5 must not straddle nwell.		–
v5.WID.1	v5 minimum width.	≥	0.02

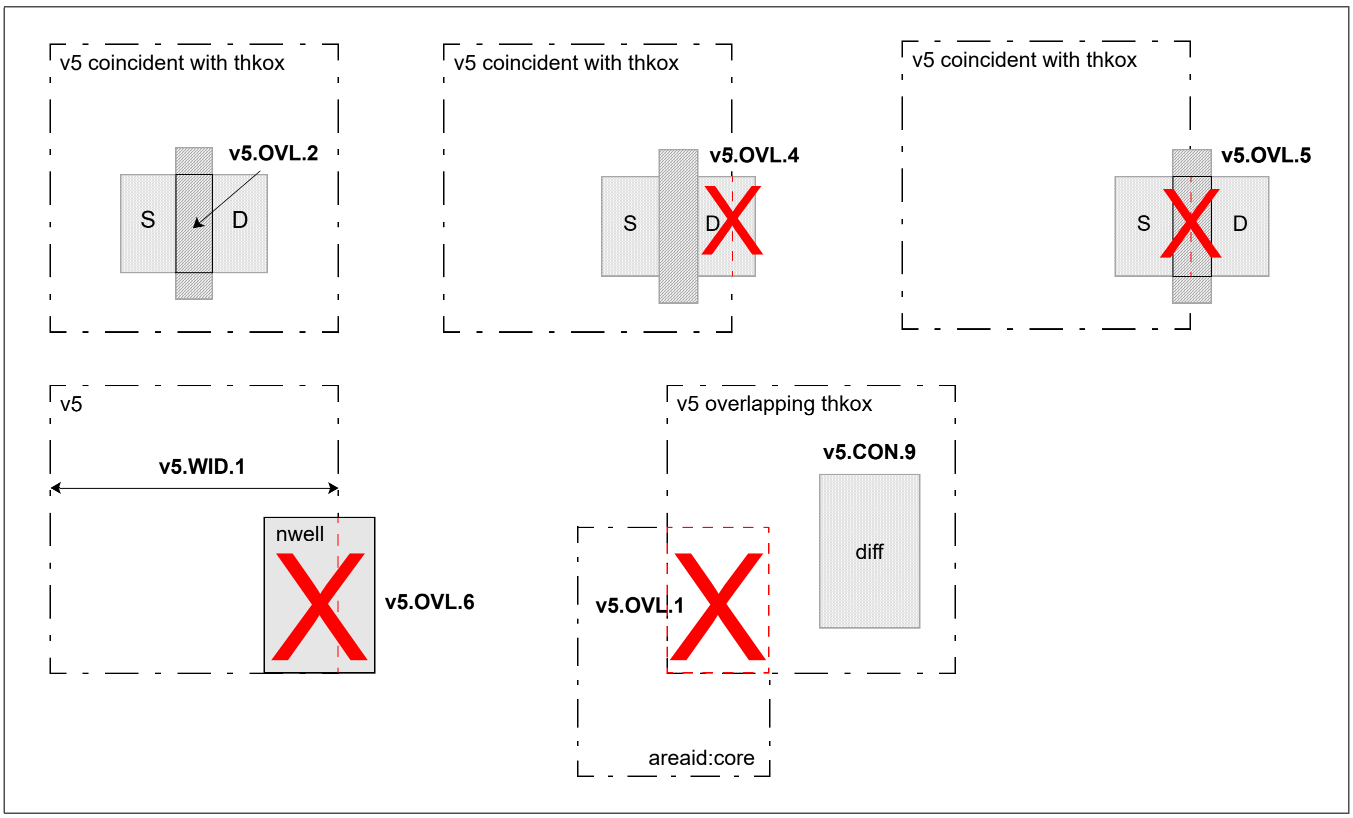


Figure 3-6. v5 Layout

3.4.7 v12 Design Rules

These design rules are used to define the v12 marker layer.

Table 3-23. v12 Design Rules

Rule No.	Description	Operator	Value (µm)
v12.ANG.3	v12 edge must be octagonal.		–
v12.CON.1	diff outside areaid:extendedDrain must not be connected to source or drain inside v12.		–
v12.CON.6	v12 must not overlap v20.		–
v12.CON.9	v12 over diff must overlap thkox.		–
v12.OVL.1	v12 must not overlap areaid:core.		–
v12.OVL.2	gate inside v12 must overlap thkox.		–
v12.OVL.3	v12 must overlap poly connected to same net as source or drain inside v12.		–
v12.OVL.5	v12 must not straddle source or drain inside v12 (exempt inside areaid:extended-Drain).		–
v12.OVL.6	v12 overlapping source or drain inside v12 must not overlap poly (exempt inside areaid:extendedDrain).		–
v12.OVL.7	v12 must not straddle poly inside v12 (exempt inside areaid:extendedDrain).		–
v12.WID.1	v12 minimum width.	≥	0.02

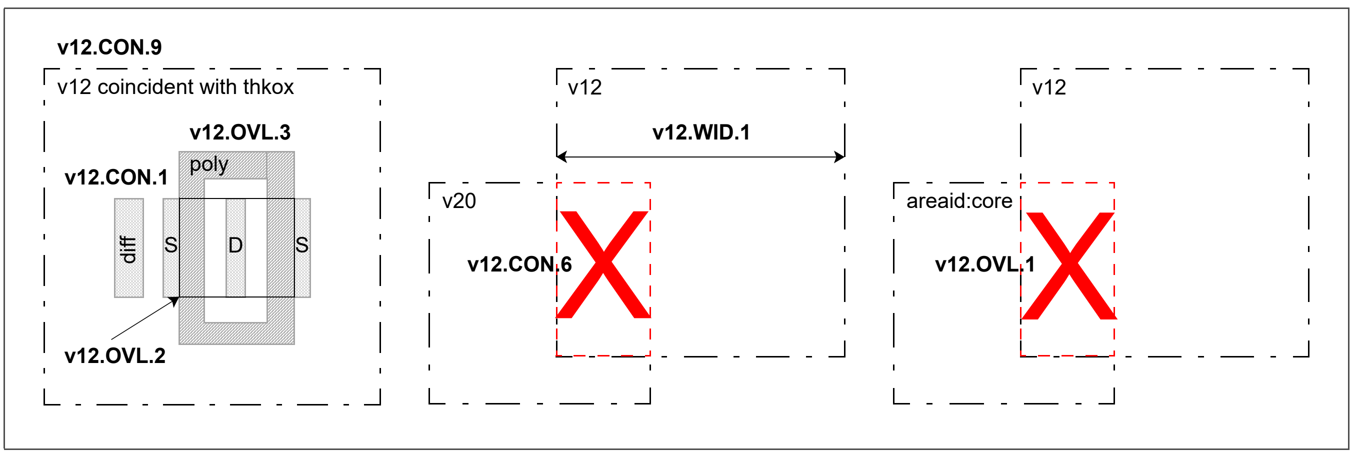


Figure 3-7. v12 Layout (Page 1 of 2)

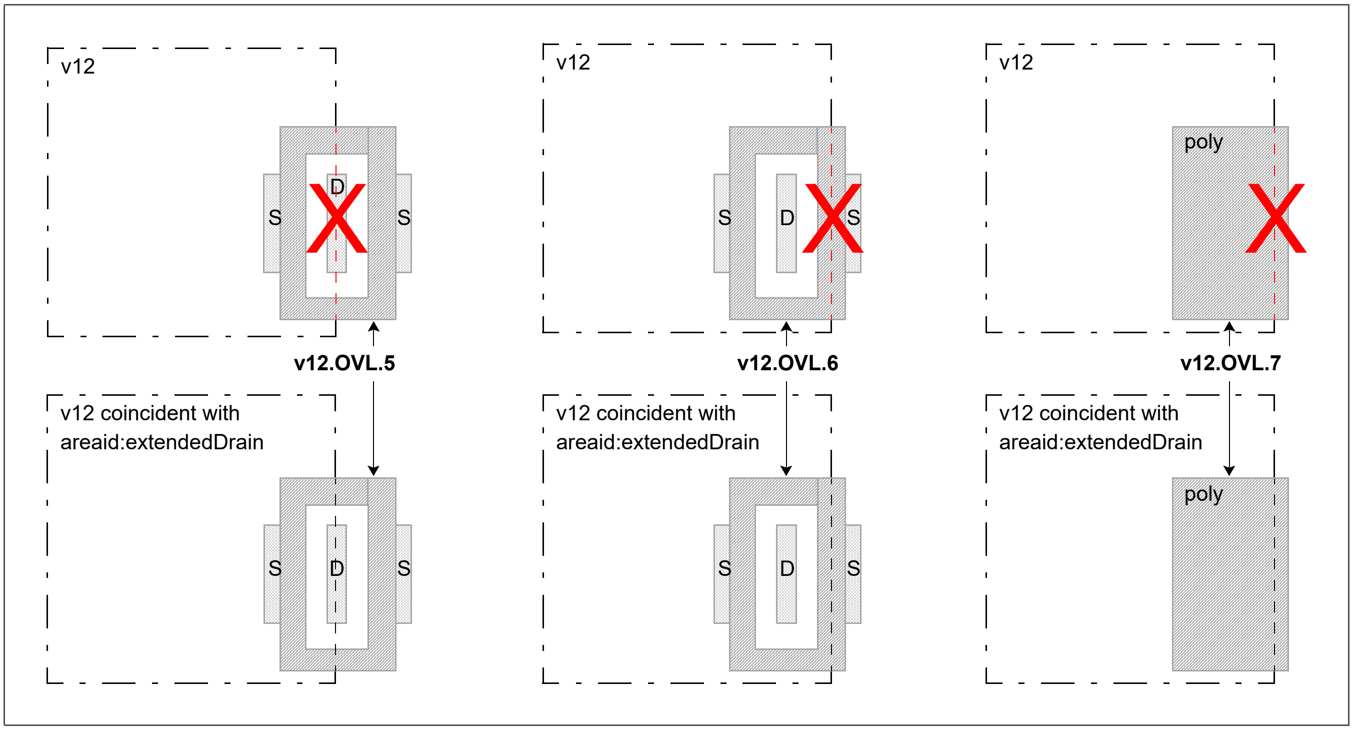


Figure 3-7. v12 Layout (Page 2 of 2)

3.4.8 v20 Design Rules

These design rules are used to define the v20 marker layer.

Table 3-24. v20 Design Rules

Rule No.	Description	Operator	Value (µm)
v20.CON.1	diff must not straddle v20.		–
v20.CON.2	poly must not straddle v20.		–
v20.CON.3	dnwell must not straddle v20.		–
v20.CON.4	areaid:low_vt must not straddle v20.		–
v20.CON.9	v20 over diff must overlap thkox.		–
v20.ENC.1	v20 must enclose pwbm outside areaid:low_vt.		–
v20.ENC.2	v20 interacting with dnwell must enclose dnwell.		–
v20.OVL.1	v20 gate must overlap thkox.		–

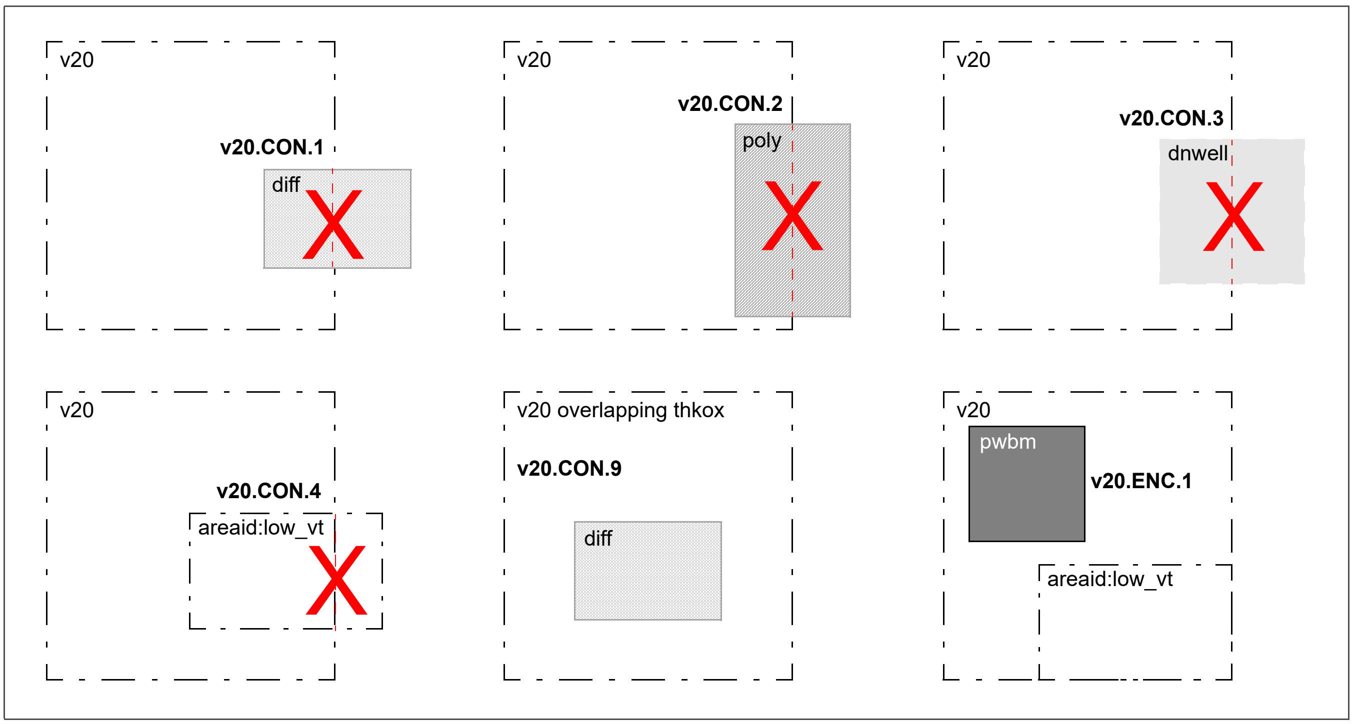


Figure 3-8. v20 Layout (Page 1 of 2)

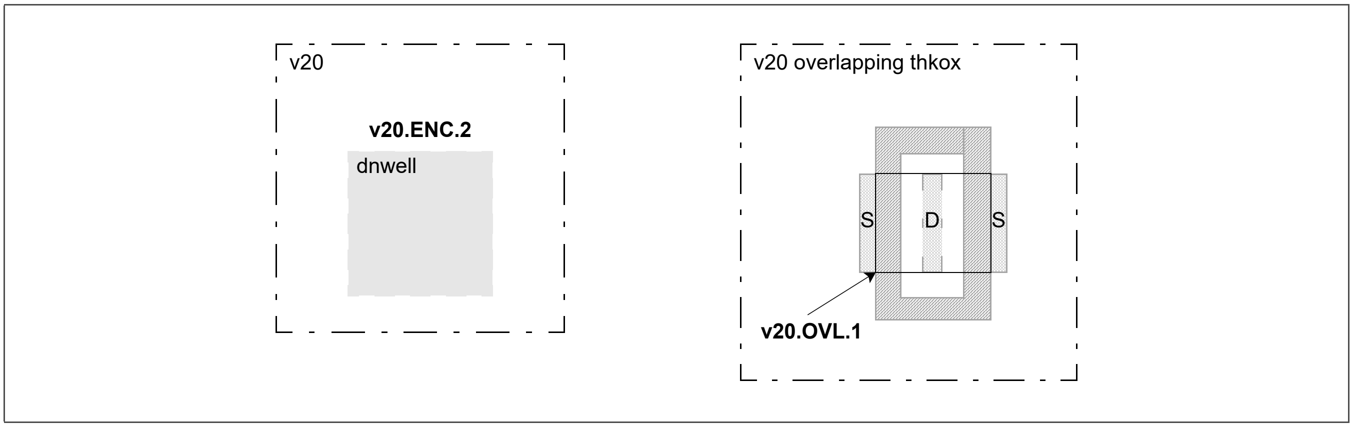


Figure 3-8. v20 Layout (Page 2 of 2)

3.4.9 High-Voltage Design Rules

These design rules are used to define 12 V and 20 V nets. High-voltage poly may be drawn over multiple diffusion regions that are *all* reverse-biased by at least 300 mV (the existence of reverse bias is not checked during DRC). High-voltage poly may also be drawn over multiple diffusion regions when all sources and drains are tied together within each region. Any layout tied to a high-voltage source or drain is considered a high-voltage feature.

Table 3-25. High-Voltage Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
hv.CON.1		hvPoly must overlap only one diff unless all sources and drains are tied together.		–
hv.CON.2		hvPoly must not straddle nwell (exempt inside a drain-extended PMOS device).		–
hv.ENC.1		nwell minimum enclosure of hvPoly, including an hvPoly resistor (exempt inside a drain-extended PMOS device).	≥	0.3
hv.ENC.2		poly minimum extension beyond hvGate (exempt for poly extending beyond diff along the source and drain direction in a drain-extended NMOS or PMOS device).	≥	0.16
hv.SP.1		hv_source/drain minimum space to diff for hv_source/drain diff edges not abutting tap.	≥	0.3
hv.SP.2		ndiff or pdiff resistor connected to hv_source/drain minimum space to diff.	≥	0.3
hv.SP.3		ndiff or pdiff diode connected to hv_source/drain minimum space to diff.	≥	0.3
hv.SP.4	1	nwell connected to hv_source/drain minimum space to ndiff.	≥	0.43
hv.SP.5		N+ hv_source/drain minimum space to nwell.	≥	0.55
hv.SP.6		ndiff resistor connected to hv_source/drain minimum space to nwell.	≥	0.55
hv.SP.7		ndiff diode connected to hv_source/drain minimum space to nwell.	≥	0.55
hv.SP.8		hvPoly, including an hvPoly resistor, minimum space to ≤ 5 V diff (exempt for hvPoly abutting ≥ 12 V diff).	≥	0.3
hv.SP.9		hvPoly, including an hvPoly resistor, minimum space to nwell (exempt for poly straddling nwell inside a drain-extended PMOS device).	≥	0.55

1. This rule is not checked for the source of a drain-extended device.

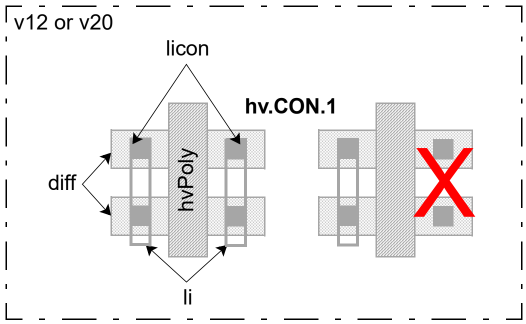


Figure 3-9. High-Voltage Layout

3.4.10 High-V_t PMOS Implant Design Rules

These design rules are used to define the threshold voltage adjustment implant region for a high-V_t 1.8 V PMOS device.

Table 3-26. High-V_t PMOS Implant Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
hvtp.ANG.3		hvtp edge must be octagonal.		–
hvtp.AR.1		hvtp minimum area (µm^2).	≥	0.265
hvtp.AR.2		hvtp hole minimum area (µm^2).	≥	0.265
hvtp.ENC.1	1	hvtp minimum enclosure of pfet outside areaid:core.	≥	0.18
hvtp.SP.1	**2**	hvtp minimum space and notch.	≥	0.38
hvtp.SP.2	1	hvtp minimum space to pfet outside areaid:core.	≥	0.18
hvtp.WID.1	**2**	hvtp minimum width.	≥	0.38

1. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
2. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

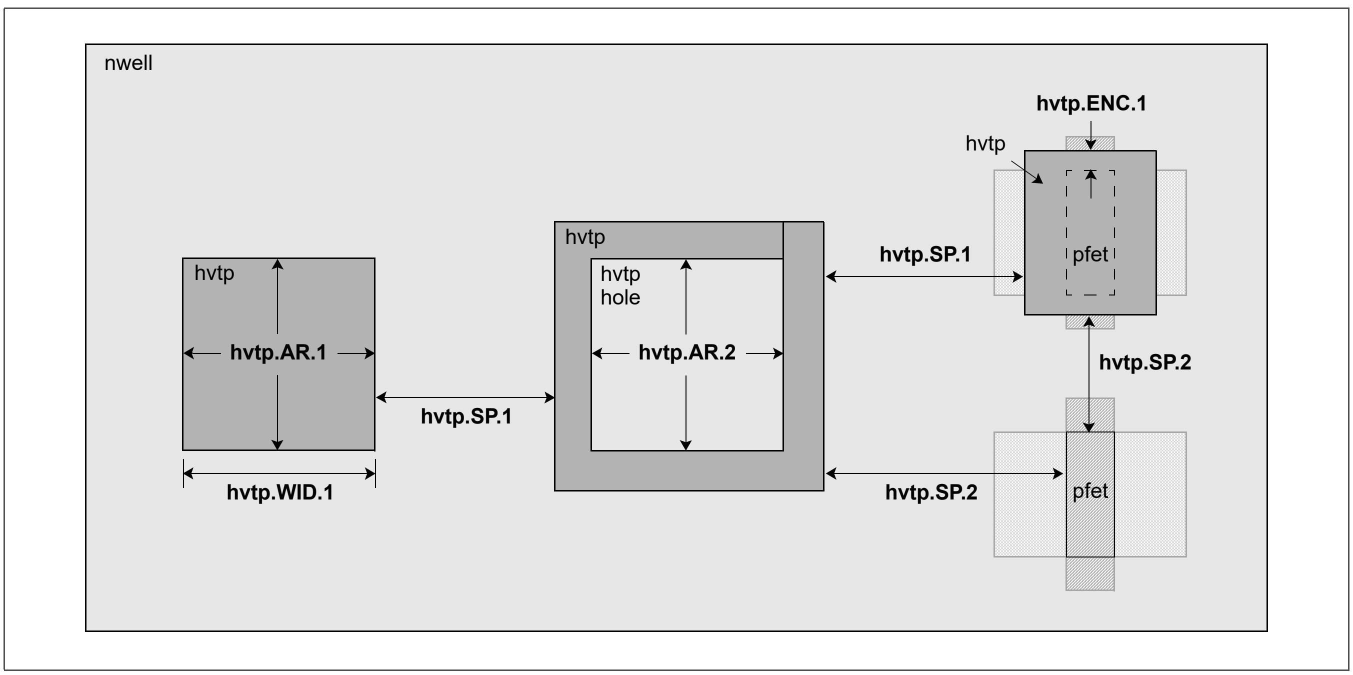


Figure 3-10. High-V_t PMOS Implant Layout

3.4.11 Low-V_t PMOS Implant Design Rules

These design rules are used to define the threshold voltage adjustment implant region for a low-V_t 1.8 V PMOS device.

Table 3-27. Low-V_t PMOS Implant Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
hvtr.ANG.3		hvtr edge must be octagonal.		–
hvtr.CON.1		hvtr must not overlap hvtp.		–
hvtr.ENC.1	1	hvtr minimum enclosure of pfet.	≥	0.18
hvtr.SP.1	**2**	hvtr minimum space to hvtp.	≥	0.38
hvtr.WID.1	**2**	hvtr minimum width.	≥	0.38

1. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
2. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

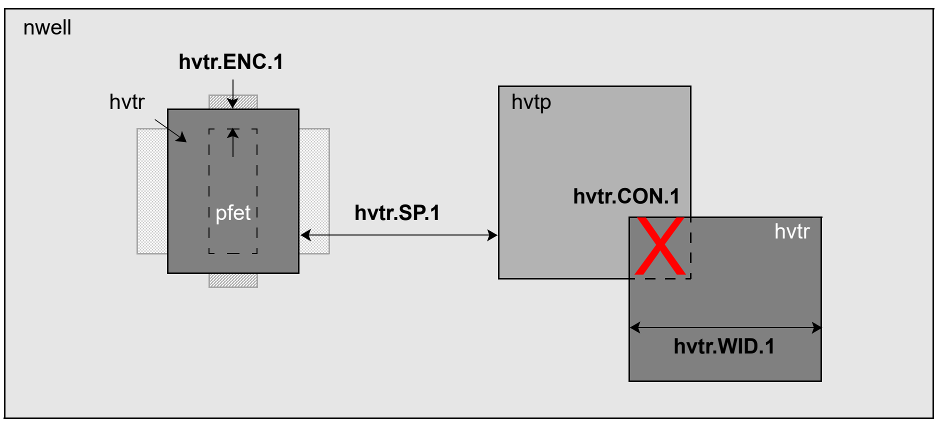


Figure 3-11. Low-V_t PMOS Implant Layout

3.4.12 Low-V_t NMOS/PMOS Implant Blocking Design Rules

These design rules are used to define the region that blocks threshold voltage adjustment implants for low-V_t 1.8 V NMOS and PMOS devices.

Table 3-28. Low-V_t NMOS/PMOS Implant Blocking Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
lvtn.ANG.3		lvtn edge must be octagonal.		–
lvtn.AR.1		lvtn minimum area (μm^2).	≥	0.265
lvtn.AR.2		lvtn hole minimum area (μm^2).	≥	0.265
lvtn.ENC.1	1	lvtn outside areaid:core minimum enclosure of gate outside areaid:core and v20.	≥	0.18
lvtn.ENC.2	2	(nwell not overlapping var_channel) minimum enclosure of lvtn excluding coincident edges.	≥	0.38
lvtn.OVL.1		lvtn must not overlap hvtp.		–
lvtn.SP.1	**3**	lvtn minimum space and notch.	≥	0.38
lvtn.SP.2	1	lvtn minimum space to gate (exempt inside v20 and outside areaid:core).	≥	0.18
lvtn.SP.3	1	lvtn outside areaid:core minimum space to pfet along the source and drain direction.	≥	0.19
lvtn.SP.4		lvtn minimum space to hvtp.	≥	0.38
lvtn.WID.1	**3**	lvtn minimum width.	≥	0.38

1. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
2. This technology does not currently support a varactor device.
3. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

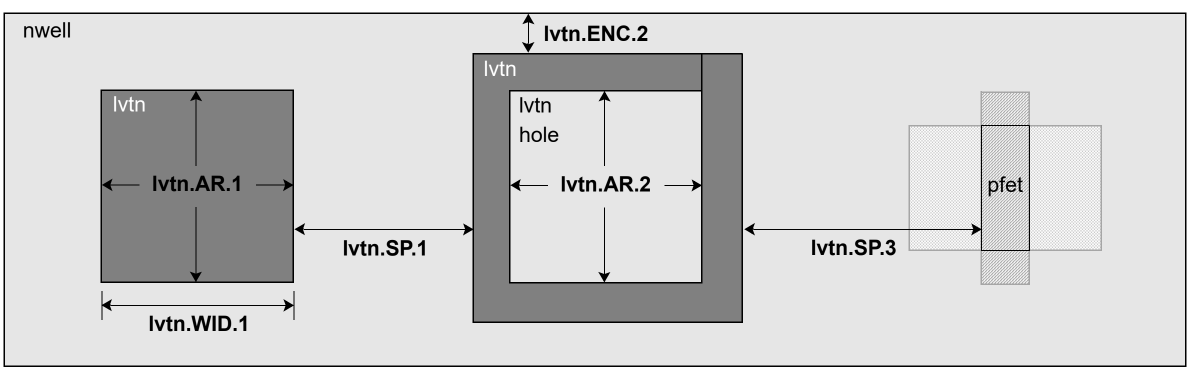


Figure 3-12. Low-V_t NMOS/PMOS Implant Blocking Layout (Page 1 of 2)

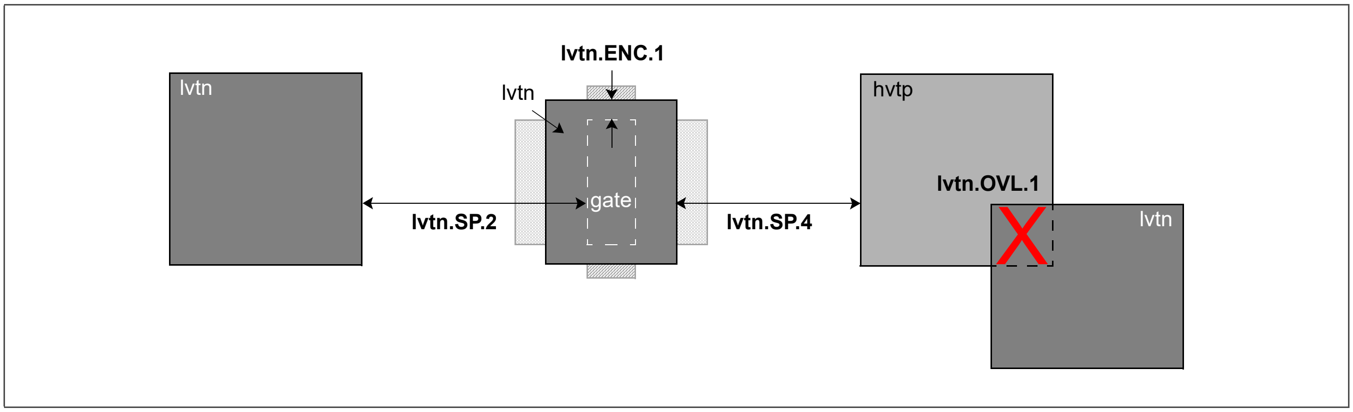


Figure 3-12. Low-V_t NMOS/PMOS Implant Blocking Layout (Page 2 of 2)

3.4.13 5 V N-Tip Implant Design Rules

These design rules are used to define a 5 V n-tip implant region.

Table 3-29. 5 V N-Tip Implant Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
hvntm.ANG.3		hvntm edge must be octagonal.		–
hvntm.CON.2		hvntm must be inside v5 and thkox.		–
hvntm.CON.3		hvntm must not overlap ndiff outside v5 and thkox.		–
hvntm.CON.4		hvntm must not overlap pdiff.		–
hvntm.CON.5	1	hvntm must not overlap ptap except along the edge abutting diff.		–
hvntm.CON.6	1	hvntm must not overlap ptap along the edge abutting diff.		–
hvntm.CON.7	1	hvntm outside areaid:core must enclose esd_nwell_tap inside v5 and thkox.		–
hvntm.CON.8		hvntm must enclose a 5 V esd_nwell_tap outside areaid:core.		–
hvntm.CON.9		hvntm must not overlap areaid:core.		–
hvntm.ENC.1	1	hvntm minimum enclosure of (ndiff inside v5 and thkox) outside areaid:core.	≥	0.185
hvntm.SP.1	1, **2**	hvntm outside areaid:core minimum space and notch.	≥	0.7
hvntm.SP.2	1	hvntm minimum space to ndiff outside v5 and thkox.	≥	0.185
hvntm.SP.3	1, 3	hvntm minimum space to pdiff.	≥	0.185
hvntm.SP.4	1	hvntm minimum space to ptap not along the edge abutting diff.	≥	0.185
hvntm.WID.1	1, **2**	hvntm outside areaid:core minimum width.	≥	0.7

1. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
2. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
3. This rule is not checked for the source of a drain-extended device.

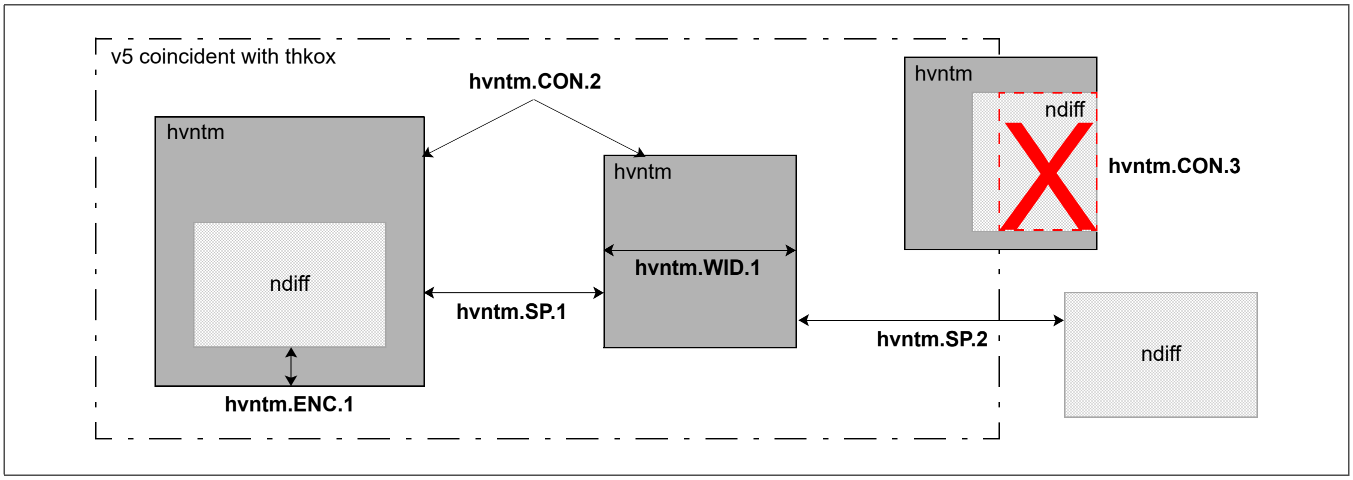


Figure 3-13. 5 V N-Tip Implant Layout

3.4.14 Diffusion Design Rules

These design rules are used to define an active region or a contact to the substrate.

Table 3-30. Diffusion Design Rules (Page 1 of 2)

Rule No.	Notes	Description	Operator	Value (µm)
diff.ANG.1		diff edge outside areaid:seal or v20 must be orthogonal.		–
diff.ANG.2		diff or tap inside areaid:analog must be rectangular.		–
diff.ANG.3		diff edge must be octagonal.		–
tap_extended_drain.ANG.3		tap edge inside areaid:extendedDrain must be octagonal.		–
tap_seal.ANG3		tap edge inside areaid:seal must be octagonal.		–
diffPin.CON.1		diff must enclose dff:pin.		–
diffres.CON.4		diff:res must not overlap licon.		–
diffres.CON.5		diff:res must be coincident inside diff and divide diff into two nets.		–
tap.CON.7		tap must not overlap areaid:seal.		–
diff.CON.8		diff must not straddle areaid:seal.		–
diff.ENC.1	1, 2, 3	nwell minimum enclosure of pdiff inside periphery outside areaid:esd or v20.	≥	0.18
diff.ENC.2	2, 3	nwell minimum enclosure of ntap outside areaid:esd or v20.	≥	0.18
diff.SP.1	**4**	diff minimum space and notch.	≥	0.27
diff.SP.2	2	diff abutting edge minimum space to diff non-coincident edge.	≥	0.13
diff.SP.3	1, 2, 3	(ndiff outside areaid:esd or v20) or (nwell outside areaid:esd) minimum spacing.	≥	0.34
diff.SP.4		ptap outside v20 minimum space to nwell.	≥	0.13
diff.SP.6		diff outside v20 minimum space to pwbm.	≥	0.50
diff.WARN.1		diff without implant is prohibited.		–
diff.WARN.2		diff or tap extension beyond abutting edge is prohibited.		–
diff.WID.1		diff crossing areaid:core minimum width.	≥	0.15
diff.WID.2	3, **4**	diff inside periphery minimum width.	≥	0.15
diff.WID.4		tap crossing areaid:core minimum width.	≥	0.15
diff.WID.5		tap inside periphery minimum width.	≥	0.15
diff.WID.7	3	gate outside areaid:standardc minimum width.	≥	0.42
diff.WID.8	3	gate inside areaid:standardc minimum width.	≥	0.36

1. This rule is not checked for the source of a drain-extended device.
2. This rule is not checked for esd_nwell_tap. A corresponding core rule does not exist.
3. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
4. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
5. This rule is not checked during DRC; use it as a guideline.

Table 3-30. Diffusion Design Rules (Page 2 of 2)

Rule No.	Notes	Description	Operator	Value (µm)
diff.WID.9		tap abutting diff minimum width.	≥	0.29
diff.WID.10	3	tap abutting and between diff inside periphery minimum width.	≥	0.40
NC	5	esd_nwell_tap abutting diff is considered a short circuit.		–

1. This rule is not checked for the source of a drain-extended device.
2. This rule is not checked for esd_nwell_tap. A corresponding core rule does not exist.
3. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
4. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
5. This rule is not checked during DRC; use it as a guideline.

3.4.15 5 V Diffusion Design Rules

These design rules are used to define a 5 V active region or contact to the substrate.

Table 3-31. 5 V Diffusion Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
diff_5v.CON.1	1	diff inside periphery must not straddle v5.		–
diff_5v.CON.1a	1	diff inside periphery must not straddle thkox.		–
diff_5v.ENC.1	2, 3	nwell inside v5 minimum enclosure of pdiff outside areaid:esd.	≥	0.33
diff_5v.ENC.2	3	nwell inside v5 minimum enclosure of ntap outside areaid:esd.	≥	0.33
diff_5v.ENC.3	1	thkox minimum enclosure of diff inside v5 (exempt inside v20).	≥	0.18
diff_5v.SP.1	1	diff inside v5 and periphery minimum space and notch.	≥	0.30
diff_5v.SP.2	1	ndiff minimum space to non-abutting ptap inside v5.	≥	0.37
diff_5v.SP.3	2, 3	ndiff outside areaid:esd minimum space to nwell inside v5.	≥	0.43
diff_5v.SP.4		ptap minimum space to nwell inside v5 (exempt inside v20 and for ptap abutting pwell:res).	≥	0.43
diff_5v.SP.5	1	diff outside thkox minimum space to thkox.	≥	0.18
diff_5v.SP.6	2, 3	(ndiff inside v5) outside v20 and areaid:esd minimum space to nwell.	≥	0.43
diff_5v.WID.1	1	diff inside v5 and periphery minimum width (exempt for pdiff:res inside v5).	≥	0.29
diff_5v.WID.2	1, 4	pdiff:res inside v5 and periphery minimum width.	≥	0.15
diff_5v.WID.3		tap abutting diff on one or two sides inside v5 minimum width (exempt inside v20).	≥	0.70
diff_5v.WID.4		tap abutting and between diff inside v5 minimum width.	≥	0.70
NC	5	ndiff inside v5 minimum space to ndiff inside v5 with non-abutting tap between the ndiff shapes.	≥	1.07

1. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
2. This rule is not checked for the source of a drain-extended device.
3. This rule is not checked for esd_nwell_tap. A corresponding core rule does not exist.
4. This technology does not currently support a 5 V pwell resistor device.
5. This rule is not checked during DRC; use it as a guideline.

3.4.16 Poly Design Rules

These design rules are used to define FET gates, interconnects, resistors, capacitor plates, and poly-bound emitters in npn bipolar junction transistors.

Table 3-32. Poly Design Rules

Rule No.	Notes	Description	Operator	Value (μm)
poly.ANG.1		poly 90° bends on diff are prohibited.		–
poly_not_in_esd.ANG.1		poly edge outside npn, areaid:esd, or anchor region must be orthogonal.		–
poly_esd.ANG.3		poly edge inside areaid:esd must be octagonal.		–
polyPin.CON.1		poly must enclose poly:pin.		–
poly.CON.2		gate must not straddle thkox.		–
polyres.CON.4		poly:res must not overlap licon.		–
polyres.CON.5		poly:res must be coincident inside poly and divide poly into two nets.		–
poly.CON.7		poly must not overlap areaid:seal.		–
poly.ENC.1	1	diff edge of an abutting tap minimum extension beyond gate edge inside periphery.	≥	0.30
poly.ENC.2	1	diff minimum extension beyond gate edge inside periphery.	≥	0.25
poly.ENC.3	1	poly minimum extension beyond gate end inside periphery.	≥	0.13
poly.LEN.1		pfet overlapping lvtn minimum channel length.	≥	0.35
poly.OVL.1		poly:res must not overlap diff.		–
poly.OVL.2		poly must not overlap any diff inner corner.		–
poly.OVL.3	1	poly outside nwell or v20 must not overlap tap inside periphery.		–
poly.OVL.4		poly must not overlap diff:res.		–
poly.SP.1	**2**	poly inside periphery minimum space and notch.	≥	0.21
poly.SP.2		poly not covered by cell sr_bltd_eq crossing areaid:core boundary minimum spacing.	≥	0.21
poly.SP.5	1	poly inside periphery minimum space to diff.	≥	0.075
poly.SP.6	1	poly inside periphery minimum space to tap.	≥	0.055
poly.SP.7		poly:res inside (rpm OR urpm) minimum space to diff.	≥	0.48
poly.SP.8		poly:res minimum space to poly.	≥	0.21
poly.WID.1		gate inside thkox inside periphery minimum width.	≥	0.50
poly.WID.2	**2**	poly minimum width.	≥	0.15
poly.WID.3		poly:res minimum width.	≥	0.33

1. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
2. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.4.17 5 V Poly Design Rules

These design rules are used to define 5 V FET gates.

Table 3-33. 5 V Poly Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
hpoly.CON.1		gate must not straddle v5.		–
hpoly.WID.1	1	gate inside v5 inside periphery minimum width.	≥	0.5

1. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.

3.4.18 Nitride Poly Cut Design Rules

These design rules are used to define a nitride opening for a poly contact, subsequently formed using local interconnect contact (licon) and local interconnect (li) layers.

Table 3-34. Nitride Poly Cut Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
npc.ANG.3		npc edge must be octagonal.		–
npc.CON.1		npc must not overlap gate.		–
npc.SP.1	**1**	npc minimum space and notch.	≥	0.27
npc.SP.2		npc minimum space to gate.	≥	0.09
npc.WID.1	**1**	npc minimum width.	≥	0.27

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.4.19 Extended Drain Design Rules

These design rules are used to define the areaid:extendedDrain marker layer.

Table 3-35. Extended Drain Design Rules

Rule No.	Description	Operator	Value (µm)
extd.CON.1	diff must not straddle areaid:extendedDrain.		–
extd.CON.2	diff enclosed by areaid:extendedDrain must have two or three coincident edges with areaid:extendedDrain.		–
extd.CON.3	poly must extend beyond overlapping diff inside areaid:extendedDrain.		–

3.4.20 N+ Source/Drain Implant Design Rules

These design rules are used to define the opening for an n+ source or drain implant.

Table 3-36. N+ Source/Drain Implant Design Rules

Rule No.	Notes	Description	Operator	Value (μm)
nsdm.ANG.3		nsdm edge must be octagonal.		–
nsdm.AR.1		nsdm minimum area (μm^2).	≥	0.265
nsdm.AR.2		nsdm hole minimum area (μm^2).	≥	0.265
nsdm.ENC.1		nsdm minimum enclosure of ndiff.	≥	0.125
nsdm.ENC.2	1	nsdm minimum enclosure of ntap inside periphery.	≥	0.125
nsdm.OVL.1	2	nsdm must not overlap pdiff or ptap (source of drain-extended FET is exempt).		–
nsdm.SP.1	1	nsdm inside periphery minimum space and notch.	≥	0.38
nsdm.SP.2	**3**	nsdm crossing areaid:core boundary minimum spacing.	≥	0.38
nsdm.SP.3		nsdm minimum space to pdiff.	≥	0.13
nsdm.WID.1	**3**	nsdm crossing areaid:core minimum width.	≥	0.38
nsdm.WID.2	1	nsdm inside periphery minimum width.	≥	0.38

1. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
2. This rule is not checked for the source of a drain-extended device.
3. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.4.21 P+ Source/Drain Implant Design Rules

These design rules are used to define the opening for a p+ source or drain implant.

Table 3-37. P+ Source/Drain Implant Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
psdm.ANG.3		psdm edge must be octagonal.		–
psdm.AR.1		psdm minimum area (μm^2).	≥	0.255
psdm.AR.2		psdm hole minimum area (μm^2).	≥	0.265
psdm.CON.6		psdm must not overlap nsdm.		–
psdm.ENC.1		psdm minimum enclosure of pdiff.	≥	0.125
psdm.ENC.2	1	psdm minimum enclosure of ptap inside periphery.	≥	0.125
psdm.OVL.1	2	psdm must not overlap ndiff or ntap (source of drain-extended FET is exempt).		–
psdm.SP.1	1	psdm inside periphery minimum space and notch.	≥	0.38
psdm.SP.2	**3**	psdm crossing areaid:core boundary minimum spacing.	≥	0.38
psdm.SP.3		psdm minimum space to ndiff.	≥	0.13
psdm.WID.1	**3**	psdm crossing areaid:core minimum width.	≥	0.38
psdm.WID.2	1	psdm inside periphery minimum width.	≥	0.38

1. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
2. This rule is not checked for the source of a drain-extended device.
3. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.4.22 Local Interconnect Contact Design Rules

These design rules are used to define the contact between poly or diff and the local interconnect (li) layer.

Table 3-38. Local Interconnect Contact Design Rules (Page 1 of 2)

Rule No.	Notes	Description	Operator	Value (µm)
licon.ANG.1		licon must be orthogonal.		–
licon.ANG.2		licon must be rectangular.		–
licon.CON.2		licon must be inside li and diff or poly.		–
licon.CON.6		licon must not overlap gate.		–
licon.CON.7	1	licon must not straddle tap.		–
licon.CON.8	1	Every source or drain diff must enclose at least one licon (exempt inside v20).		–
licon.CON.9	1	Every tap must enclose at least one licon (exempt inside v20).		–
licon.CON.10	**2**	licon must not overlap gate.		–
licon.CON.11		poly_licon must be inside npc.		–
licon.ENC.1	1	(diff NOT tap) minimum enclosure of licon.	≥	0.04
licon.ENC.2	1	diff minimum enclosure of licon on at least two opposite sides.	≥	0.06
licon.ENC.4	1	isolated_tap minimum enclosure of licon on at least two opposite sides.	≥	0.12
licon.ENC.5	1	poly minimum enclosure of licon.	≥	0.05
licon.ENC.6	1	poly minimum enclosure of licon on at least two opposite sides.	≥	0.08
licon.ENC.7	1	npc minimum enclosure of poly_licon outside areaid:core.	≥	0.10
licon.LEN.1		licon (non-bar) maximum length.	≤	0.17
licon.SP.1	1, **2**	licon (non-bar) minimum spacing.	≥	0.17
licon.SP.2		licon bar end to end minimum spacing.	≥	0.35
licon.SP.3		licon bar side to side minimum spacing.	≥	0.51
licon.SP.4		licon bar minimum space to licon square.	≥	0.51
licon.SP.5	1	tap_licon minimum space to tap edge abutting diff.	≥	0.06
licon.SP.6	1	poly_licon minimum space to psdm (overlap prohibited).	≥	0.11
licon.SP.7	1, 3	licon on (tap inside low-voltage nwell) minimum space to var_channel.	≥	0.25
licon.SP.8	1	licon on diff minimum space to gate outside areaid:standardc (overlap prohibited).	≥	0.055
licon.SP.9	1	licon on diff minimum space to gate inside areaid:standardc (exempt for 0.15 µm high-V_t pfet).	≥	0.05
licon.SP.10	1	licon on diff minimum space to gate inside areaid:standardc for high-V_t pfet.	≥	0.05

1. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
2. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
3. This technology does not currently support a varactor device.
4. This rule is not checked during DRC; use it as a guideline.

Table 3-38. Local Interconnect Contact Design Rules (Page 2 of 2)

Rule No.	Notes	Description	Operator	Value (µm)
licon.SP.11	1, **2**	licon on diff minimum space to npc (overlap prohibited).	≥	0.09
licon.SP.12	1	poly_licon minimum space to diff.	≥	0.19
licon.WID.1	**2**	licon (non-bar) minimum width.	≥	0.17
licon.WID.2		licon bar exact size: 0.19 µm × 2.0 µm		–
licon.WID.3		licon inside areaid:seal minimum width.	≥	0.17
licon.WID.4		licon inside areaid:seal maximum width.	≤	0.175
NC	4	source or drain without licon maximum width.	≤	5.7

1. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
2. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
3. This technology does not currently support a varactor device.
4. This rule is not checked during DRC; use it as a guideline.

3.4.23 Local Interconnect Design Rules

These design rules are used to define the local interconnect (li) layer to diff and poly.

Table 3-39. Local Interconnect Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
li.ANG.1		li must be orthogonal.		–
li.AR.1	1, **2**	li minimum area (μm^2).	≥	0.0561
li_block.CON.1		li must not overlap li:block.		–
liPin.CON.1		li must enclose li:pin.		–
lires.CON.5		li:res must be coincident inside li and divide li into two nets.		–
li.CON.7		li must not overlap areaid:seal.		–
li.ENC.1	1, **2**	li minimum enclosure of licon on at least two opposite sides.	≥	0.08
li_block.SP.1		li minimum space to li:block.	≥	0.17
li.SP.1	1, **2**	li crossing or outside areaid:core minimum spacing (exempt for vertical parallel plate [VPP] capacitors).	≥	0.17
li.SP.2	1, 3	li in vppcap minimum spacing.	≥	0.14
li.WID.1	1, **2**	li crossing or outside areaid:core minimum width (exempt for VPP capacitors).	≥	0.17
li.WID.2	1, 3	li in vppcap minimum width.	≥	0.14
li.WID.4		li:res minimum width.	≥	0.29
NC	4	li without licon or mcon maximum aspect ratio (length to width).	≤	10:1

1. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
2. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
3. This technology does not currently support a VPP capacitor.
4. This rule is not checked during DRC; use it as a guideline.

3.4.24 Metal Contact Design Rules

These design rules are used to define the contact between the local interconnect (li) and first metal (met1) layers.

Table 3-40. Metal Contact Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
mcon.ANG.1		mcon must be orthogonal.		–
mcon.ANG.2		mcon must be rectangular.		–
mcon.CON.2		mcon must be inside met1 and li.		–
mcon.CON.10	**1**	li must enclose mcon.		–
mcon.LEN.1	**1**, 2	mcon maximum length.	≤	0.17
mcon.SP.1	**1**, 2	mcon minimum spacing.	≥	0.19
mcon.WID.1	**1**, 2	mcon minimum width.	≥	0.17

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
2. The drawn dimension does not equal the final dimension on silicon.

3.4.25 First Metal Layer Design Rules

These design rules are used to define the first layer of metal interconnects, buses, and so on.

Table 3-41. First Metal Layer Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
met1.ANG.3		met1 edge must be octagonal.		–
met1.AR.1	**1**	met1 minimum area (μm^2).	≥	0.083
met1.AR.2	**1**	met1 hole minimum area (μm^2).	≥	0.14
met1_block.CON.1		met1 must not overlap met1:block.		–
met1Pin.CON.1		met1 must enclose met1:pin.		–
met1res.CON.5		met1:res must be coincident inside met1 and divide met1 into two nets.		–
met1.CON.7		met1 must not overlap areaid:seal.		–
met1.ENC.1	**1**, 2	met1 minimum enclosure of mcon (except inside areaid:core).	≥	0.03
met1.ENC.2	**1**, 2, 3	met1 minimum enclosure of mcon adjacent sides inside periphery.	≥	0.06
met1_block.SP.1		met1 minimum space to met1:block.	≥	0.14
met1_block.SP.2		met1:block minimum space to met1:routing.	≥	0.145
met1.SP.1	**1**	met1 minimum space and notch.	≥	0.14
met1.SP.2	**1**	wide_met1 minimum space to met1.	≥	0.28
met1.WID.1	**1**	met1 minimum width.	≥	0.14

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
2. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
3. This rule applies to an aluminum (Al) physical design flow only.

3.4.26 First Via Contact Design Rules

These design rules are used to define the contact between the first metal (met1) and second metal (met2) layers.

Table 3-42. First Via Contact Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
via1.ANG.1		via1 must be orthogonal.		–
via1.ANG.2		via1 must be rectangular.		–
via1.CON.2		via1 must be inside met1 and met2.		–
via1.CON.10	**1**	via1 outside areaid:moduleCut must be an orthogonal rectangle.		–
via1.ENC.1	**1**	met1 outside areaid:moduleCut minimum enclosure of via1.	≥	0.055
via1.ENC.2	2	met1 inside areaid:moduleCut minimum enclosure of 0.23 µm via1.	≥	0.03
via1.ENC.3	2	met1 inside areaid:moduleCut minimum enclosure of 0.28 µm via1.	≥	0
via1.ENC.4	**1**	met1 minimum enclosure of 0.15 µm via1 on at least two opposite sides.	≥	0.085
via1.ENC.5	2	met1 minimum enclosure of 0.23 µm via1 on at least two opposite sides.	≥	0.085
via1.LEN.1	**1**, 2	via1 outside areaid:moduleCut maximum length.	≤	0.15
via1.SP.1	**1**, 2	via1 minimum spacing.	≥	0.17
via1.WID.1	**1**, 2	via1 outside areaid:moduleCut minimum width.	≥	0.15
via1.WID.2	2	Only 0.15 µm, 0.23 µm, and 0.28 µm square via1 contacts are permitted inside areaid:moduleCut.		–

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
2. This rule applies to an aluminum (Al) physical design flow only.

3.4.27 Second Metal Layer Design Rules

These design rules are used to define the second layer of metal interconnects, buses, and so on.

Table 3-43. Second Metal Layer Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
met2.ANG.3		met2 edge must be octagonal.		–
met2.AR.1	**1**	met2 minimum area (μm^2).	≥	0.0676
met2.AR.2	**1**	met2 hole minimum area (μm^2).	≥	0.14
met2_block.CON.1		met2 must not overlap met2:block.		–
met2Pin.CON.1		met2 must enclose met2:pin.		–
met2res.CON.5		met2:res must be coincident inside met2 and divide met2 into two nets.		–
met2.CON.7		met2 must not overlap areaid:seal.		–
met2.ENC.1	**1**, 2, 3	met2 minimum enclosure of via1.	≥	0.055
met2.ENC.2	**1**, 2, 3	met2 minimum enclosure of via1 adjacent sides inside periphery.	≥	0.085
met2_block.SP.1		met2 minimum space to met2:block.	≥	0.14
met2_block.SP.2		met2:block minimum space to met2:routing.	≥	0.145
met2.SP.1	**1**	met2 minimum space and notch.	≥	0.14
met2.SP.2	**1**	wide_met2 minimum space to met2.	≥	0.28
met2.WID.1	**1**	met2 minimum width.	≥	0.14

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
2. This rule applies only to the periphery, that is, outside areaid:core. A corresponding core rule might exist.
3. This rule applies to an aluminum (Al) physical design flow only.

3.4.28 Second Via Contact Design Rules

These design rules are used to define the contact between the second metal (met2) and third metal (met3) layers.

Table 3-44. Second Via Contact Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
via2.ANG.1		via2 must be orthogonal.		–
via2.ANG.2		via2 must be rectangular.		–
via2.CON.2		via2 must be inside met2 and met3.		–
via2.CON.10	**1**	via2 outside areaid:moduleCut must be an orthogonal rectangle.		–
via2.ENC.1	**1**, 2	met2 minimum enclosure of via2.	≥	0.04
via2.ENC.2		met2 inside areaid:moduleCut minimum enclosure of 1.5 µm via2.	≥	0.14
via2.ENC.3	**1**, 2	met2 minimum enclosure of via2 on at least two opposite sides.	≥	0.085
via2.LEN.1	**1**, 2	via2 outside areaid:moduleCut maximum length.	≤	0.20
via2.SP.1	**1**, 2	via2 minimum spacing.	≥	0.20
via2.WID.1	**1**, 2	via2 outside areaid:moduleCut minimum width.	≥	0.20
via2.WID.2	2	Only 0.20 µm, 0.28 µm, 1.2 µm, and 1.5 µm square via2 contacts are permitted inside areaid:moduleCut.		–

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
2. This rule applies to an aluminum (Al) physical design flow only.

3.4.29 Third Metal Layer Design Rules

These design rules are used to define the third layer of metal interconnects, buses, and so on.

Table 3-45. Third Metal Layer Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
met3.ANG.3		met3 edge must be octagonal.		–
met3.AR.1	**1**	met3 minimum area (μm^2).	≥	0.24
met3.AR.2	**1**	met3 hole minimum area (μm^2).	≥	0.2
met3_block.CON.1		met3 must not overlap met3:block.		–
met3Pin.CON.1		met3 must enclose met3:pin.		–
met3res.CON.5		met3:res must be coincident inside met3 and divide met3 into two nets.		–
met3.CON.7		met3 must not overlap areaid:seal.		–
met3.ENC.1	**1**, 2	met3 minimum enclosure of via2.	≥	0.065
met3_block.SP.1		met3 minimum space to met3:block.	≥	0.3
met3_block.SP.2		met3:block minimum space to met3:routing.	≥	0.305
met3.SP.1	**1**	met3 minimum space and notch.	≥	0.3
met3.SP.2	**1**	wide_met3 minimum space to met3.	≥	0.4
met3.WID.1	**1**	met3 minimum width.	≥	0.3

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
2. This rule applies to an aluminum (Al) physical design flow only.

3.4.30 Third Via Contact Design Rules

These design rules are used to define the contact between the third metal (met3) and fourth metal (met4) layers.

Table 3-46. Third Via Contact Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
via3.ANG.1		via3 must be orthogonal.		–
via3.ANG.2		via3 must be rectangular.		–
via3.CON.2		via3 must be inside met3 and met4.		–
via3.CON.10	**1**	via3 outside areaid:moduleCut must be an orthogonal rectangle.		–
via3.ENC.1	**1**, 2	met3 minimum enclosure of via3.	≥	0.06
via3.ENC.2	**1**, 2	met3 minimum enclosure of via3 on at least two opposite sides.	≥	0.09
via3.LEN.1	**1**, 2	via3 outside areaid:moduleCut maximum length.	≤	0.20
via3.SP.1	**1**, 2	via3 minimum spacing.	≥	0.20
via3.WID.1	**1**, 2	via3 outside areaid:moduleCut minimum width.	≥	0.20
via3.WID.2	2	Only 0.20 µm and 0.80 µm square via3 contacts are permitted inside areaid:moduleCut.		–

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
2. This rule applies to an aluminum (Al) physical design flow only.

3.4.31 Fourth Metal Layer Design Rules

These design rules are used to define the fourth layer of metal interconnects, buses, and so on.

Table 3-47. Fourth Metal Layer Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
met4.ANG.3		met4 edge must be octagonal.		–
met4.AR.1	**1**	met4 minimum area (µm^2).	≥	0.24
met4.AR.2	**1**	met4 hole minimum area (µm^2).	≥	0.2
met4_block.CON.1		met4 must not overlap met4:block.		–
met4Pin.CON.1		met4 must enclose met4:pin.		–
met4res.CON.5		met4:res must be coincident inside met4 and divide met4 into two nets.		–
met4.CON.7		met4 must not overlap areaid:seal.		–
met4.ENC.1	**1**, 2	met4 minimum enclosure of via3.	≥	0.065
met4_block.SP.1		met4 minimum space to met4:block.	≥	0.3
met4_block.SP.2		met4:block minimum space to met4:routing.	≥	0.305
met4.SP.1	**1**	met4 minimum space and notch.	≥	0.3
met4.SP.2	**1**	wide_met4 minimum space to met4.	≥	0.4
met4.WID.1	**1**	met4 minimum width.	≥	0.3

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.
2. This rule applies to an aluminum (Al) physical design flow only.

3.4.32 Fourth Via Contact Design Rules

These design rules are used to define the contact between the fourth metal (met4) and fifth metal (met5) layers.

Table 3-48. Fourth Via Contact Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
via4.ANG.1		via4 must be orthogonal.		–
via4.ANG.2		via4 must be rectangular.		–
via4.CON.2		via4 must be inside met4 and met5.		–
via4.CON.10	**1**	via4 outside areaid:moduleCut must be an orthogonal rectangle.		–
via4.ENC.1	**1**	met4 minimum enclosure of via4.	≥	0.06
via4.LEN.1	**1**	via4 outside areaid:moduleCut maximum length.	≤	0.80
via4.SP.1	**1**	via4 minimum spacing.	≥	0.80
via4.WID.1	**1**	via4 outside areaid:moduleCut minimum width.	≥	0.80

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.4.33 Fifth Metal Layer Design Rules

These design rules are used to define the fifth layer of metal interconnects, buses, and so on.

Table 3-49. Fifth Metal Layer Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
met5.ANG.3		met5 edge must be octagonal.		–
met5.AR.1	**1**	met5 minimum area (μm^2).	≥	4.0
met5.AR.2	**1**	met5 hole minimum area (μm^2).	≥	0.14
met5_block.CON.1		met5 must not overlap met5:block.		–
met5Pin.CON.1		met5 must enclose met5:pin.		–
met5res.CON.5		met5:res must be coincident inside met5 and divide met5 into two nets.		–
met5.CON.7		met5 must not overlap areaid:seal.		–
met5.ENC.1	**1**	met5 minimum enclosure of via4.	≥	0.31
met5_block.SP.1		met5 minimum space to met5:block.	≥	1.6
met5_block.SP.2		met5:block minimum space to met5:routing.	≥	1.605
met5.SP.1	**1**	met5 minimum space and notch.	≥	1.6
met5.WID.1	**1**	met5 minimum width.	≥	1.6

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.4.34 Seal Ring Design Rules

These design rules are used to define the areaid:seal marker layer.

Table 3-50. Seal Ring Design Rules

Rule No.	Description	Operator	Value (µm)
seal.CON.1	Lower-left corner of areaid:seal must coincide with the design grid origin (0,0).		–
seal.WID.1	areaid:seal minimum width.	≥	6.0

3.4.35 Nitride Seal Mask Design Rules

These design rules are used to define the nitride seal mask drawing layer.

Table 3-51. Nitride Seal Mask Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
nsm.ANG.3		nsm edge must be octagonal.		–
nsm.ENC.2		areaid:frameRect minimum enclosure of diff.	≥	3.0
nsm.ENC.3		areaid:frameRect minimum enclosure of diff_fill.	≥	3.0
nsm.ENC.5		areaid:frameRect minimum enclosure of poly.	≥	3.0
nsm.ENC.7		areaid:frameRect minimum enclosure of li.	≥	3.0
nsm.ENC.9		areaid:frameRect minimum enclosure of met1.	≥	3.0
nsm.ENC.11		areaid:frameRect minimum enclosure of met2.	≥	3.0
nsm.ENC.13		areaid:frameRect minimum enclosure of met3.	≥	3.0
nsm.ENC.15		areaid:frameRect minimum enclosure of met4.	≥	3.0
nsm.ENC.17		areaid:frameRect minimum enclosure of met5.	≥	3.0
nsm.OVL.2		diff must not overlap nsm.		–
nsm.OVL.3		diff_fill must not overlap nsm.		–
nsm.OVL.5		poly must not overlap nsm.		–
nsm.OVL.7		li must not overlap nsm.		–
nsm.OVL.9		met1 must not overlap nsm.		–
nsm.OVL.11		met2 must not overlap nsm.		–
nsm.OVL.13		met3 must not overlap nsm.		–
nsm.OVL.15		met4 must not overlap nsm.		–
nsm.OVL.17		met5 must not overlap nsm.		–
nsm.SP.1	**1**	nsm minimum spacing.	≥	4.0
nsm.SP.2		nsm minimum space to non-exempt diff.	≥	1.0
nsm.SP.3		nsm minimum space to non-exempt diff_fill.	≥	1.0
nsm.SP.5		nsm minimum space to non-exempt poly.	≥	1.0
nsm.SP.7		nsm minimum space to non-exempt li.	≥	1.0
nsm.SP.9		nsm minimum space to non-exempt met1.	≥	1.0
nsm.SP.11		nsm minimum space to non-exempt met2.	≥	1.0
nsm.SP.13		nsm minimum space to non-exempt met3.	≥	1.0
nsm.SP.15		nsm minimum space to non-exempt met4.	≥	1.0
nsm.SP.17		nsm minimum space to non-exempt met5.	≥	1.0
nsm.WID.1	**1**	nsm minimum width.	≥	3.0

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.4.36 Bond Pad Design Rules

These design rules are used to define the passivation opening.

Table 3-52. Bond Pad Design Rules (Page 1 of 3)

Rule No.	Notes	Description	Operator	Value (µm)
pad.ANG.3		pad edge must be octagonal.		–
pad.CON.1		bondpad must use pad pcells.		–
pad.CON.3		Hermetic package pads are not supported in this physical design flow.		–
pad.CON.4		bondpad 90° corners are prohibited.		–
pad.CON.5		bondpad missing a 45° corner is prohibited.		–
pad.CON.6		bondpad must have four chamfered 45° corners.		–
pad.CON.7		bondpad must have only four orthogonal edges.		–
pad.CON.10	1	pmm must enclose pad opening inside inductor.		–
pad.CON.11	1	met5 must enclose pad opening inside inductor.		–
pad.CON.12		Only one areaid:pad_pwr, areaid:pad_io, or areaid:pad_gnd may be used on a single pad.		–
pad.CON.13		areaid:pad_pwr, areaid:pad_io, and areaid:pad_gnd must be inside pad.		–
pad.CON.14		met4 inside pad is prohibited.		–
pad.ENC.1		met5 minimum enclosure of bondpad.	≥	0.27
pad.ENC.2		Solid seal minimum enclosure of any plastic pad.	≥	16.99
pad.ENC.3	1	pmm minimum enclosure of pad opening inside inductor.	≥	0
pad.ENC.4	1	met5 minimum enclosure of pad opening inside inductor.	≥	2.7
pad.ENC.5	1	rdl minimum enclosure of pmm inside inductor.	≥	10.75
pad.ENC.6	1	rdl must enclose pmm inside inductor.		–
pad.LEN.1		45° bevel on bondpad minimum length.	≥	7.0
pad.LEN.2		45° bevel on bondpad maximum length.	≤	8.8
pad.LEN.3		Fine-pitch pad in y-direction minimum length.	≥	60.0
pad.LEN.4		Staggered pad in y-direction minimum length.	≥	60.0
pad.LEN.5		High-parallel pad in y-direction minimum length.	≥	60.0
pad.LEN.7		Wafer-level burn-in pad in y-direction minimum length.	≥	60.0
pad.LEN.9		bondpad maximum length or width.	≤	150.0
pad.OVL.1		pad opening (not probe) must not overlap met4.		–
pad.SP.1		Ring-shaped met4 inside bondpad must be coincident with and outside pad layer of pad cell.		–

1. This technology does not currently support an inductor device.
2. Small bondpad width is 30–60 µm.
3. Large bondpad width is 61–150 µm.
4. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

Table 3-52. Bond Pad Design Rules (Page 2 of 3)

Rule No.	Notes	Description	Operator	Value (µm)
pad.SP.2	2	pad opening inside a group of small bondpads in the y-direction minimum space to met4 or met5 outside bondpad.	≥	5.0
pad.SP.3	3	pad opening inside a group of large bondpads in the y-direction minimum space to met4 or met5 outside bondpad.	≥	10.0
pad.SP.4	2	pad opening inside a group of small bondpads in the x-direction minimum space to met4 or met5 outside bondpad.	≥	5.0
pad.SP.5	3	pad opening inside a group of large bondpads in the x-direction minimum space to met4 or met5 outside bondpad.	≥	10.0
pad.SP.6	**4**	pad minimum spacing.	≥	1.27
pad.SP.10		bondpad opening maximum space to scribe_line.	≤	500.0
pad.SP.11		met1 under pad metal minimum space and notch.	≥	1.5
pad.SP.12		met2 under pad metal minimum space and notch.	≥	1.5
pad.SP.13		met3 under pad metal minimum space and notch.	≥	1.5
pad.SP.14		Fine-pitch pad in x-direction minimum space to fine-pitch, high-parallel, staggered, or wafer-level burn-in pad.	≥	8.0
pad.SP.15		High-parallel pad in x-direction minimum space to high-parallel, staggered, or wafer-level burn-in pad.	≥	15.0
pad.SP.17		Wafer-level burn-in pad in x-direction minimum space to staggered or wafer-level burn-in pad.	≥	50.0
pad.SP.18		Staggered pad in x-direction minimum spacing.	≥	30.0
pad.SP.19		Staggered pad to adjacent row in x-direction minimum pitch.	≥	40.0
pad.SP.20		Staggered pad to adjacent row in y-direction minimum pitch.	≥	40.0
pad.SP.21		Staggered pad to adjacent row in y-direction minimum spacing.	≥	9.0
pad.SP.22		Staggered pad opening minimum space to adjacent scribe_line in x-direction.	≥	200.0
pad.SP.23		Fine-pitch pad opening minimum space to adjacent scribe_line in x-direction.	≥	200.0
pad.SP.24		Wafer-level burn-in pad opening minimum space to adjacent scribe_ line in x-direction.	≥	200.0
pad.WID.1		met1 under pad metal minimum width.	≥	0.14
pad.WID.2		met2 under pad metal minimum width.	≥	0.14
pad.WID.3		met3 under pad metal minimum width.	≥	0.30
pad.WID.4		met1 under pad metal maximum width.	≤	0.25
pad.WID.5		met2 under pad metal maximum width.	≤	0.25

1. This technology does not currently support an inductor device.
2. Small bondpad width is 30–60 µm.
3. Large bondpad width is 61–150 µm.
4. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

Table 3-52. Bond Pad Design Rules (Page 3 of 3)

Rule No.	Notes	Description	Operator	Value (µm)
pad.WID.6		met3 under pad metal maximum width.	≤	0.60
pad.WID.7		Fine-pitch pad in x-direction minimum width.	≥	60.0
pad.WID.8		Staggered pad in x-direction minimum width.	≥	60.0
pad.WID.9		High-parallel pad in x-direction minimum width.	≥	60.0
pad.WID.12		Wafer-level burn-in pad in x-direction minimum width.	≥	50.0
pad.WID.14	1	Pad opening inside inductor minimum width.	≥	5.0

1. This technology does not currently support an inductor device.
2. Small bondpad width is 30–60 µm.
3. Large bondpad width is 61–150 µm.
4. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.4.37 Antenna Design Rules

These design rules are used to define an antenna.

Table 3-53. Antenna Design Rules

Rule No.	Notes	Description	Operator	Value
li.ANT.1		Maximum ratio of li perimeter area to connected gate area.	≤	75
licon.ANT.1		Maximum ratio of licon area to connected gate area.	≤	3
mcon.ANT.1		Maximum ratio of mcon area to connected gate area.	≤	3
metX.ANT.1		Maximum ratio of metX perimeter area, where metX = met1 or met 2, to connected gate area.	≤	400
met2.ANT.2	1	met2 exact spacing between (met2 areas with met2-to-via2 surface area ratio ≥ 20.0) and (met2 areas with met2-to-via2 surface area ratio ≤ 31.25).	==	0.14
metX.ANT.1		Maximum ratio of metX perimeter area, where metX = met3–met5, to connected gate area.	≤	400
poly.ANT.1		Maximum ratio of poly perimeter area to connected gate area.	≤	50
viaY.ANT.1		Maximum ratio of viaY area, where viaY = via1–via4, to connected gate area.	≤	6

1. This rule is recommended at any IP level.

3.4.38 Bipolar Junction Transistor Design Rules

These design rules are used to define SkyWater-provided npn and pnp bipolar junction transistors. Placement and surrounding guard rings of bipolar junction transistors must adhere to the design rules described in *Section 3.4.54 Latchup Protection Design Rules* on page 91.

Note: The S130 PDK does not support the use of npn and pnp devices other than those provided by SkyWater.

Table 3-54. Bipolar Junction Transistor Design Rules

Rule No.	Description	Operator	Value
npn.CON.1	npn must be used inside pcell npn_1x1, npn_1x1_v5, or npn_1x2 only.		–
pnp.CON.1	pnp must be used inside pcell pnp2 or pnp_5x only.		–

3.4.39 12 V Drain-Extended NMOS Device Design Rules

These design rules are used to define a 12 V drain-extended NMOS device.

Table 3-55. 12 V Drain-Extended NMOS Device Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
denmos.CON.1		90° angles are prohibited for nwell overlapping de_nfet_drain.		–
denmos.CON.2		nsdm must enclose de_nfet_source.		–
denmos.ENC.1		de_nfet_source minimum/maximum extension beyond nwell.	==	0.225
denmos.ENC.2		nwell minimum enclosure of de_nfet_drain.	≥	0.66
denmos.ENC.3		nsdm minimum enclosure of de_nfet_source.	≥	0.13
denmos.SP.1		de_nfet_source minimum/maximum space to de_nfet_drain.	==	1.585
denmos.SP.2		ptap minimum space to (nwell overlapping de_nfet_drain).	≥	0.86
denmos.SP.3		(nwell overlapping de_nfet_drain) minimum spacing.	≥	2.4
denmos.WID.1		de_nfet_gate minimum width.	≥	1.055
denmos.WID.2		(de_nfet_source not overlapping poly) minimum width.	≥	0.28
denmos.WID.3		(de_nfet_source overlapping poly) minimum width.	≥	0.925
denmos.WID.4		de_nfet_drain minimum width.	≥	0.17
denmos.WID.5		de_nfet_gate minimum channel width.	≥	5.0
NC	1	nwell bevels must be 45° angles and 0.43 µm from corners.		–
NC	1	de_nfet_drain bevels must be 45° angles and 0.05 µm from corners.		–

1. This rule is not checked during DRC; use it as a guideline.

3.4.40 12 V Drain-Extended PMOS Device Design Rules

These design rules are used to define a 12 V drain-extended PMOS device.

Table 3-56. 12 V Drain-Extended PMOS Device Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
depmos.CON.1		90° angles are prohibited for nwell hole overlapping de_pfet_drain.		–
depmos.CON.2		psdm must enclose de_pfet_source.		–
depmos.ENC.1		nwell hole minimum enclosure of de_pfet_drain.	≥	0.86
depmos.ENC.2		psdm minimum enclosure of de_pfet_source.	≥	0.13
depmos.EXT.1		de_pfet_source minimum/maximum extension beyond nwell.	==	0.26
depmos.SP.2		de_pfet_source minimum/maximum space to de_pfet_drain.	==	1.19
depmos.SP.3		ntap minimum space to (nwell hole enclosing de_pfet_drain).	≥	0.66
depmos.WID.1		de_pfet_gate minimum width.	≥	1.05
depmos.WID.2		(de_pfet_source not overlapping poly) minimum width.	≥	0.28
depmos.WID.3		(de_pfet_source overlapping poly) minimum width.	≥	0.92
depmos.WID.4		de_pfet_drain minimum width.	≥	0.17
depmos.WID.5		de_pfet_gate minimum channel width.	≥	5.0
NC	1	nwell hole bevels must be 45° angles and 0.43 µm from corners.		–
NC	1	de_pfet_drain bevels must be 45° angles and 0.05 µm from corners.		–

1. This rule is not checked during DRC; use it as a guideline.

3.4.41 20 V Drain-Extended NMOS Device Design Rules

These design rules are used to define a 20 V drain-extended NMOS device.

Table 3-57. 20 V Drain-Extended NMOS Device Design Rules (Page 1 of 3)

Rule No.	Description	Operator	Value (µm)
denmos_20.ANG.1	90° corners on drain_diff of standard 20 V drain-extended NMOS device are prohibited.		–
denmos_20.ANG.2	90° corners on drain_diff of native 20 V drain-extended NMOS device are prohibited.		–
denmos_20.ANG.3	90° corners on drain_diff of zero-V_t 20 V drain-extended NMOS device are prohibited.		–
denmos_20.ANG.4	90° corners on drain_diff of isolated 20 V drain-extended NMOS device are prohibited.		–
denmos_20.ENC.1	dnwell minimum enclosure of drain tap in the direction of current flow within standard 20 V drain-extended NMOS device.	≥	3.5
denmos_20.ENC.2	pwbm minimum enclosure of dnwell of standard 20 V drain-extended NMOS device.	≥	0.5
denmos_20.ENC.3	nwell minimum enclosure of drain of standard 20 V drain-extended NMOS device.	≥	0.05
denmos_20.ENC.4	dnwell minimum enclosure of drain tap in the direction of current flow within native 20 V drain-extended NMOS device.	≥	3.5

Table 3-57. 20 V Drain-Extended NMOS Device Design Rules (Page 2 of 3)

Rule No.	Description	Operator	Value (μm)
denmos_20.ENC.5	pwbm minimum enclosure of dnwell of native 20 V drain-extended NMOS device.	≥	0.5
denmos_20.ENC.6	nwell minimum enclosure of drain of native 20 V drain-extended NMOS device.	≥	0.05
denmos_20.ENC.7	dnwell minimum enclosure of drain tap in the direction of current flow within zero-V_t 20 V drain-extended NMOS device.	≥	3.0
denmos_20.ENC.8	pwbm minimum enclosure of dnwell of zero-V_t 20 V drain-extended NMOS device.	≥	6.0
denmos_20.ENC.9	nwell minimum enclosure of drain of zero-V_t 20 V drain-extended NMOS device.	≥	0.05
denmos_20.ENC.10	nwell minimum enclosure of drain of isolated 20 V drain-extended NMOS device.	≥	0.05
denmos_20.EXT.1	poly field minimum extension beyond diff of standard 20 V drain-extended NMOS device.	≥	1.0
denmos_20.EXT.2	poly field minimum extension beyond diff of native 20 V drain-extended NMOS device.	≥	1.5
denmos_20.EXT.3	poly field minimum extension beyond diff of zero-V_t 20 V drain-extended NMOS device.	≥	1.0
denmos_20.EXT.4	poly field minimum extension beyond diff of isolated 20 V drain-extended NMOS device.	≥	1.0
denmos_20.LEN.1	Standard 20 V drain-extended NMOS device minimum channel width.	≥	30.0
denmos_20.LEN.2	Native 20 V drain-extended NMOS device minimum channel width.	≥	30.0
denmos_20.LEN.3	Zero-V_t 20 V drain-extended NMOS device minimum channel width.	≥	30.0
denmos_20.LEN.4	Isolated 20 V drain-extended NMOS device minimum channel width.	≥	30.0
denmos_20.OVL.1	dnwell minimum extension beyond standard 20 V drain-extended NMOS device channel.	≥	0.5
denmos_20.OVL.2	dnwell minimum extension beyond native 20 V drain-extended NMOS device channel.	≥	0.5
denmos_20.OVL.3	lvtn must enclose native 20 V drain-extended NMOS device.		–
denmos_20.OVL.4	dnwell minimum extension beyond zero-V_t 20 V drain-extended NMOS device channel.	≥	0.5
denmos_20.OVL.5	pwbm minimum extension beyond isolated 20 V drain-extended NMOS device channel.	≥	1.0
denmos_20.SP.1	drain_diff minimum space to gate or source_diff of standard 20 V drain-extended NMOS device.	≥	3.0
denmos_20.SP.2	ptap minimum space to source of standard 20 V drain-extended NMOS device.	≥	0.5
denmos_20.SP.3	drain_diff minimum space to gate or source_diff of native 20 V drain-extended NMOS device.	≥	3.0
denmos_20.SP.4	ptap minimum space to source of native 20 V drain-extended NMOS device.	≥	0.5
denmos_20.SP.5	drain_diff minimum space to gate or source_diff of zero-V_t 20 V drain-extended NMOS device.	≥	2.0
denmos_20.SP.6	ptap minimum space to source of zero-V_t 20 V drain-extended NMOS device.	≥	0.5
denmos_20.SP.7	drain_diff minimum space to gate or source_diff of isolated 20 V drain-extended NMOS device.	≥	2.0
denmos_20.SP.8	ptap minimum space to source of isolated 20 V drain-extended NMOS device.	≥	0.5

Table 3-57. 20 V Drain-Extended NMOS Device Design Rules (Page 3 of 3)

Rule No.	Description	Operator	Value (µm)
denmos_20.WID.1	gate_poly in standard 20 V drain-extended NMOS device minimum width.	≥	3.0
denmos_20.WID.2	source in standard 20 V drain-extended NMOS device minimum width.	≥	0.29
denmos_20.WID.3	gate of standard 20 V drain-extended NMOS device minimum width.	≥	1.5
denmos_20.WID.4	drain of standard 20 V drain-extended NMOS device minimum width.	≥	0.75
denmos_20.WID.5	Standard 20 V drain-extended NMOS device minimum channel length.	≥	0.5
denmos_20.WID.6	gate_poly in native 20 V drain-extended NMOS device minimum width.	≥	3.0
denmos_20.WID.7	source in native 20 V drain-extended NMOS device minimum width.	≥	0.29
denmos_20.WID.8	gate of native 20 V drain-extended NMOS device minimum width.	≥	1.5
denmos_20.WID.9	drain of native 20 V drain-extended NMOS device minimum width.	≥	0.75
denmos_20.WID.10	Native 20 V drain-extended NMOS device minimum channel length.	≥	0.5
denmos_20.WID.11	gate_poly in zero-V_t 20 V drain-extended NMOS device minimum width.	≥	7.0
denmos_20.WID.12	source in zero-V_t 20 V drain-extended NMOS device minimum width.	≥	0.29
denmos_20.WID.13	gate of zero-V_t 20 V drain-extended NMOS device minimum width.	≥	6.0
denmos_20.WID.14	drain of zero-V_t 20 V drain-extended NMOS device minimum width.	≥	0.75
denmos_20.WID.15	Zero-V_t 20 V drain-extended NMOS device minimum channel length.	≥	5.5
denmos_20.WID.16	gate_poly in isolated 20 V drain-extended NMOS device minimum width.	≥	2.5
denmos_20.WID.17	source in isolated 20 V drain-extended NMOS device minimum width.	≥	0.63
denmos_20.WID.18	gate of isolated 20 V drain-extended NMOS device minimum width.	≥	1.5
denmos_20.WID.19	drain of isolated 20 V drain-extended NMOS device minimum width.	≥	1.5
denmos_20.WID.20	Isolated 20 V drain-extended NMOS device minimum channel length.	≥	0.5
denmos_20.XOR.1	lvtn must be coincident with pwbm of standard 20 V drain-extended NMOS device.		–
denmos_20.XOR.2	lvtn must be coincident with pwbm of zero-V_t 20 V drain-extended NMOS device.		–
denmos_20.XOR.3	lvtn inside poly must be coincident with pwbm inside poly of isolated 20 V drain-extended NMOS device.		–

3.4.42 20 V Drain-Extended PMOS Device Design Rules

These design rules are used to define a 20 V drain-extended PMOS device.

Table 3-58. 20 V Drain-Extended PMOS Device Design Rules

Rule No.	Description	Operator	Value (µm)
depmos_20.ANG.1	90° corners on drain_diff of 20 V drain-extended PMOS device are prohibited.		–
depmos_20.ENC.1	pwbm minimum enclosure of drain tap in the direction of current flow within 20 V drain-extended PMOS device.	≥	3.0
depmos_20.ENC.2	pwde minimum enclosure of drain tap in the direction of current flow within 20 V drain-extended PMOS device.	≥	2.5
depmos_20.ENC.3	pwdm of 20 V drain-extended PMOS device minimum enclosure of pwde.	≥	0.5
depmos_20.EXT.1	poly field minimum extension beyond diff of 20 V drain-extended PMOS device.	≥	0.5
depmos_20.LEN.1	20 V drain-extended PMOS device minimum channel width.	≥	30.0
depmos_20.SP.1	drain_diff minimum space to gate or source_diff of 20 V drain-extended PMOS device.	≥	1.5
depmos_20.SP.2	ntap minimum space to source of 20 V drain-extended PMOS device.	≥	0.29
depmos_20.WID.1	gate_poly in 20 V drain-extended PMOS device minimum width.	≥	2.0
depmos_20.WID.2	source in 20 V drain-extended PMOS device minimum width.	≥	0.29
depmos_20.WID.3	gate of 20 V drain-extended PMOS device minimum width.	≥	1.5
depmos_20.WID.4	drain of 20 V drain-extended PMOS device minimum width.	≥	0.75
depmos_20.WID.5	(pwde AND poly AND diff) minimum width.	≥	0.5
depmos_20.WID.6	20 V drain-extended PMOS device minimum channel length.	≥	0.5

3.4.43 Metal Fuse Design Rules

These design rules are used to define a metal fuse on the fourth metal (met4) layer.

Note: The S130 PDK does not currently support the metal fuse. Contact SkyWater Technology for more information.

Table 3-59. Metal Fuse Design Rules (Page 1 of 2)

Rule No.	Notes	Description	Operator	Value (µm)
fuse.ANG.1		fuse must be rectangular.		–
fuse.CON.1		target must not overlap met1.		–
fuse.CON.2		target must not overlap li.		–
fuse.CON.3		target must not overlap poly.		–
fuse.CON.4		target must not overlap diff.		–
fuse.CON.5		target must not overlap nwell.		–

1. This rule is not checked during DRC; use it as a guideline.

Table 3-59. Metal Fuse Design Rules (Page 2 of 2)

Rule No.	Notes	Description	Operator	Value (µm)
fuse.CON.6		target must not overlap met2.		–
fuse.CON.7		fuse_shield must be 2.4 µm × 0.5 µm.		–
fuse.CON.8		fuse_shield is permitted for non-isolated_fuse_edges only.		–
fuse.CON.9		fuse_shield is required between perimeter metal and non-isolated_fuse_edges.		–
fuse.CON.10		Only one fuse per met4 line is permitted.		–
fuse.CON.11		target must not overlap met5.		–
fuse.ENC.1		met4 maximum extension beyond target boundary.	≤	0.83
fuse.LEN.1		fuse inside met4 minimum length.	≥	7.2
fuse.LEN.2		fuse inside met4 maximum length.	≤	7.2
fuse.SP.1		target minimum spacing.	≥	2.75
fuse.SP.2		met4 connection to fuse minimum spacing.	≥	1.96
fuse.SP.3		target minimum space to met1.	≥	3.295
fuse.SP.4		target minimum space to li.	≥	3.295
fuse.SP.5		target minimum space to poly.	≥	2.655
fuse.SP.6		target minimum space to tap.	≥	2.635
fuse.SP.7		target minimum space to diff.	≥	3.245
fuse.SP.8		target minimum space to nwell.	≥	3.315
fuse.SP.9		target minimum space to met2.	≥	3.295
fuse.SP.10		target minimum space to (met4 NOT fuse).	≥	3.295
fuse.SP.11		target minimum space to fuse_shield.	≥	2.195
fuse.SP.12		target maximum space to fuse_shield.	≤	3.30
fuse.SP.13		fuse_shield minimum/maximum space to met4.	==	0.6
fuse.SP.14		target minimum space to met5.	≥	3.295
fuse.WID.1		fuse inside met4 minimum width.	≥	0.8
fuse.WID.2		fuse inside met4 maximum width.	≤	0.8
NC	1	fuse_shield center space to fuse center.	==	0

1. This rule is not checked during DRC; use it as a guideline.

3.4.44 MIM Capacitor (over met3) Design Rules

These design rules are used to define a metal-insulator-metal (MIM) capacitor placed between the third metal (met3) and fourth metal (met4) layers.

Table 3-60. MIM Capacitor (over met3) Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
capm.ANG.3		capm edge must be octagonal.		–
capm.AR.1		capm maximum area (μm^2).	≤	1×10^7
capm.CON.1		ccapm mask is prohibited.		–
capm.CON.2		capm without met3 or met4 is prohibited.		–
capm.CON.3		capm without via3 is prohibited.		–
capm.CON.4		capm maximum aspect ratio (length to width).	≤	20:1
capm.CON.5		capm must be rectangular.		–
capm.CON.6		capm must not overlap via2.		–
capm.ENC.1	**1**	met3 minimum enclosure of capm.	≥	0.14
capm.ENC.2	**1**	capm minimum enclosure of via3.	≥	0.14
capm.SP.1	**1**	capm minimum spacing.	≥	0.84
capm.SP.2	**1**	capm bottom plate minimum spacing.	≥	1.2
capm.SP.3	**1**	capm minimum space to via3.	≥	0.14
capm.SP.4		capm minimum space to via2.	≥	0.14
capm.SP.5		capm minimum space to (met3 not overlapping capm).	≥	0.50
capm.WID.1	**1**	capm minimum width.	≥	1.0

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.4.45 MIM Capacitor (over met4) Design Rules

These design rules are used to define a metal-insulator-metal (MIM) capacitor placed between the fourth metal (met4) and fifth metal (met5) layers.

Table 3-61. MIM Capacitor (over met4) Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
cap2m.ANG.3		cap2m edge must be octagonal.		–
cap2m.AR.1		cap2m maximum area (μm^2).	≤	1×10^7
cap2m.CON.1		cap2m without met4 or met5 is prohibited.		–
cap2m.CON.2		cap2m without via4 is prohibited.		–
cap2m.CON.3		cap2m maximum aspect ratio (length to width).	≤	20:1
cap2m.CON.4		cap2m must be rectangular.		–
cap2m.CON.5		cap2m must not overlap via3.		–
cap2m.ENC.1	**1**	met4 minimum enclosure of cap2m.	≥	0.14
cap2m.ENC.2	**1**	cap2m minimum enclosure of via4.	≥	0.20
cap2m.SP.1	**1**	cap2m minimum spacing.	≥	0.84
cap2m.SP.2	**1**	cap2m bottom plate minimum spacing.	≥	1.2
cap2m.SP.3	**1**	cap2m minimum space to via4.	≥	0.20
cap2m.SP.4		cap2m minimum space to via3.	≥	0.14
cap2m.SP.5		cap2m minimum space to (met4 not overlapping cap2m).	≥	0.50
cap2m.WID.1	**1**	cap2m minimum width.	≥	1.0

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.4.46 Photodiode Design Rules

These design rules are used to define a photodiode for sensing light.

Note: The S130 PDK does not currently support the photodiode. Contact SkyWater Technology for more information.

Table 3-62. Photodiode Design Rules

Rule No.	Description	Operator	Value (µm)
photo.CON.1	Photodiode edges must be coincident with areaid:photo.		–
photo.CON.2	dnwell ring must enclose areaid:photo.		–
photo.CON.3	ptap ring must enclose areaid:photo.		–
photo.ENC.1	areaid:photo minimum/maximum enclosure of nwell.	==	1.08
photo.ENC.2	nwell inside areaid:photo minimum/maximum enclosure of tap.	==	0.215
photo.SP.1	areaid:photo minimum space and notch.	≥	5.0
photo.SP.2	areaid:photo minimum space to dnwell.	≥	5.3
photo.WID.1	areaid:photo minimum/maximum width.	==	3.0
photo.WID.2	nwell inside areaid:photo minimum/maximum width.	==	0.84
photo.WID.3	tap inside areaid:photo minimum/maximum width.	==	0.41

3.4.47 P+ Poly Resistor Protection Design Rules

The rpm drawn layer is used to define a p+ poly resistor (330 Ω/sq.). The rpm created layer is used to protect both p+ poly and p- poly resistor bodies from n+ poly gate implants.

Table 3-63. P+ Poly Resistor Protection Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
rpm.ANG.3		rpm edge must be octagonal.		–
rpm.ENC.1		rpm minimum enclosure of prec_resistor.	≥	0.20
rpm.OVL.1		rpm must not overlap nsdm.		–
rpm.OVL.2		poly must not straddle rpm.		–
rpm.OVL.3		rpm must not overlap pwbm.		–
rpm.OVL.4		rpm must overlap poly.		–
rpm.SP.1	**1**	rpm minimum space and notch.	≥	0.84
rpm.SP.2		rpm minimum space to nsdm.	≥	0.20
rpm.SP.3		rpm minimum space to poly.	≥	0.20
rpm.SP.4		rpm minimum space to pwbm.	≥	0.20
rpm.WID.1	**1**	rpm minimum width.	≥	1.27

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.4.48 P- Poly Resistor Protection Design Rules

These design rules are used to define an ultrahigh sheet rho p- poly resistor (2 kΩ/sq.). The urpm drawn layer defines the implant region, and it's merged with the rpm drawn layer to generate the rpm created layer that protects the p+ poly and p- poly resistors from n-type poly implants.

Table 3-64. P- Poly Resistor Protection Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
urpm.ANG.3		urpm edge must be octagonal.		–
urpm.ENC.1		urpm minimum enclosure of prec_resistor.	≥	0.20
urpm.OVL.1		urpm must not overlap nsdm.		–
urpm.OVL.2		poly must not straddle urpm.		–
urpm.OVL.3		urpm must overlap poly.		–
urpm.SP.1	**1**	urpm minimum space and notch.	≥	0.84
urpm.SP.2		urpm minimum space to nsdm.	≥	0.20
urpm.SP.3		urpm minimum space to poly.	≥	0.20
urpm.WID.1	**1**	urpm minimum width.	≥	1.27

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.4.49 Precision Resistor Design Rules

These design rules are used to define a precision resistor.

Table 3-65. Precision Resistor Design Rules

Rule No.	Description	Operator	Value (µm)
prec_res.CON.1	psdm must enclose prec_resistor.		–
prec_res.CON.2	npc must enclose prec_resistor.		–
prec_res.ENC.1	psdm minimum enclosure of prec_resistor.	≥	0.11
prec_res.ENC.2	npc minimum enclosure of prec_resistor.	≥	0.095
prec_res.OVL.1	prec_resistor must not overlap hvntm.		–
prec_res.SP.1	prec_resistor minimum space to hvntm.	≥	0.185

3.4.50 P-Well Resistor Design Rules

These design rules are used to define a p-well resistor. A p-well resistor is placed over an n-well opening enclosed by a deep n-well.

Table 3-66. P-Well Resistor Design Rules

Rule No.	Description	Operator	Value (µm)
pwres.CON.1	dnwell and nwell hole must enclose pwell:res.		–
pwres.CON.2	pwell:res enclosed by dnwell must be rectangular.		–
pwres.CON.3	pwell:res width must not exceed ptap width.		–
pwres.CON.4	ptap abutting pwell resistor must enclose 12 licon shapes.		–
pwres.CON.5	pwell:res must be coincident inside nwell hole and divide nwell hole into two nets.		–
pwres.CON.5a	diff or poly inside pwell:res is prohibited.		–
pwres.CON.6	ntap inside nwell ring of pwell resistor must have metal straps.		–
pwres.CON.7	pwell:res must abut pwell resistor terminals on opposite and parallel edges.		–
pwres.CON.8	pwell:res must abut nwell edges on opposite sides.		–
pwres.LEN.1	pwell resistor minimum length.	≥	26.5
pwres.LEN.2	pwell resistor maximum length.	≤	265
pwres.SP.1	ptap abutting pwell:res minimum/maximum space to nwell.	==	0.22
pwres.WID.1	pwell:res minimum/maximum width.	==	2.65
pwres.WID.2	pwell resistor terminal minimum width.	≥	0.53

3.4.51 Chip Pattern Density Design Rules

These design rules are used to define chip pattern density.

Table 3-67. Chip Pattern Density Design Rules (Page 1 of 4)

Rule No.	Notes	Description	Operator	Value
diff.chip.high.DEN.1	**1**	diff maximum chip pattern density (%) using 500 µm^2 window stepped at 100 µm.	≤	62
diff.chip.low.DEN.1	**1**	diff minimum chip pattern density (%) using 500 µm^2 window stepped at 100 µm.	≥	28
poly.chip.high.DEN.1		poly maximum chip pattern density (%) using 500 µm^2 window stepped at 100 µm.	≤	52
poly.chip.low.DEN.1		poly minimum chip pattern density (%) using 500 µm^2 window stepped at 100 µm.	≥	30
met1.chip.high.DEN.4		met1 maximum chip pattern density (%) using 700 µm^2 window stepped at 70 µm.	≤	80
met1.chip.low.DEN.3		met1 minimum chip pattern density (%) using 700 µm^2 window stepped at 70 µm.	≥	30

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

Table 3-67. Chip Pattern Density Design Rules (Page 2 of 4)

Rule No.	Notes	Description	Operator	Value
met2.chip.high.DEN.4		met2 maximum chip pattern density (%) using 700 μm^2 window stepped at 70 µm.	≤	80
met2.chip.low.DEN.3		met2 minimum chip pattern density (%) using 700 μm^2 window stepped at 70 µm.	≥	30
met3.chip.high.DEN.4		met3 maximum chip pattern density (%) using 700 μm^2 window stepped at 70 µm.	≤	80
met3.chip.low.DEN.3		met3 minimum chip pattern density (%) using 700 μm^2 window stepped at 70 µm.	≥	30
met4.chip.high.DEN.4		met4 maximum chip pattern density (%) using 700 μm^2 window stepped at 70 µm.	≤	80
met4.chip.low.DEN.3		met4 minimum chip pattern density (%) using 700 μm^2 window stepped at 70 µm.	≥	30
met5.chip.high.DEN.4		met5 maximum chip pattern density (%) using 700 μm^2 window stepped at 70 µm.	≤	60
met5.chip.low.DEN.3		met5 minimum chip pattern density (%) using 700 μm^2 window stepped at 70 µm.	≥	19
met1.low.DEN_under_30_ fill_block.1_CHIP		met1 minimum chip pattern density (%) when 700 μm^2 window is covered 30–40% by met1_fill block.	≥	30
met1.low.DEN_under_30_ fill_block.1_IP		met1 minimum IP pattern density (%) when 700 μm^2 window is covered 30–40% by met1_fill block.	≥	30
met1.low.DEN_under_40_ fill_block.1_CHIP		met1 minimum chip pattern density (%) when 700 μm^2 window is covered 40–50% by met1_fill block.	≥	40
met1.low.DEN_under_40_ fill_block.1_IP		met1 minimum IP pattern density (%) when 700 μm^2 window is covered 40–50% by met1_fill block.	≥	40
met1.low.DEN_under_50_ fill_block.1_CHIP		met1 minimum chip pattern density (%) when 700 μm^2 window is covered 50–60% by met1_fill block.	≥	50
met1.low.DEN_under_50_ fill_block.1_IP		met1 minimum IP pattern density (%) when 700 μm^2 window is covered 50–60% by met1_fill block.	≥	50
met1.low.DEN_under_60_ fill_block.1_CHIP		met1 minimum chip pattern density (%) when 700 μm^2 window is covered 60–80% by met1_fill block.	≥	60
met1.low.DEN_under_60_ fill_block.1_IP		met1 minimum IP pattern density (%) when 700 μm^2 window is covered 60–80% by met1_fill block.	≥	60
met1.low.DEN_under_80_ fill_block.1_CHIP		met1 minimum chip pattern density (%) when 700 μm^2 window is covered 80–100% by met1_fill block.	≥	65
met1.low.DEN_under_80_ fill_block.1_IP		met1 minimum IP pattern density (%) when 700 μm^2 window is covered 80–100% by met1_fill block.	≥	65
met1.low.DEN_under_100_ fill_block.1_CHIP		met1 minimum chip pattern density (%) when 700 μm^2 window is covered 100% by met1_fill block.	≥	70
met1.low.DEN_under_100_ fill_block.1_IP		met1 minimum IP pattern density (%) when 700 μm^2 window is covered 100% by met1_fill block.	≥	70

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

Table 3-67. Chip Pattern Density Design Rules (Page 3 of 4)

Rule No.	Notes	Description	Operator	Value
met2.low.DEN_under_30_fill_block.1_CHIP		met2 minimum chip pattern density (%) when 700 μm^2 window is covered 30–40% by met2_fill block.	≥	30
met2.low.DEN_under_30_fill_block.1_IP		met2 minimum IP pattern density (%) when 700 μm^2 window is covered 30–40% by met2_fill block.	≥	30
met2.low.DEN_under_40_fill_block.1_CHIP		met2 minimum chip pattern density (%) when 700 μm^2 window is covered 40–50% by met2_fill block.	≥	40
met2.low.DEN_under_40_fill_block.1_IP		met2 minimum IP pattern density (%) when 700 μm^2 window is covered 40–50% by met2_fill block.	≥	40
met2.low.DEN_under_50_fill_block.1_CHIP		met2 minimum chip pattern density (%) when 700 μm^2 window is covered 50–60% by met2_fill block.	≥	50
met2.low.DEN_under_50_fill_block.1_IP		met2 minimum IP pattern density (%) when 700 μm^2 window is covered 50–60% by met2_fill block.	≥	50
met2.low.DEN_under_60_fill_block.1_CHIP		met2 minimum chip pattern density (%) when 700 μm^2 window is covered 60–80% by met2_fill block.	≥	60
met2.low.DEN_under_60_fill_block.1_IP		met2 minimum IP pattern density (%) when 700 μm^2 window is covered 60–80% by met2_fill block.	≥	60
met2.low.DEN_under_80_fill_block.1_CHIP		met2 minimum chip pattern density (%) when 700 μm^2 window is covered 80–100% by met2_fill block.	≥	65
met2.low.DEN_under_80_fill_block.1_IP		met2 minimum IP pattern density (%) when 700 μm^2 window is covered 80–100% by met2_fill block.	≥	65
met2.low.DEN_under_100_fill_block.1_CHIP		met2 minimum chip pattern density (%) when 700 μm^2 window is covered 100% by met2_fill block.	≥	70
met2.low.DEN_under_100_fill_block.1_IP		met2 minimum IP pattern density (%) when 700 μm^2 window is covered 100% by met2_fill block.	≥	70
met3.low.DEN_under_30_fill_block.1_CHIP		met3 minimum chip pattern density (%) when 700 μm^2 window is covered 30–40% by met3_fill block.	≥	30
met3.low.DEN_under_30_fill_block.1_IP		met3 minimum IP pattern density (%) when 700 μm^2 window is covered 30–40% by met3_fill block.	≥	30
met3.low.DEN_under_40_fill_block.1_CHIP		met3 minimum chip pattern density (%) when 700 μm^2 window is covered 40–50% by met3_fill block.	≥	40
met3.low.DEN_under_40_fill_block.1_IP		met3 minimum IP pattern density (%) when 700 μm^2 window is covered 40–50% by met3_fill block.	≥	40
met3.low.DEN_under_50_fill_block.1_CHIP		met3 minimum chip pattern density (%) when 700 μm^2 window is covered 50–60% by met3_fill block.	≥	50
met3.low.DEN_under_50_fill_block.1_IP		met3 minimum IP pattern density (%) when 700 μm^2 window is covered 50–60% by met3_fill block.	≥	50
met3.low.DEN_under_60_fill_block.1_CHIP		met3 minimum chip pattern density (%) when 700 μm^2 window is covered 60–80% by met3_fill block.	≥	60
met3.low.DEN_under_60_fill_block.1_IP		met3 minimum IP pattern density (%) when 700 μm^2 window is covered 60–80% by met3_fill block.	≥	60

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

Table 3-67. Chip Pattern Density Design Rules (Page 4 of 4)

Rule No.	Notes	Description	Operator	Value
met3.low.DEN_under_80_ fill_block.1_CHIP		met3 minimum chip pattern density (%) when 700 μm^2 window is covered 80–100% by met3_fill block.	≥	65
met3.low.DEN_under_80_ fill_block.1_IP		met3 minimum IP pattern density (%) when 700 μm^2 window is covered 80–100% by met3_fill block.	≥	65
met3.low.DEN_under_100_ fill_block.1_CHIP		met3 minimum chip pattern density (%) when 700 μm^2 window is covered 100% by met3_fill block.	≥	70
met3.low.DEN_under_100_ fill_block.1_IP		met3 minimum IP pattern density (%) when 700 μm^2 window is covered 100% by met3_fill block.	≥	70
met4.low.DEN_under_30_ fill_block.1_CHIP		met4 minimum chip pattern density (%) when 700 μm^2 window is covered 30–40% by met4_fill block.	≥	30
met4.low.DEN_under_30_ fill_block.1_IP		met4 minimum IP pattern density (%) when 700 μm^2 window is covered 30–40% by met4_fill block.	≥	30
met4.low.DEN_under_40_ fill_block.1_CHIP		met4 minimum chip pattern density (%) when 700 μm^2 window is covered 40–50% by met4_fill block.	≥	40
met4.low.DEN_under_40_ fill_block.1_IP		met4 minimum IP pattern density (%) when 700 μm^2 window is covered 40–50% by met4_fill block.	≥	40
met4.low.DEN_under_50_ fill_block.1_CHIP		met4 minimum chip pattern density (%) when 700 μm^2 window is covered 50–60% by met4_fill block.	≥	50
met4.low.DEN_under_50_ fill_block.1_IP		met4 minimum IP pattern density (%) when 700 μm^2 window is covered 50–60% by met4_fill block.	≥	50
met4.low.DEN_under_60_ fill_block.1_CHIP		met4 minimum chip pattern density (%) when 700 μm^2 window is covered 60–80% by met4_fill block.	≥	60
met4.low.DEN_under_60_ fill_block.1_IP		met4 minimum IP pattern density (%) when 700 μm^2 window is covered 60–80% by met4_fill block.	≥	60
met4.low.DEN_under_80_ fill_block.1_CHIP		met4 minimum chip pattern density (%) when 700 μm^2 window is covered 80–100% by met4_fill block.	≥	65
met4.low.DEN_under_80_ fill_block.1_IP		met4 minimum IP pattern density (%) when 700 μm^2 window is covered 80–100% by met4_fill block.	≥	65
met4.low.DEN_under_100_ fill_block.1_CHIP		met4 minimum chip pattern density (%) when 700 μm^2 window is covered 100% by met4_fill block.	≥	70
met4.low.DEN_under_100_ fill_block.1_IP		met4 minimum IP pattern density (%) when 700 μm^2 window is covered 100% by met4_fill block.	≥	70

1. This **manufacturing rule** must not be waived without written approval from the SkyWater fab management team.

3.4.52 Local Pattern Density Design Rules

These design rules are used to define local pattern density. Local pattern density checks are used only to isolate chip-level pattern density violations. Local pattern density violations are not necessarily true pattern density violations. Examine and resolve local pattern density violations only when chip-level pattern density violations of the same name occur.

Table 3-68. Local Pattern Density Design Rules

Rule No.	Description	Operator	Value (µm)
diff.local.high.DEN.1	diff maximum local pattern density (%) using 50 μm^2 window stepped at 25 µm.	≤	62
diff.local.low.DEN.1	diff minimum local pattern density (%) using 50 μm^2 window stepped at 25 µm.	≥	28
poly.local.high.DEN.1	poly maximum local pattern density (%) using 50 μm^2 window stepped at 25 µm.	≤	40
poly.local.low.DEN.1	poly minimum local pattern density (%) using 50 μm^2 window stepped at 25 µm.	≥	30
met1.local.high.DEN.2	met1 maximum local pattern density (%) using 200 μm^2 window stepped at 100 µm.	≤	80
met1.local.low.DEN.1	met1 minimum local pattern density (%) using 200 μm^2 window stepped at 100 µm.	≥	30
met2.local.high.DEN.2	met2 maximum local pattern density (%) using 200 μm^2 window stepped at 100 µm.	≤	80
met2.local.low.DEN.1	met2 minimum local pattern density (%) using 200 μm^2 window stepped at 100 µm.	≥	30
met3.local.high.DEN.2	met3 maximum local pattern density (%) using 200 μm^2 window stepped at 100 µm.	≤	80
met3.local.low.DEN.1	met3 minimum local pattern density (%) using 200 μm^2 window stepped at 100 µm.	≥	30
met4.local.high.DEN.2	met4 maximum local pattern density (%) using 200 μm^2 window stepped at 100 µm.	≤	80
met4.local.low.DEN.1	met4 minimum local pattern density (%) using 200 μm^2 window stepped at 100 µm.	≥	30
met5.local.high.DEN.2	met5 maximum local pattern density (%) using 200 μm^2 window stepped at 100 µm.	≤	60
met5.local.low.DEN.1	met5 minimum local pattern density (%) using 200 μm^2 window stepped at 100 µm.	≥	19

3.4.53 Fill Pattern Design Rules

These design rules are used to define fill patterns for meeting chip pattern density requirements.

Table 3-69. Fill Pattern Design Rules (Page 1 of 2)

Rule No.	Description	Operator	Value (µm)
diff_fill.CON.1	diff_fill must not overlap areaid:seal.		–
diff_fill.CON.2	diff_fill must not overlap fuse.		–
diff_fill.CON.3	diff_fill must not overlap nsdm.		–
diff_fill.CON.4	diff_fill must not overlap psdm.		–
diff_fill.ENC.1	nwell minimum enclosure of diff_fill.	≥	0.18
diff_fill.ENC.2	areaid:hvnwell minimum enclosure of diff_fill.	≥	0.43
diff_fill.ENC.3	areaid:frame minimum enclosure of diff_fill.	≥	0.50
diff_fill.SP.1	diff_fill minimum space and notch.	≥	0.40
diff_fill.SP.2	diff_fill minimum space to areaid:seal.	≥	1.00

Table 3-69. Fill Pattern Design Rules (Page 2 of 2)

Rule No.	Description	Operator	Value (µm)
diff_fill.SP.3	diff_fill minimum space to fuse.	≥	3.25
diff_fill.SP.4	diff_fill minimum space to nsdm.	≥	0.13
diff_fill.SP.5	diff_fill minimum space to psdm.	≥	0.13
diff_fill.SP.6	diff_fill minimum space to nwell.	≥	0.34
diff_fill.SP.7	diff_fill minimum space to hvnwell.	≥	0.33
diff_fill.SP.8	diff_fill minimum space to areaid:dieCut.	≥	0.50
diff_fill.WID.1	diff_fill minimum width.	≥	0.50
diff_fill.WID.2	diff_fill maximum width.	≤	25.00
fill.CON.1	diff_fill must float.		–
fill.CON.2	poly_fill must float.		–
fill.CON.3	li_fill must float.		–
fill.CON.4	met1_fill must float.		–
fill.CON.5	met2_fill must float.		–
fill.CON.6	met3_fill must float.		–
fill.CON.7	met4_fill must float.		–
fill.CON.8	met5_fill must float.		–
fill.OVL.1	diff_fill must be outside diff.		–
fill.OVL.2	poly_fill must be outside poly.		–
fill.OVL.3	li_fill must be outside li.		–
fill.OVL.4	met1_fill must be outside met1.		–
fill.OVL.5	met2_fill must be outside met2.		–
fill.OVL.6	met3_fill must be outside met3.		–
fill.OVL.7	met4_fill must be outside met4.		–
fill.OVL.8	met5_fill must be outside met5.		–

3.4.54 Latchup Protection Design Rules

These design rules are used to reduce or prevent latchup occurrences.

Table 3-70. Latchup Protection Design Rules (Page 1 of 3)

Rule No.	Notes	Description	Operator	Value (µm)
latchup.generic.2.1a_b		Center of ptap licon maximum space to ndiff inside pwell where dnwell or an abutting nwell does not enclose pdiff connected to a power supply pad.	≤	6
latchup.generic.2a		Center of ptap licon maximum space to ndiff inside the same ptub or pwell placed less than 50 µm from diff connected to a signal pad.	≤	6
latchup.generic.2b		Center of ptap licon maximum space to ndiff inside the same ptub or pwell placed more than 50 µm from diff connected to a signal pad.	≤	15
latchup.generic.3a		Center of ntap licon maximum space to pdiff inside the same nwell or dnwell placed less than 50 µm from diff connected to a signal pad.	≤	6
latchup.generic.3b		Center of ntap licon maximum space to pdiff inside the same nwell or dnwell placed more than 50 µm from diff connected to a signal pad.	≤	15
latchup.generic.4		diff connected to a signal pad minimum space to areaid:core.	≥	50
latchup.generic.5a		ndiff minimum space (not notch) inside a common ptub or pwell connected to a different net.	≥	3
latchup.generic.5b		pdiff minimum space (not notch) inside a common nwell or dnwell connected to a different net.	≥	3
latchup.misc.4		nfet or pfet with source connected to one pad and drain connected to another pad minimum width.	≥	2000
latchup.shv.1		dnwell must connect to ground through a resistance ≥ 1000 Ω when dnwell contains active nfets or pfets connected to shv_nets.		–
latchup.shv.2		nwell or dnwell connected to an shv signal pad minimum space to pdiff connected to a power supply pad.	≥	70
latchup.shv.3		pwell connected to an shv signal pad minimum space to ndiff, ntap, nwell, or dnwell connected to ground.	≥	100
latchup.signal.2		A pair of guard rings must enclose nwell connected to a signal pad.		–
latchup.signal.2.1a		dnwell connected to a signal pad is prohibited.		–
latchup.signal.2.1b		dnwell must connect to ground through a resistance ≥ 250 Ω. Back-to-back diodes that protect separate ground supplies are exempt.		–
latchup.signal.3		A pair of guard rings must enclose pdiff or ptap connected to a signal pad.		–
latchup.signal.3.1a		ptap connected to a signal pad must have an nwell guard ring connected to a power supply pad.		–
latchup.signal.3.1b		ptap or p+ source/drain connected to a signal pad must have a ptap guard ring connected to ground.		–
latchup.signal.12a		diff connected to a signal pad minimum space to ndiff connected to ground.	≥	27
latchup.signal.12b		pwell connected to a signal pad minimum space to ndiff connected to ground.	≥	40

1. This rule is not checked during DRC; use it as a guideline.

Table 3-70. Latchup Protection Design Rules (Page 2 of 3)

Rule No.	Notes	Description	Operator	Value (µm)
latchup.signal.12c		pdiff connected to a signal pad minimum space to nwell connected to ground.	≥	40
latchup.signal.12d		pwell connected to a signal pad minimum space to nwell connected to ground.	≥	40
latchup.signal.12e		ndiff connected to a signal pad minimum space to pdiff connected to a power supply pad (exempt for ndiff inside pwell connected to the same signal pad and for ndiff inside ptub).	≥	27
latchup.signal.12f		nwell connected to a signal pad minimum space to pdiff connected to a power supply pad.	≥	40
latchup.signal.12g		ndiff connected to a signal pad minimum space to pwell connected to a power supply pad.	≥	40
latchup.signal.12h		nwell connected to a signal pad minimum space to pwell connected to a power supply pad.	≥	40
latchup.signal.12i		pdiff connected to a signal pad minimum space to ndiff connected to another signal pad.	≥	27
latchup.signal.12j		pdiff connected to a signal pad minimum space to nwell connected to another signal pad.	≥	40
latchup.signal.12k		pwell connected to a signal pad minimum space to ndiff connected to another signal pad.	≥	40
latchup.signal.12l		pwell connected to a signal pad minimum space to nwell connected to another signal pad.	≥	40
latchup.signal.12m		ndiff connected to a signal pad minimum space to pdiff inside an at_risk_non_Vcc_nwell.	≥	33
latchup.signal.12n		ndiff connected to a signal pad minimum space to an at_risk_non_Vcc_nwell.	≥	16.75
latchup.signal.12o		pdiff connected to a signal pad minimum space to nwell connected to a ≤ 1.8 V power supply pad.	≥	27
latchup.signal.12p		pwell connected to a signal pad minimum space to nwell connected to a ≤ 1.8 V power supply pad.	≥	40
latchup.signal.13		A PMOS device placed between the inner p+ and outer n+/nwell guard rings or inside the shared well of the outer n+/nwell guard ring is prohibited.		–
latchup.signal.14		diff:res inside diff connected to a signal pad or in the shared well of the outer n+/nwell guard ring is prohibited.		–
latchup.signal.15	1	Any circuitry within 100 µm of any diff or tap connected to a signal pad must have each FET source and body directly connected in metal.		–
latchup.special.1a.1		The inner ptap connected to ground must enclose an at_risk_non_Vcc_nwell.		–
latchup.special.1a.2		ptap enclosing an at_risk_non_Vcc_nwell must have a continuous li metal strap.		–

1. This rule is not checked during DRC; use it as a guideline.

Table 3-70. Latchup Protection Design Rules (Page 3 of 3)

Rule No.	Notes	Description	Operator	Value (µm)
latchup.special.1b.1		The outer n+/nwell guard ring connected to a power supply pad must enclose an at_risk_non_Vcc_nwell.		–
latchup.special.1b.2		ntap enclosing an at_risk_non_Vcc_nwell must have a continuous li metal strap.		–
latchup.special.1c.1		licon inside inner ptap enclosing at_risk_non_Vcc_nwell maximum spacing.	≤	2
latchup.special.1c.2		licon inside outer ntap enclosing at_risk_Vcc_non_nwell maximum spacing.	≤	2
latchup.special.2.a		The inner ptap guard ring connected to ground must enclose pnp.		–
latchup.special.2.b		The outer ntap or nwell guard ring connected to a power supply pad must enclose pnp.		–
latchup.special.2.c		licon inside inner ptap guard ring enclosing pnp maximum spacing.	≤	2
latchup.special.2.d		licon inside outer ntap guard ring enclosing pnp maximum spacing.	≤	2
latchup.special.6		ndiff placed between pdiff inside an at_risk_non_Vcc_nwell and the ptap guard ring is prohibited.		–
latchup.special.7.a		A ptap guard ring must enclose nwell connected to ground.		–
latchup.special.7.b		A ptap guard ring enclosing nwell connected to ground must also be connected to ground.		–
latchup.WARN.1		pad outside areaid:pad_io, pad_pwr, or pad_gnd is prohibited.		–

1. This rule is not checked during DRC; use it as a guideline.

3.4.55 Anchor Region Design Rules

These design rules are used to reinforce large brittle dielectric regions of the die.

Table 3-71. Anchor Region Design Rules

Rule No.	Description	Operator	Value (µm)
anchor.CON.1	Open area anchor regions are required in any 50 µm × 50 µm open window inside areaid:critCorner or areaid:critSid.		–
metX.anchor.connect.CON.1	metX inside anchor region, where metX = met1–met5, must not connect to any other metal bus.		–
licon.mcon.anchor.CON.2	licon must not overlap mcon within anchor region.		–
licon.viaY.anchor.CON.2	licon must not overlap viaY within anchor region, where viaY = via1–via4.		–
via1.mcon.anchor.CON.2	via1 must not overlap mcon within anchor region.		–
via1.viaY.anchor.CON.2	via1 must not overlap viaY within anchor region, where viaY = via2–via4.		–
via2.mcon.anchor.CON.2	via2 must not overlap mcon within anchor region.		–
via2.viaY.anchor.CON.2	via2 must not overlap viaY within anchor region, where viaY = via3 or via4.		–
via3.mcon.anchor.CON.2	via3 must not overlap mcon within anchor region.		–
via3.via4.anchor.CON.2	via3 must not overlap via4 within anchor region.		–
via4.mcon.anchor.CON.2	via4 must not overlap mcon within anchor region.		–
anchor.SP.1	Anchor region minimum space and notch.	≥	5.0
licon.anchor.SP.1	licon overlapping an anchor region minimum spacing.	≥	2.93
mcon.anchor.SP.1	mcon overlapping an anchor region minimum spacing.	≥	2.93
via1.anchor.SP.1	via1 overlapping an anchor region minimum spacing.	≥	2.95
via2.anchor.SP.1	via2 overlapping an anchor region minimum spacing.	≥	2.9
via3.anchor.SP.1	via3 overlapping an anchor region minimum spacing.	≥	2.9
via4.anchor.SP.1	via4 overlapping an anchor region minimum spacing.	≥	2.3
li.licon.anchor.WARN.1	This li anchor region must contain additional licon contacts.		–
li.mcon.anchor.WARN.1	This li anchor region must contain additional mcon contacts.		–
met1.mcon.anchor.WARN.1	This met1 anchor region must contain additional mcon contacts.		–
met1.via1.anchor.WARN.1	This met1 anchor region must contain additional via1 contacts.		–
met2.via1.anchor.WARN.1	This met2 anchor region must contain additional via1 contacts.		–
met2.via2.anchor.WARN.1	This met2 anchor region must contain additional via2 contacts.		–
met3.via2.anchor.WARN.1	This met3 anchor region must contain additional via2 contacts.		–
met3.via3.anchor.WARN.1	This met3 anchor region must contain additional via3 contacts.		–
poly.licon.anchor.WARN.1	This poly anchor region must contain additional licon contacts.		–
li.anchor.WID.1	li overlapping an anchor region minimum width.	≥	3.0
metX.anchor.WID.1	metX overlapping an anchor region, where metX = met1–met5, minimum width.	≥	3.0
poly.anchor.WID.1	poly overlapping an anchor region minimum width.	≥	3.0

3.4.56 Slotted Metal Design Rules

These design rules are used to minimize metal volume and reduce residual tensile stress.

Table 3-72. Slotted Metal Design Rules

Rule No.	Description	Operator	Value (µm)
metX.slot.CON.1	metX wider than 25 µm inside areaid:critCorner or areaid:critSid, where metX = met1–met5, must contain a slot.		–
metX.slot.CON.2	Start and end points of metX slots spaced ≤ 25 µm apart in adjacent rows, where metX = met1–met5, must be offset.		–
metX.slot.CON.9	metX in lower slotted stack must enclose met(X + 1) in upper slotted stack, where metX = met1–met4.		–
metX.slot.DEN.1	Slot minimum density (%) on wide_metX bus, where metX = met1–met5.	≥	7.5
metX.slot.LEN.1	metX slot maximum length, where metX = met1–met5.	≤	600
metX.slot.SP.1	metX slot minimum spacing, where metX = met1–met5.	≥	2.3
metX.slot.WID.1	metX slot minimum width, where metX = met1–met5.	≥	2.3
metX.slot.WID.2	metX slot maximum width, where metX = met1–met5.	≤	10

3.4.57 Metal Passivation Stress Reduction Design Rules

These design rules are used to define metal passivation stress reduction requirements.

Table 3-73. Metal Passivation Stress Reduction Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
diff.stress.1		diff outside areaid:seal must not overlap areaid:deadZon.		–
poly.stress.1		poly must be outside areaid:deadZon unless poly is within anchor region.		–
metXjstress.CON.1		90° bend of inner side of 5 µm wide metX inside areaid:critSid, where metX = met1–met5, is prohibited.		–
metX.stress.CON.2		90° bend of inner side of metX inside areaid:critCorner, where metX = met1–met5, is prohibited.		–
metX.stress.CON.3		Minimum length of inner 45° edge of ≥ 5 µm wide metX inside areaid: critCorner, where metX = met1–met5.	≥	5.0
metX.stress.CON.4		Minimum length of inner 45° edge of > 1 µm to < 5 µm wide metX inside areaid:critCorner, where metX = met1–met5.	≥	1.0
metX.stress.CON.5		Minimum length of inner 45° edge of < 1 µm wide metX inside areaid: critCorner, where metX = met1–met5.	≥	0.17
metX.stress.CON.6		90° bend at connection point of metX bus connecting pad, where metX = met4 or met5, is prohibited.		–
stress.CON.7		Minimum distance (A, B) between three consecutive 45° edges of a shape inside areaid:critCorner or areaid:critSid.	≥	2.3
metX.stress.DEN.1		viaY minimum density (%) in a wide_metX and wide_met(X + 1) bus_stack, where viaY = via1–via4 and metX = met1–met4.	≥	3
metX.stress.ENC.1		metX in slotted stack minimum enclosure of met(X + 1) in slotted stack, where metX = met1–met4.	≥	1.0
metX.stress.ENC.2		metX bus minimum enclosure of met(X – 1) in a bus_stack, where metX = met2–met5.	≥	1.0
metX.stress.SP.1		metX bus (≥ 5 µm wide by ≥ 10 µm long) minimum space to non-bus metX, where metX = met1–met5.	≥	0.54
metX.stress.SP.2		metX bus (≥ 5 µm wide by ≥ 10 µm long) minimum space and notch, where metX = met1–met5.	≥	0.54
metX.stress.SP.3	1	Standalone metX maximum space to non-standalone metX inside areaid: critCorner or areaid:critSid, where metX = met1–met5.	≤	10.0
metX.stress.WID.1		metX inside areaid:deadZon, where metX = met1–met5, minimum width.	≥	8.0

1. A standalone metal layer has no connectivity to an upper or lower metal layer through a contact or via.

3.4.58 LVS Exclusion Design Rules

These design rules are used to define the LVS_exclude marker layer.

Table 3-74. LVS Exclusion Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
LVS_exclude.CON.1		LVS_exclude must not straddle gate.		–
LVS_exclude.CON.2		LVS_exclude must not straddle n+ source or drain.		–
LVS_exclude.CON.3		LVS_exclude must not straddle p+ source or drain.		–
LVS_exclude.CON.4		LVS_exclude must not straddle capm.		–
LVS_exclude.CON.5		LVS_exclude must not straddle cap2m.		–
LVS_exclude.CON.6		LVS_exclude must not straddle pwell:res.		–
LVS_exclude.CON.7		LVS_exclude must not straddle diff:res.		–
LVS_exclude.CON.8		LVS_exclude must not straddle poly:res.		–
LVS_exclude.CON.9		LVS_exclude must not straddle li:res.		–
LVS_exclude.CON.10		LVS_exclude must not straddle met1:res.		–
LVS_exclude.CON.11		LVS_exclude must not straddle met2:res.		–
LVS_exclude.CON.12		LVS_exclude must not straddle met3:res.		–
LVS_exclude.CON.13		LVS_excludediff_fill.CON.2, diff_fill.SP.3, and LVS_exclude.CON.15 must not straddle met4:res.		–
LVS_exclude.CON.14		LVS_exclude must not straddle met5:res.		–
LVS_exclude.CON.15	1	LVS_exclude must not straddle fuse.		–
LVS_exclude.CON.16		LVS_exclude must not straddle areaid:diode.		–
LVS_exclude.CON.17		LVS_exclude must not straddle pnp.		–
LVS_exclude.CON.18		LVS_exclude must not straddle npn.		–
LVS_exclude.CON.19	1	LVS_exclude must not straddle areaid:photo.		–
LVS_exclude.CON.20		LVS_exclude must not straddle pad.		–
LVS_exclude.OVL.1		LVS_exclude must be inside areaid:moduleCut.		–
LVS_exclude.WARN.1		LVS_exclude must enclose a device.		–

1. This technology does not currently support this device.

3.5 Wafer-Level Chip Scale Packaging Design Rules

3.5.1 Redistribution Layer Design Rules

These design rules are used to define the layer that connects the top metal layer to the wafer-level chip scale package (WLCSP) solder bumps.

Table 3-75. Redistribution Layer Design Rules

Rule No.	Description	Operator	Value (µm)
rdl.CON.1	rdl or ccu1m mask must not overlap areaid:frameRect.		–
rdl.ENC.1	rdl outside bump minimum enclosure of pad.	≥	10.75
rdl.SP.1	rdl minimum spacing.	≥	10
rdl.SP.2	rdl minimum space to scribe_line.	≥	15
rdl.SP.3	rdl outside bump minimum space to pad.	≥	19.66
rdl.WID.1	rdl minimum width.	≥	10

3.5.2 Under Solder Bump Metal Design Rules

These design rules are used to define the metal layer placed underneath WLCSP solder bumps.

Table 3-76. Under Solder Bump Metal Design Rules

Rule No.	Description	Operator	Value (µm)
ubm.CON.1	ubm must not straddle areaid:moduleCut.		–
ubm.CON.2	ubm must be inside rdl.		–
ubm.ENC.1	rdl minimum enclosure of ubm.	≥	10
ubm.SP.1	Center of 400 µm pitch ubm minimum space to scribe_line.	≥	155
ubm.SP.2	Center of 500 µm pitch ubm minimum space to scribe_line.	≥	155
ubm.WID.1	400 µm pitch ubm minimum width (opposite parallel edges only).	≥	215
ubm.WID.2	500 µm pitch ubm minimum width (opposite parallel edges only).	≥	250

3.5.3 Solder Bump Design Rules

These design rules are used to define WLCSP solder bumps.

Table 3-77. Solder Bump Design Rules

Rule No.	Notes	Description	Operator	Value (µm)
bump.CON.1		bump must not straddle areaid:moduleCut.		–
bump.CON.2		chip_extent with 500 µm pitch bumps minimum size.	≥	1000
bump.CON.3		chip_extent with 500 µm pitch bumps maximum size.	≤	6800
bump.CON.4		chip_extent with 400 µm pitch bumps minimum size.	≥	750 × 1000
bump.CON.5		chip_extent with 400 µm pitch bumps maximum size.	≤	6800
bump.CON.6		Chip may contain either 400 µm pitch bumps or 500 µm pitch bumps, but not both.		–
bump.ENC.1		scribe_line minimum enclosure of bump.	≥	25
bump.SP.1		Minimum/maximum bump pitch spacing must be either 400 µm or 500 µm.		–
bump.WID.1		400 µm pitch bump minimum width.	≥	261
bump.WID.2		500 µm pitch bump minimum width.	≥	310
NC	1	Minimum/maximum bump pitch spacing must be either 400 µm or 500 µm across scribe_line.		–

1. This rule is not checked during DRC; use it as a guideline.

3.6 Recommended Design Rules

These design rules are used to identify soft-connection violations during layout versus schematic (LVS) checking and to support design best practices.

Note: Recommended design rules are optional in DRC.

Table 3-78. Recommended Design Rules

Rule No.	Description	Operator	Value (µm)
floating.net.li.R	Connect floating li net to a defined device or pad.		–
floating.net.met1.R	Connect floating met1 net to a defined device or pad.		–
floating.net.met2.R	Connect floating met2 net to a defined device or pad.		–
floating.net.met3.R	Connect floating met3 net to a defined device or pad.		–
floating.net.met4.R	Connect floating met4 net to a defined device or pad.		–
floating.net.met5.R	Connect floating met5 net to a defined device or pad.		–
floating.net.ntap.R	Connect floating ntap to pad.		–
floating.net.poly.R	Connect floating poly net to a defined device or pad.		–
floating.net.ptap.R	Connect floating ptap to pad.		–
lonely.diff.licon.R	Add more licon contacts to diff and/or li if space allows.		–
lonely.diff.licon.a.R	Increase diff and/or li to accommodate more licon contacts.		–
lonely.mcon.R	Add more mcon contacts to li and/or met1 if space allows.		–
lonely.mcon.a.R	Increase li and/or met1 to accommodate more mcon contacts.		–
lonely.poly.licon.R	Add more licon contacts to poly and/or li if space allows.		–
lonely.poly.licon.a.R	Increase poly and/or li to accommodate more licon contacts.		–
lonely.via1.R	Add more via1 contacts to met1 and/or met2 if space allows.		–
lonely.via1.a.R	Increase met1 and/or met2 to accommodate more via1 contacts.		–
lonely.via2.R	Add more via2 contacts to met2 and/or met3 if space allows.		–
lonely.via2.a.R	Increase met2 and/or met3 to accommodate more via2 contacts.		–
lonely.via3.R	Add more via3 contacts to met3 and/or met4 if space allows.		–
lonely.via3.a.R	Increase met3 and/or met4 to accommodate more via3 contacts.		–
lonely.via4.R	Add more via4 contacts to met4 and/or met5 if space allows.		–
lonely.via4.a.R	Increase met4 and/or met5 to accommodate more via4 contacts.		–

4 Company Contact Information

Get your design enablement questions answered through the SkyWater Technology online IP distribution and support portal.

For all other questions about SkyWater, send us a message through the company website: Contact Us | SkyWater Technology Foundry.